ongaku

Ongaku Studio

User Manual

This manual explains how to make music with Ongaku Studio, from your first project to mixing, session players and the Ongaku plug-ins. Each chapter explains what a feature is and when you'd use it, then shows you how, step by step.

If you're new to music apps like this, start with Chapter 1. It introduces the words the rest of the manual uses: project, track, region, playhead, cycle and so on. After that, read the chapters in any order.

Contents

Chapter	What it covers
1. Getting Started	Projects, tracks and regions; a tour of the window; creating and saving projects; the transport, zoom and undo
2. Arranging and Editing	Selecting, moving, nudging, copying, trimming, splitting and joining regions; the tools; the Region Inspector; fades, crossfades, loops, markers, the Arrangement, Tempo and Signature tracks; track stacks; importing audio
3. Recording, Takes and Comping	Audio and MIDI inputs, input monitoring, recording, the metronome and count-in, cycle recording into take folders, renaming and deleting takes, and comping the best parts of each take
4. MIDI Editing	Choosing instruments, the Piano Roll and its tools, quantize, the Step Editor, the Step Sequencer and MIDI Draw lanes
5. Session Players	The Drummer, the Chord track, and the Bass and Keyboard players that follow it
6. Flex Time, Flex Pitch and Smart Tempo	Moving the timing and pitch of recorded audio, and matching recordings and the project to each other's tempo
7. Mixing and Automation	The Mixer, plug-in inserts (including mastering effects on Stereo Out), sends and aux tracks, named mixer groups, freezing tracks and automation
8. Ongaku Instruments, Effects and MIDI FX	Every Ongaku effect and instrument with its controls and presets, Mastering, and the MIDI FX
9. Library, Bounce, MIDI Files and Settings	Finding sounds, bouncing your song to an audio file, MIDI file import and export, and every setting
10. Key Commands	Every keyboard shortcut and mouse gesture, by area
11. Making Music with AI Agents	Letting an AI agent such as Claude build, mix and master a song with you over MCP: setting it up, what it can do, and working with it

Conventions

- **Bold** text is a label you'll see on screen. **Track › New Audio Track** means: open the **Track** menu and choose **New Audio Track**.
- Keyboard shortcuts use the Mac's symbols: ⌘ Command, ⌥ Option, ⇧ Shift, ⌃ Control, ⌫ Delete. ⌥⌘U means hold Option and Command and press U.
- Many single-key shortcuts, such as Space to play or E to show the editor, are shown in the menus like any other shortcut, but only work while the project window is in front and you aren't typing in a text field. Chapter 10 explains how this works.

- **Settings › General** means the General tab of the Settings window. Open it with **Ongaku Studio › Settings…** (⌘,).

1. Getting Started

Ongaku Studio is a music-making app for the Mac, laid out like Apple's Logic Pro: you build a song from tracks, and each track holds blocks of music called regions that you can move, copy and edit. This chapter shows you around the window, explains how projects are created and saved, and covers the controls you'll use every few seconds: the transport, zoom and undo.

If you have never used a program like this, read this chapter from start to finish once. Everything else in the manual builds on the words and places introduced here.

What Ongaku Studio is

A song in Ongaku is a **project**. A project holds:

- **Tracks**, stacked one above the other. Each track is one part of the song: a drum kit, a bass, a vocal, a piano.
- **Regions** on those tracks. A region is a block of music laid along the timeline: either recorded or imported audio, or MIDI notes that an instrument plays. Chapter 2 is all about regions.
- **The song's tempo, time signature and key**, which decide where the bars and beats fall.
- **A mix**: each track's volume, pan, effects and where its sound goes (see Chapter 7).

Time runs from left to right, measured in bars and beats rather than seconds, so a region placed "at bar 9" stays at bar 9 even if you later change the tempo.

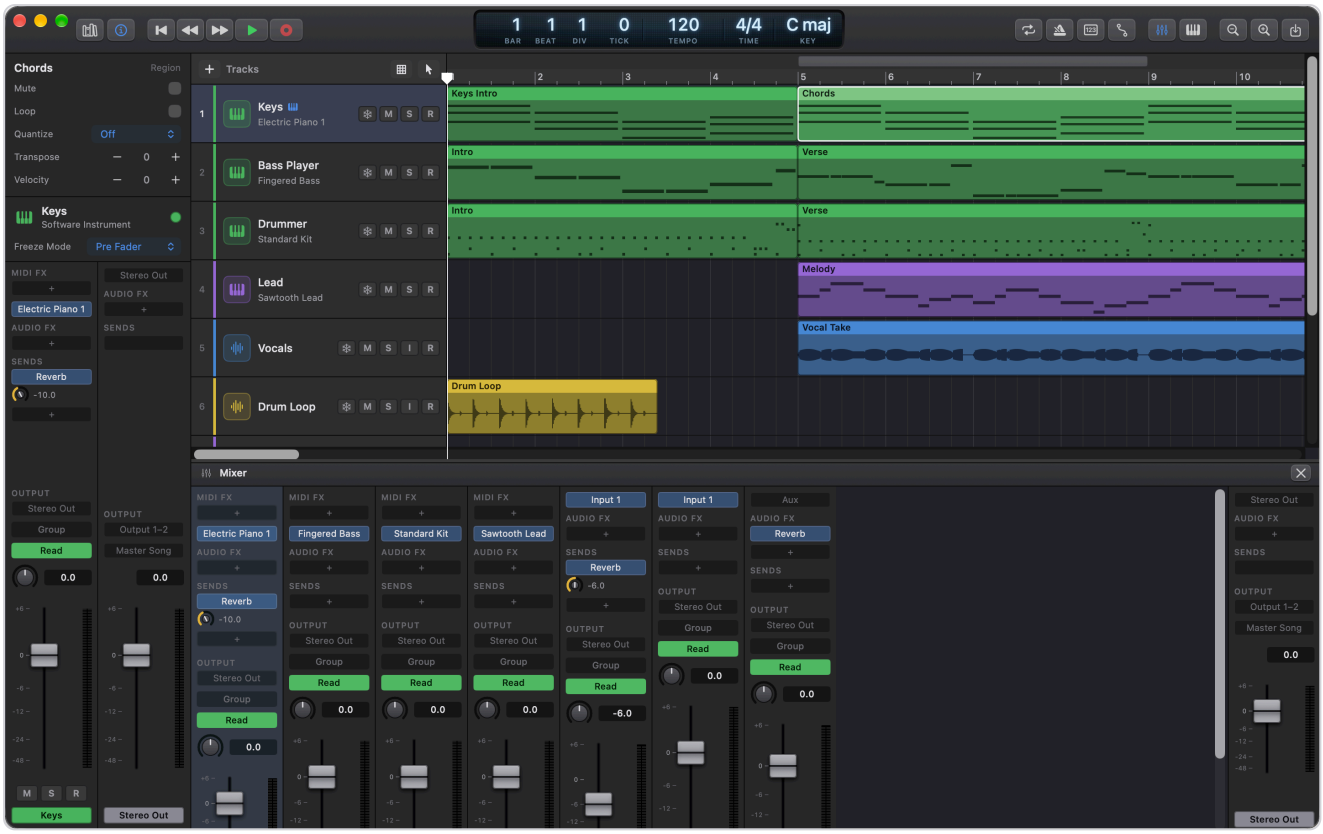
There are three kinds of track:

Track kind	What it holds	Typical use
Software Instrument	MIDI regions (notes), played by an instrument such as a piano, synth or drum kit	Anything you play on a keyboard, or draw in note by note
Audio	Audio regions (recordings and imported sound files)	Vocals, guitars, samples, loops
Aux	No regions: it only mixes the sound other tracks send to it	A shared reverb, or a group of tracks you want to control with one fader

Drummer, Bass Player and Keyboard Player tracks are software instrument tracks that come with a virtual musician already playing; they are described in Chapter 5.

A tour of the window

Each project opens in its own window. Ongaku always uses a dark appearance, whatever your Mac is set to.



The project window: the Inspector on the left, the control bar across the top, the tracks, and the Mixer below

From top to bottom and left to right, the window is made of these areas:

Area	Where it is	What it's for	Show or hide
Control bar	Across the top	Transport buttons, the LCD display (position, tempo, time signature, key) and buttons that show other areas	Always shown
Inspector	Down the left	The selected track's details and channel strip, and the Region Inspector when regions are selected	**I**, or the **Inspector** button
Library	Down the left, beside the Inspector	Sounds for the selected software instrument track	**Y**, or the **Library** button
Track headers	Left of the tracks	Each track's number, name, instrument and buttons	Always shown
Ruler	Above the tracks	Bar numbers, the cycle range and the playhead	Always shown
Global tracks	Under the ruler	Arrangement, Chord, Marker, Signature and Tempo tracks	**G**, or **Track › Show Global Tracks**
Tracks area	The large middle area	Where regions sit and are edited	Always shown
Editor pane	Below the tracks	The Piano Roll and the other editors	**E**, or the **Editor** button
Mixer pane	Below the tracks (in place of the editor)	A channel strip for every track	**X**, or the **Mixer** button

The editor and the mixer share the space below the tracks, so only one of them shows at a time.

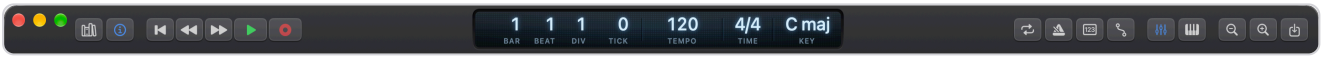
Ongaku's preferences are in the **Settings** window: choose **Settings…** from the app menu (the menu named after the app, next to the Apple menu), or press ⌘,. This manual writes its pages as, for example, **Settings › General**. All the settings are described in Chapter 9.

The control bar

The control bar replaces the normal window title bar. You can drag any empty part of it to move the window. From left to right it holds:

1. **Library** and **Inspector** buttons. Each lights up while its area is showing.
2. **The transport**: **Go to Beginning**, **Back One Bar**, **Forward One Bar**, **Play** (which turns into **Stop** while playing) and **Record**.
3. **The LCD**, the display in the middle (see The LCD display below).
4. **Cycle**, **Metronome**, **Count-in** and **Automation** buttons. These light up in their own colours while they are on.
5. **Mixer** and **Editor** buttons, which show the mixer or editor pane below the tracks.
6. **Zoom Out**, **Zoom In** and **Import Audio**.

Hold the pointer over any button to see its name and keyboard shortcut. The shortcuts are also shown next to the matching commands in the **Transport** and **View** menus.



The control bar

The LCD display

The LCD is the glowing display in the middle of the control bar. It shows four things, each with a small caption underneath:

Caption	What it shows	How to change it
BAR, **BEAT**, **DIV**, **TICK**	Where the playhead is: bar, beat, division of the beat, and tick (a very fine subdivision)	Move the playhead (see The playhead)
TEMPO	The tempo at the playhead, in beats per minute (BPM)	Double-click it and type a new value, then press Return
TIME	The time signature at the playhead	Click it and choose from the list
KEY	The key at the playhead	Click it and choose a major or minor key

Some things to know:

- **Tempo** accepts values from 5 to 990 BPM and can include decimals (for example 92.5). Typing a new tempo changes whichever tempo is in effect at the playhead: the song's starting tempo, or the most recent tempo change before the playhead if your song has any. Press Escape to cancel typing.

- **Time signature** offers the common meters: 2/4, 3/4, 4/4, 5/4, 6/4, 7/4, 2/2, 3/8, 5/8, 6/8, 7/8, 9/8 and 12/8. As with tempo, choosing one changes the time signature in effect at the playhead. To put a change of meter part-way through a song, use the Signature track (Chapter 2).
- **Key** is the key at the playhead. As with tempo and time signature, choosing one changes the key in effect there: the song's key, or a key change. To change key part-way through a song, add a key change on the Signature track (Chapter 2). The key is used when Ongaku suggests chords on the Chord track, and by the session players where there are no chords (Chapter 5). It doesn't transpose anything.

You can change the LCD's colour (blue, amber, green or white) in **Settings › Display** (Chapter 9).

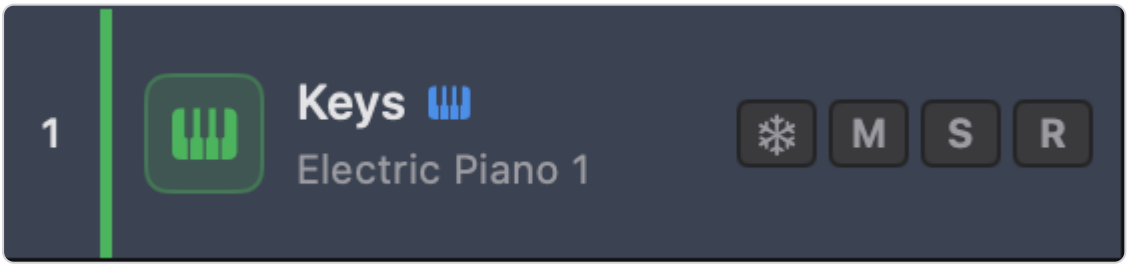
Track headers

Each track has a header on the left of its lane. From left to right:

- **Track number** and a thin **colour stripe** in the track's colour.
- **Icon**: a keyboard for software instrument tracks, a waveform for audio tracks, merging arrows for aux tracks, and a stack symbol for summing stacks.
- **Name**. Double-click it to rename the track; press Return to finish or Escape to cancel. A small keyboard icon after the name means your MIDI keyboard plays this track (see Selecting tracks); hold the pointer over it to see why.
- **Instrument name** (software instrument tracks only). Click it to choose a different instrument. See Chapter 8.
- **Freeze** button (the snowflake), on audio and instrument tracks. Right-click it to choose the track's **Freeze Mode**. Freezing is covered in Chapter 7.
- **M** (Mute) silences the track. **S** (Solo) plays only the soloed tracks.
- **I** (Input Monitoring), on audio tracks only, lets you hear what's coming into the track's input live, through its plug-ins. It turns orange while it's on. See Chapter 3.
- **R** (Record Enable) arms the track for recording. Aux tracks don't have one. See Chapter 3.

Right-click (or Control-click) a header for more: **Rename Track**, **Create Summing Stack** or **Flatten Stack**, **Freeze Track**, **Instrument**, **Color** and **Delete Track**. If the track you right-click is one of several selected tracks, the last item is called **Delete Tracks** and deletes all of them.

You can make the headers wider or narrower by dragging their right-hand edge. Double-click that edge to go back to the standard width. Ongaku remembers the width for next time.



A track header: number, icon, name and instrument, then Freeze, Mute, Solo and Record

Selecting tracks

The **selected track** is the one you clicked last. Its header and lane are highlighted. The selected track matters because:

- the Inspector and Library show it;
- new tracks are added just below it;
- pasting, and splitting with nothing selected, act on it;
- a MIDI keyboard plays it, if it's a software instrument track and no instrument track is record-enabled. (While instrument tracks are record-enabled, the keyboard plays all of those instead. See Chapter 3.)

Click a header to select that track. **Shift-click** another header to select every track in between, or **⌘-click** to add or remove one track at a time. The same clicks work on the channel strips in the Mixer. Selecting several tracks is mainly useful for creating a summing stack (Chapter 2) or a mixer group (Chapter 7).

Clicking an empty part of a track's lane also selects that track.

The corner above the track headers

Above the track headers, level with the ruler, is a small area with four things in it:

- a **+** button that adds a new track (see Adding tracks);
- the word **Tracks**;
- the **Snap** button (a grid icon), whose menu sets what regions snap to when you drag them. Its icon dims while snapping is off (see Chapter 2);
- the **Tool** button, which shows the icon of the tool you're using in the tracks area (see Chapter 2).

When the global tracks are showing, their names and their own **+** buttons appear in this corner too.

The ruler

The ruler runs across the top of the tracks area and shows bar numbers, with small marks for each beat. When you are zoomed far out, only every second, fourth or eighth bar is numbered.

The ruler has two parts:

- **The upper strip** holds the **cycle range**, a coloured bar that marks the part of the song that repeats when cycling is on (see Cycle).
- **The lower part**, with the bar numbers, moves the playhead when you click it.

How long the song is

A song is always at least 128 bars long, and it grows by itself as you need more room. After every edit, import or recording, Ongaku makes sure the song runs to 8 bars past whatever ends last: a region (or its repeats, if you've dragged its loop out to a set length), a marker or an arrangement section. So you can always scroll a little past the end of your music and keep going. While you record, the tracks area widens ahead of the playhead, so recording never runs out of room.

Undoing an edit also takes back the extra length it added. The song never gets shorter by itself, though: deleting the last region leaves the length as it was.

The Inspector

The Inspector runs down the left of the window and is shown when you open a project. From top to bottom it shows:

- **The Region Inspector**, but only while regions are selected. It changes how the selected regions play, such as their gain, transposition or looping (Chapter 2).
- **The selected track's name and kind**, with a coloured dot you can click to change the track's colour.
- **Freeze Mode** (audio and instrument tracks) and **Flex** (audio tracks). See Chapter 7 and Chapter 6.
- **Two channel strips**: the selected track's, and **Stereo Out** (the main output that everything plays through). These let you mix the track without opening the Mixer.

If no track is selected, the Inspector says **No Track Selected**.

The Library

The Library (press **Y**) lists sounds for the selected software instrument track. Clicking a sound loads it straight away, so you can play through sounds on a MIDI keyboard until you find one you like. See Chapter 9.

The editor and mixer panes

Below the tracks area there is room for one of two panes:

- **The editor** (press **E**, click **Editor** in the control bar, or choose **View › Show Editor**) shows whichever editor suits the region you're working on: the Piano Roll for a MIDI region, or the Step Editor, Step Sequencer, Drummer, Player, Takes or Flex Pitch editor. See Chapter 4. Double-clicking a MIDI region opens it here.
- **The Mixer** (press **X**, or click **Mixer**) shows a channel strip for every track. See Chapter 7.

To change the height of the pane, drag the line between it and the tracks. While you drag, a highlighted line shows where it will end up, with the new height written on it; the pane resizes when you let go. Double-click the line to go back to the standard height. Ongaku remembers the height of each pane separately.

Which area your keys act on. While the editor is showing, some keys (Delete, ⌘C, ⌘V and the like) act on the tracks area or the editor, whichever you clicked last. That area has a thin blue outline, so you can see at a glance where Delete will go. Click in the other area to move the outline there. See Chapter 10.

Projects

Creating a project

1. Choose **File › New** (⌘N). If Ongaku shows the standard macOS Open window when it starts, you can click **New Document** there instead.
2. A new project window opens.

A new project starts at 120 BPM, in 4/4, in C major, with the cycle range over bars 1 to 5. What tracks it starts with depends on **Settings › General › New projects start with**:

Setting	A new project starts with
One software instrument track (the standard choice)	A track called "Inst 1"
One audio track	A track called "Audio 1"
No tracks	An empty project

Ongaku opens each project in its own window; it doesn't use window tabs.

Saving a project

1. Choose **File › Save** (⌘S).
2. The first time, choose a name and a place for the project, then click **Save**.

After the first save, macOS saves your changes automatically as you work, as it does for documents in Pages or TextEdit. The usual **File** menu commands for documents (such as Duplicate, Rename and Revert To) also work, because Ongaku uses the standard macOS document system.

Save early. Until a project has been saved once, audio you import or record is kept in a temporary folder. It is copied into the project when you first save.

What's inside a project

An Ongaku project is saved as a single item ending in **.ongaku**. In the Finder it looks like one file, but it is really a folder (a "package") so that it can keep your audio with it. To look inside, right-click it in the Finder and choose **Show Package Contents**. You will find:

Inside the project	What it holds
The project file	Your tracks, regions, notes, mix and settings
Audio Files folder	A copy of every audio file you've imported or recorded
Freeze Files folder	Audio rendered for frozen tracks (only while tracks are frozen)
Plug-in Settings folder	Settings of plug-ins that are too big to keep in the project file (only if needed)

Things worth knowing:

- **Importing copies the audio.** The original file you imported is never changed or moved, and you can delete it: the project has its own copy. This also means you can move or email a project and its audio goes with it.
- **Audio is kept even after you delete the region that used it**, so that undo can always bring it back. A project can therefore grow as you work.
- **Freeze files are thrown away** when you unfreeze a track, because they can always be made again.
- Don't change the contents of the package by hand.

Opening a project

Choose **File › Open** (⌘O) and pick a **.ongaku** project, or double-click one in the Finder. **File › Open Recent** lists projects you've used lately.

To bring in a Standard MIDI File, use **File › Import MIDI File…** instead (Chapter 9).

Adding tracks

You can add tracks from the **Track** menu or from the **+** button above the track headers. A new track goes just below the selected track (or at the bottom if no track is selected) and becomes the selected track.

To add a track:

1. Click the track you want the new one to appear under.
2. Choose one of these:

Track menu	+ menu	Shortcut	What you get
New Audio Track	**Audio**	⌥⌘A	An empty audio track, named "Audio 1", "Audio 2"…
New Software Instrument Track	**Software Instrument**	⌥⌘S	An empty instrument track, named "Inst 1", "Inst 2"…
New Aux Track	**Aux**		An aux track, named "Aux 1"…
New Drummer Track	**Drummer**	⌥⌘U	A drum track with a drummer already playing
New Drummer Track (Drum Machine)	**Drummer (Drum Machine)**		The same, with an electronic drum machine
New Bass Player Track	**Bass Player**		A bass track that follows the Chord track
New Keyboard Player Track	**Keyboard Player**		A keys track that follows the Chord track

Drummer and player tracks are explained in Chapter 5.

Tips:

- New tracks get a standard colour for their kind: green for instruments, blue for audio, purple for aux. You can choose different colours in **Settings › Display › Colors of New Tracks**. Tracks you already have keep their colours.
- If the selected track is a summing stack, or is inside one, the new track joins that stack.
- Importing an audio file can create audio tracks for you (Chapter 2).
- To make a copy of a track with everything on it, use **Track › Duplicate Track** (⌘D), described in Chapter 2.

Deleting a track

Select the track and choose **Track › Delete Track** (⌘⌫), or right-click the track header and choose **Delete Track**.

To delete several tracks at once, select them (Shift-click or ⌘-click their headers), then choose **Track › Delete Tracks** (⌘⌫), or right-click one of the selected headers and choose **Delete Tracks**. One **Edit › Undo** (⌘Z) brings them all back.

The **Delete** key on its own never deletes tracks. It deletes the selected regions (or notes), and does nothing if nothing is selected.

Deleting a summing stack also deletes the tracks inside it.

The transport

"Transport" is the name for the play, stop and record controls, after the buttons on a tape machine.

Action	Control bar	Key	Transport menu
Play or stop	**Play** / **Stop**	Space	**Play** / **Stop**
Go to the start of the song	**Go to Beginning**	Return (or Enter)	**Go to Beginning**
Go back a bar	**Back One Bar**	, (comma)	**Back One Bar**
Go forward a bar	**Forward One Bar**	. (full stop)	**Forward One Bar**
Start or stop recording	**Record**	R	**Record** / **Stop Recording**
Turn cycling on or off	**Cycle**	C	**Turn Cycle On** / **Turn Cycle Off**
Turn the metronome on or off	**Metronome**	K	**Turn Metronome On** / **Turn Metronome Off**
Turn the count-in on or off	**Count-in**		

Back One Bar goes to the start of the bar you're in first, if you're part-way through one, as rewinding does on a tape machine. Press it again to go back another bar.

The single-key shortcuts (Space, R, C and so on) are shown next to their commands in the **Transport** menu. They work wherever you last clicked in the project window, but not while you are typing in a text field, such as the tempo display or a track name. Click in the tracks area to stop typing, and the keys

work again.

The playhead

The playhead is the white marker in the ruler, with a line down through the tracks. It shows where playback is, or will start from.

To move the playhead:

- click in the lower part of the ruler (it lands on the nearest grid line; see Snapping);
- use **Go to Beginning**, **Back One Bar** or **Forward One Bar**;
- click a marker on the Marker track, or a chord on the Chord track;
- use **Transport › Go to Previous Marker** (⌥,) or **Go to Next Marker** (⌥.).

During playback the tracks area scrolls along to keep the playhead in view, a page at a time. If you'd rather it didn't, turn off **Scroll to follow the playhead during playback** in **Settings › General**. When the playhead jumps while stopped, the tracks area always scrolls to show it.

Recording

Clicking **Record** (or pressing **R**) records onto every track whose **R** button is on. Recording is explained fully in Chapter 3.

Cycle

Cycle makes playback repeat one part of the song over and over, which is handy when you're practising a part, recording several takes or working on a section of the mix. The part that repeats is the **cycle range**, the coloured bar in the upper strip of the ruler. It is yellow while cycling is on and grey while it's off.

To set the cycle range:

1. Drag along the upper strip of the ruler, from where you want the cycle to start to where you want it to end. The ends snap to the grid.
2. Let go. Cycling turns on.

To turn cycling on or off, click the upper strip of the ruler without dragging, click **Cycle** in the control bar, or press **C**.



The ruler with a cycle range over bars 5 to 9

The locators

In Logic and in Ongaku, the start and end of the cycle range are also called the **locators**. Some editing commands work on "the section between the locators", for example cutting four bars out of the whole song. They use the cycle range whether cycling is on or off, so you can set the range and then turn cycling off again. See Chapter 2.

The metronome

The metronome clicks on every beat so you can keep time. Turn it on or off with **Metronome** in the control bar or the **K** key. Its sound and volume are set in **Settings › Metronome** (Chapter 9).

Whether the metronome is on is saved with the project, as the cycle is, so a song you saved with the click on opens with it on. Turning it on or off is an undo step called **Metronome Change**.

The **Count-in** button, just right of **Metronome**, gives you a bar of clicks before recording starts. It's described in Chapter 3.

Zooming and scrolling

Zooming makes the timeline wider or narrower, so you can see the whole song or a few beats in detail.

To	Do this
Zoom in	Click **Zoom In** in the control bar, choose **View › Zoom In**, or press ⌘=
Zoom out	Click **Zoom Out**, choose **View › Zoom Out**, or press ⌘–

Each step zooms by half as much again. Zooming changes only the horizontal scale; track heights stay the same.

Zooming also changes how finely things snap when you drag them (with the standard snap setting): zoomed out, regions snap to whole bars; zoomed in, to beats, then eighth notes, then sixteenth notes. See Snapping to the grid.

Scrolling works as in any Mac app: swipe with two fingers on a trackpad, or use a mouse's scroll wheel (hold Shift to scroll sideways on most mice). The ruler stays at the top and the track headers stay at the left while you scroll.

Undo and redo

Almost every change you make to the song can be undone, one step at a time, right back to when you opened the project.

To	Choose	Shortcut
Undo the last change	**Edit › Undo**	⌘Z
Redo a change you undid	**Edit › Redo**	⇧⌘Z

The Edit menu names the change it will undo, for example **Undo Move Region** or **Undo Split Regions**, so you can see what you're about to undo.

What counts as a change: anything that alters the song, such as moving, adding or deleting regions and tracks, renaming, changing tempo or key, mixer settings, and turning the cycle or metronome on or off. Things that only change your view are not undo steps: zooming, scrolling, selecting, choosing a tool, and showing or hiding areas of the window.

If an edit can't be done (for example, dragging an audio region onto an instrument track), Ongaku shows a message titled **Something went wrong** explaining why, and nothing changes.

Limitations

- The song's length can't be set by hand. It grows by itself (see How long the song is), but never shrinks.
- Track heights can't be changed, and there is no vertical zoom.

Quick reference

Action	How
New project	**File** › **New** (⌘N)
Open a project	**File** › **Open** (⌘O)
Save	**File** › **Save** (⌘S)
New audio track	**Track** › **New Audio Track** (⌥⌘A)
New software instrument track	**Track** › **New Software Instrument Track** (⌥⌘S)
New drummer track	**Track** › **New Drummer Track** (⌥⌘U)
Delete the selected tracks	**Track** › **Delete Track** (⌘⌫)
Play or stop	Space
Go to beginning	Return
Back or forward one bar	, or .
Record	R
Cycle on or off	C, or click the top strip of the ruler
Set the cycle range	Drag in the top strip of the ruler
Metronome on or off	K
Count-in on or off	**Count-in** in the control bar
Move the playhead	Click the lower part of the ruler
Show or hide the Inspector	I
Show or hide the Library	Y
Show or hide the editor	E
Show or hide the Mixer	X
Show or hide automation	A
Show or hide the global tracks	G
Zoom in or out	⌘= or ⌘–
Undo or redo	⌘Z or ⇧⌘Z

Action	How
Change the tempo	Double-click **TEMPO** on the LCD and type
Change the time signature or key	Click **TIME** or **KEY** on the LCD
Rename a track	Double-click its name
Resize the track headers	Drag their right edge (double-click to reset)
Resize the editor or Mixer	Drag the line above it (double-click to reset)

2. Arranging and Editing

Arranging means shaping a song from its parts: putting regions in the right places, repeating a chorus, cutting a verse shorter, fading a guitar in. This chapter explains what regions are and every way of editing them in the tracks area, then covers the tools for whole sections of a song: markers, the Arrangement track, tempo and time signature changes, track stacks, and importing audio.

You'll need the words from Chapter 1: track, region, playhead, cycle range and locators.

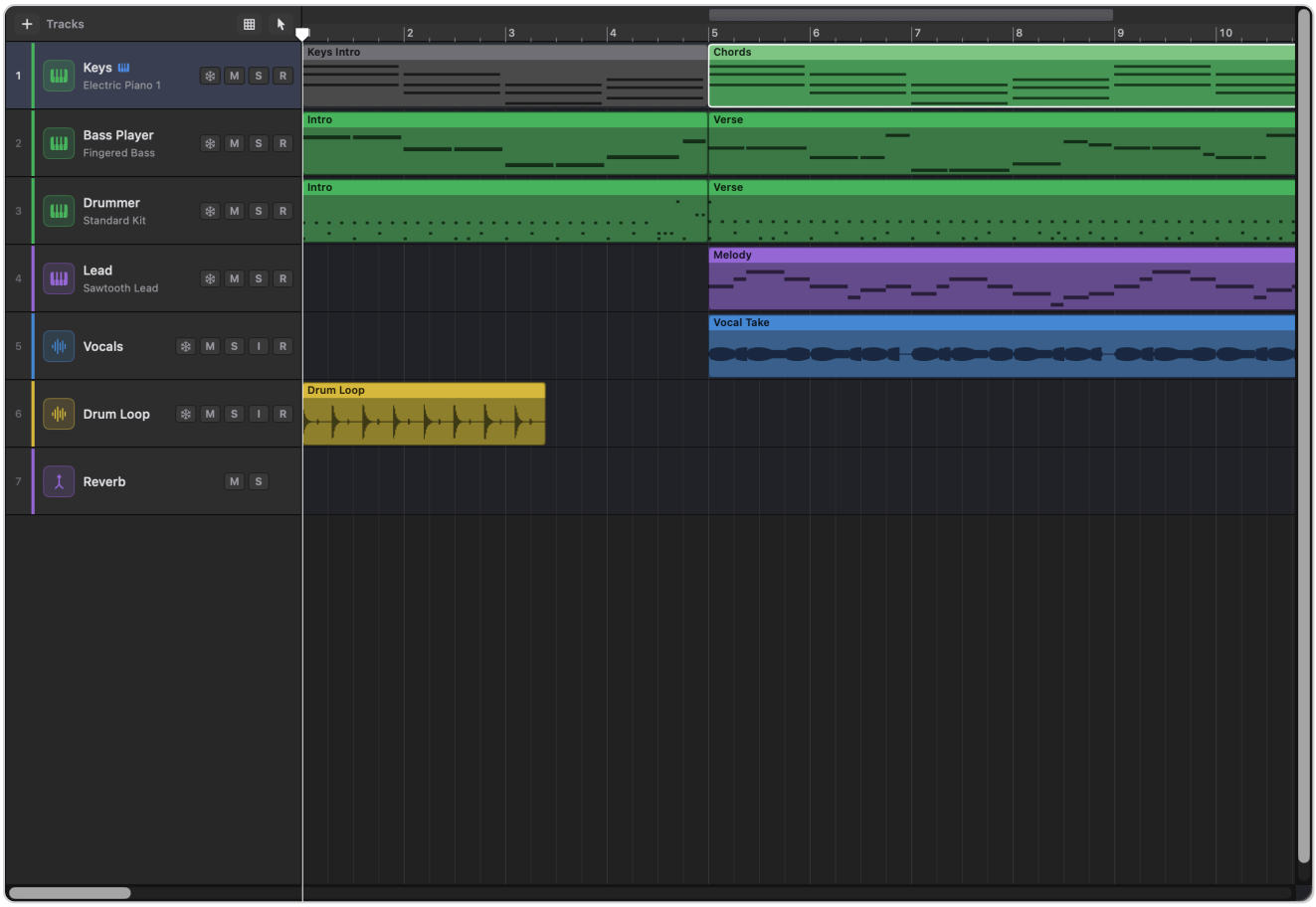
Regions

A **region** is a block of music on a track, shown as a coloured rectangle with its name in a strip along the top. There are two kinds:

Region	Lives on	Shows	Comes from
MIDI region	Software instrument tracks	Small dark bars, one for each note	Recording from a MIDI keyboard, drawing notes, importing a MIDI file, or a session player
Audio region	Audio tracks	A waveform	Recording from a microphone or instrument, or importing an audio file

An audio region doesn't hold the sound itself: it is a window onto an audio file stored in the project. When you shorten an audio region, the audio file isn't cut; the window just shows less of it, and you can drag the edge back out later. That's why most region edits are safe to experiment with.

Regions take their track's colour. Selected regions are brighter, with a white outline. Muted regions turn grey.



MIDI and audio regions: Chords on the Keys track is selected (outlined), and Keys Intro is muted

Snapping to the grid

When you move, resize or split a region, or click the ruler, the position "snaps" to the nearest line of a musical grid, so things land neatly on the beat. Automation points snap to the same grid (Chapter 7).

You choose the grid with the **Snap** button, the grid icon above the track headers, just left of the Tool button. Click it and pick a setting from its menu. The same choice is in **Settings › General › Snap regions to**; the two are the same setting, so changing one changes the other.

Setting	Snaps to
Automatic (follows the zoom) (the standard choice)	A grid that gets finer as you zoom in: bars when zoomed out, then beats, eighth notes and sixteenth notes
Bars	Whole bars
Beats	Beats
1/8 Triplets	Eighth-note triplets
1/16 Triplets	Sixteenth-note triplets
Off	Nothing: positions follow the pointer exactly

The Snap button's icon dims while snapping is **Off**, as a reminder that things will land exactly where you drop them. Hold the pointer over the button to see the current setting.

Moving freely for one drag. To place something off the grid just this once, hold **Control (^)** while you drag. Press Control *after* you've started dragging, not before: a Control-click is a right-click, so pressing it first opens a menu instead. Let go of the mouse while still holding Control and the region stays exactly where you dropped it. This works for regions, the Piano Roll's notes and automation points.

Tip: with **Automatic**, if a region won't land where you want it, zoom in (⌘=) and drag again.

Selecting regions

Most edits act on the **selected** regions, so selecting is the first step of nearly everything.

To	Do this
Select one region	Click it
Add a region to the selection, or take it out	Shift-click it
Select several regions in an area	With the Pointer tool, drag across empty space; every region the rectangle touches is selected (hold Shift to add to the selection)
Select every region	**Edit › Select All** (⌘A)
Select every region on one track	Right-click the track's lane and choose **Select All on Track**
Select a region and everything after it on its track	Right-click the region and choose **Select All Following**
Select the regions in a song section	Click the section on the Arrangement track (see The Arrangement track)
Deselect everything	Click an empty part of a lane, or below the last track

Selecting a region also selects its track. When you select several regions, the Region Inspector shows the first one's settings, and changes you make there apply to all of them.

Moving, copying and resizing regions

Moving

To move regions:

1. Select the region (or several regions).
2. Drag any one of them by its middle. All the selected regions move together, snapping to the grid.
3. To move them to another track, drag up or down as well.

Regions can only move to tracks of the same kind: MIDI regions onto instrument tracks, audio regions onto audio tracks. If any region in the selection would land on the wrong kind of track, nothing moves and Ongaku tells you why. Regions can't be moved before the start of the song.

Nudging with the keyboard

To move the selected regions by a small, exact amount, press ⌥← (earlier) or ⌥→ (later). Each press moves them one step of the snap grid (see Snapping to the grid): a bar, a beat or a smaller division, whatever regions would snap to at this zoom. With **Snap** set to **Off**, the step is the grid that **Automatic** would use at this zoom. Each press is one undo step, called **Nudge Region** (or **Nudge Regions**).

The regions keep their spacing, and they stop at the start of the song. These keys work while the tracks area was the last place you clicked; if you clicked in the Piano Roll last, ⌥← and ⌥→ nudge the selected notes instead (Chapter 4).

Copying with Option-drag

To make a copy as you drag:

1. Select the regions you want to copy.
2. Drag them to the new place.
3. Hold the Option (⌥) key **as you let go**. Copies land where you dropped and the originals stay put. The copies become the selection.

You can press Option at any point during the drag; what counts is whether it's held when you let go. While it's held, the pointer shows a small **+** (the copy pointer) and the original regions stay where they are on screen, with the copies following the pointer. Let go of Option before you let go of the mouse and the regions are moved instead.

Resizing (trimming)

Trimming shortens or lengthens a region from either end without moving the music inside it. A note at bar 3 stays at bar 3; trimming just decides how much of the region plays.

To trim a region:

1. Move the pointer over the left edge, or the **lower half** of the right edge, of the region. The pointer turns into a resize arrow.
2. Drag the edge. It snaps to the grid.

Gotchas:

- The **upper half** of the right edge is the loop handle, not a resize handle (see Looping regions). Watch for the circular-arrow pointer.
- Very narrow regions (when you're zoomed out) have no edge handles; zoom in to trim them.
- An audio region can't be stretched to start before the beginning of its audio file.
- If you shorten a MIDI region, notes outside it stay in the region but don't play; lengthen it again and they come back.

Tools

The **Tool** decides what a click in the tracks area does. Most of the time you'll use the Pointer, but other tools make some jobs one click instead of several.

Choosing a tool

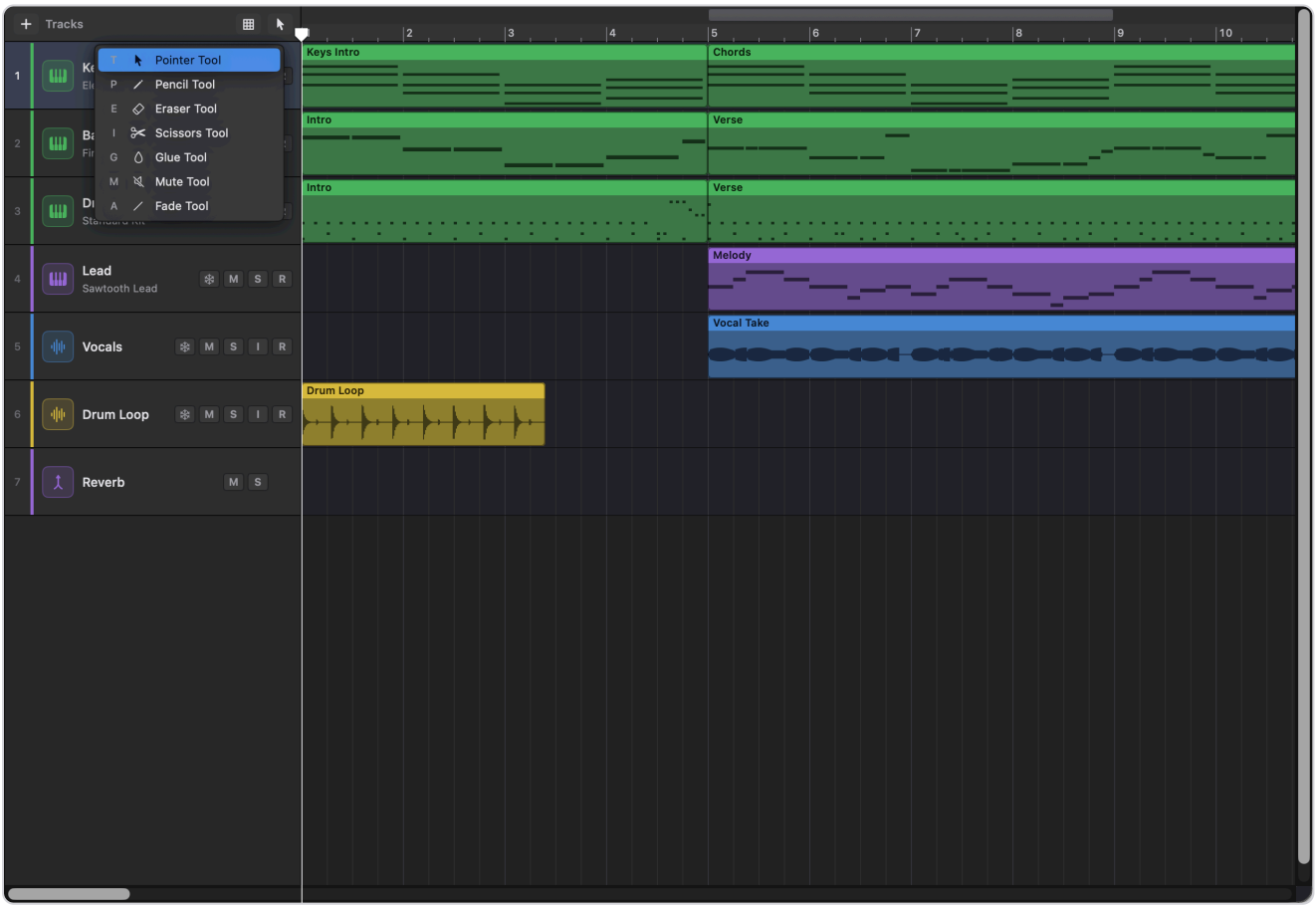
There are two ways:

- **The Tool button**: click the tool icon above the track headers (to the right of the word **Tracks**) and choose a tool from the menu.
- **The T key**: with the pointer over the tracks area, press **T**. The Tool menu appears under the Tool button, listing each tool with its key. Press that key to choose the tool, or press **Escape** to close the menu without changing. You can also click a tool in the menu.

So **T** then **T** always takes you back to the Pointer, and **T** then **I** gives you the Scissors.

If the pointer is over the editor pane when you press T, you get the editor's own tools instead (see Chapter 4).

The Tool button shows the current tool's icon. It's white for the Pointer and lit in the accent colour for any other tool, as a reminder that clicks will do something different.



The Tool menu, open under the Tool button after pressing T

The tools

Tool	Key (after T)	What a click on a region does	Can move and resize regions?
Pointer Tool	T	Selects it	Yes
Pencil Tool	P	Selects it. Clicking an empty part of an instrument lane creates a new MIDI region	Yes
Eraser Tool	E	Deletes it	No
Scissors Tool	I	Splits it where you click (snapped to the grid)	No
Glue Tool	G	Joins the selected regions	No
Mute Tool	M	Mutes it, or unmutes it if it was muted	No
Fade Tool	A	Nothing; you drag with it to make fades (see Fades and crossfades)	No

With every tool except the Pointer and Pencil, **clicking a region that is part of the selection acts on all the selected regions**. For example, select four regions, choose the Eraser, and click one of them: all four are deleted. Clicking a region that isn't selected acts on that region alone.

Hold the pointer over the Tool button to see a reminder of what the current tool does.

Limitations

- There is one tool at a time. Logic's second tool (on the ⌘ key) and right-button tool aren't available.
- The Mute Tool mutes whole regions only.

Creating MIDI regions

A new, empty MIDI region is one bar long and starts at the bar you clicked in. It's named after its track, and opens in the Piano Roll straight away so you can add notes (Chapter 4).

To create one, do any of these on a software instrument track's lane:

- double-click an empty spot (with any tool);
- click an empty spot with the Pencil Tool;
- right-click an empty spot and choose **New MIDI Region**.

The lane's right-click menu also has **New Pattern Region** (for the Step Sequencer, Chapter 4), **New Drummer Region** and **New Player Region** (Chapter 5). **Track › New Pattern Region** makes a pattern region from the menu bar too: on the selected instrument track, in the bar containing the playhead.

Double-clicking a MIDI region opens it in the editor. Double-clicking an ordinary audio region does nothing (unless its track uses Flex Pitch, Chapter 6, or it's a take folder, Chapter 3). Selecting a region and pressing **E** opens the same editor.

Splitting, joining and deleting

Splitting

Splitting cuts a region in two at a point. Both halves end up selected, so you can move or delete either straight away.

To split at the playhead:

1. Put the playhead where you want the cut.
2. Select the regions to split. (If nothing is selected, every region under the playhead on the selected track is split.)
3. Choose **Edit › Split Regions at Playhead** (⌘T), or right-click and choose **Split at Playhead**.

To split where you click:

1. Press **T**, then **I**, to choose the Scissors Tool.
2. Click the region where you want to cut. The cut snaps to the grid. If the region was selected, every selected region that crosses that point is cut there too.

What happens to the contents:

- **MIDI**: each note goes to the half it starts in. A long note that crosses the cut stays in the left half.
- **Audio**: a fade-in stays on the left half and a fade-out on the right half.
- **Looped regions**: the repeats carry on after the right-hand half.
- **Drummer and player regions** stay drummer and player regions, each half playing for its own length with the same settings. A drummer's crash stays at the start of the left half and its fill at the end of the right half; a player's lead-in to the next section stays on the right half.
- **Pattern regions** become ordinary MIDI regions, keeping the notes they were playing.

Joining (glue)

Joining turns several regions on a track into one, running from the start of the first to the end of the last, with the first one's name. It's useful for tidying up after lots of splitting.

To join regions:

1. Select two or more regions on the same track. (If you select regions on several tracks, each track's regions are joined separately.)
2. Choose **Edit › Join Regions** (⌘J), or right-click and choose **Join Selected Regions**, or click one of the selected regions with the Glue Tool (**T** then **G**).

What can be joined:

- **MIDI regions**: always. The joined region keeps the notes that were actually playing: notes hidden by trimming, and muted notes, are left out.
- **Drummer or player regions with the same settings** (for example the two halves of one you split): they join into one drummer or player region that plays over the whole length, with the first one's crash and the last one's fill or lead-in.

- **Drummer or player regions with different settings, and pattern regions**: they're joined as ordinary MIDI regions holding the notes they were playing. The Edit menu then calls the step **Undo Join and Convert to MIDI**, so you can see that's what happened.
- **Audio regions**: only when they play one unbroken stretch of the same audio file, such as the pieces of a recording you split. Joining different files would need a new audio file, which Ongaku doesn't make.

Deleting

Select the regions and press **Delete**, choose **Edit › Delete Selection** (its shortcut, Delete, is shown in the menu), right-click and choose **Delete**, or click them with the Eraser Tool.

With no regions selected, Delete does nothing. It never deletes tracks: to delete a track, use **Track › Delete Track** (⌘⌫) (see Chapter 1).

If the editor pane is open and you clicked in the Piano Roll last, Delete removes the selected notes there instead of regions. If you last clicked a chord on the Chord track, Delete removes that chord instead (Chapter 5). The area Delete will act on has a thin blue outline (see Chapter 10).

When an edit is refused

Some regions can't be resized, split, joined or crossfaded until you've changed them back to plain regions:

- **Take folders** (from recording in cycle, Chapter 3): flatten them first.
- **Audio regions with flex markers, retuned notes or Smart Tempo stretch** (Chapter 6): remove those first, with the **Remove Flex Markers**, **Reset Retuned Notes** or **Remove Stretch** button in the Region Inspector.
- **Regions on frozen tracks** can't be edited at all until the track is unfrozen (Chapter 7). Frozen tracks' lanes have an icy tint.

Ongaku explains which of these is in the way when it refuses. For a flexed region, the message names the button that fixes it, for example: "To resize, split, join or crossfade “Vocal”, first remove its flex markers (Remove Flex Markers): select it and use the button in the Region Inspector."

Muting regions

Muting a region silences it without deleting it, so you can try the song with and without a part. Muted regions turn grey.

To mute or unmute regions, do one of these:

- right-click and choose **Mute or Unmute** (if any of the selected regions are playing, they're all muted; if they're all muted, they're all unmuted);
- click them with the Mute Tool (**T** then **M**);
- turn on **Mute** in the Region Inspector.

To silence a whole track, use the track's **M** button instead.

Cut, copy, paste and repeat

Ongaku has its own clipboard for regions. It's shared between all open projects, so you can copy MIDI regions from one song into another.

Command	Menu	Shortcut	What it does
Cut	**Edit › Cut**	⌘X	Copies the selected regions and removes them
Copy	**Edit › Copy**	⌘C	Copies the selected regions
Paste	**Edit › Paste**	⌘V	Pastes at the playhead, onto the selected track
Repeat	**Edit › Repeat Regions**	⌘R	Places a copy of the selection straight after it

These commands act on regions when you last clicked in the tracks area, and on notes when you last clicked in the Piano Roll. While you're typing in a text field they work on text as usual.

How pasting works:

1. Copy one or more regions.
2. Click the track you want them on (that selects it) and put the playhead where the first region should start.
3. Choose **Edit › Paste** (⌘V).

The pasted regions keep their spacing in time and between tracks: the topmost copied region lands on the selected track and the others land on the tracks below, in the same pattern. The pasted regions become the selection.

You can also right-click a lane and choose **Paste**: the regions are pasted at the spot you right-clicked, on that track.

Gotchas:

- The tracks they land on must be the right kind (MIDI onto instrument tracks, audio onto audio tracks), and there must be enough tracks below.
- **Audio regions can only be pasted into the project they were copied from.**

Repeat

Repeat is a quick way to make a part play again: select a two-bar drum pattern, press ⌘R three times, and you have eight bars. The copies are laid end to end straight after the selection, keeping the spacing between regions, and the copies become the selection (so pressing ⌘R again carries on).

Right-click a region and choose **Repeat** to make several copies at once: **Once**, **2 Times**, **3 Times**, **4 Times** or **8 Times**.

Duplicating a track

Track › Duplicate Track (⌘D) makes a copy of the selected track just below it, with all its regions, its channel strip settings, plug-ins and automation. The copy gets a numbered name, such as "Bass 2", and becomes the selected track. It's a good way to try a variation of a part while keeping the original.

Summing stacks can't be duplicated yet; duplicate the tracks inside them instead.

Editing a section between the locators

Sometimes you want to change the length of the whole song: cut out a bar that drags, add two bars of space before the chorus, or play a section twice. These commands work on **every track at once**, between the **locators** (the start and end of the cycle range, whether cycling is on or off).

The commands are in **Edit › Section Between Locators**, and in the right-click menu of the tracks area:

Command	Shortcut	What it does
Cut Section	^⌘X	Removes everything between the locators and closes the gap: everything after moves earlier
Insert Silence	^⌘Z	Opens up empty time between the locators: everything from the left locator onwards moves later
Repeat Section	^⌘R	Plays everything between the locators twice: a copy goes straight after, and everything later moves to make room

To cut two bars out of a song:

1. Drag in the top strip of the ruler over the bars you want to remove (this sets the locators).
2. Press **C** if you don't want cycling on.
3. Choose **Edit › Section Between Locators › Cut Section** (^⌘X).

"Everything" means regions, automation, markers and Arrangement sections. Regions that cross a locator are split there first, so the cut is clean.

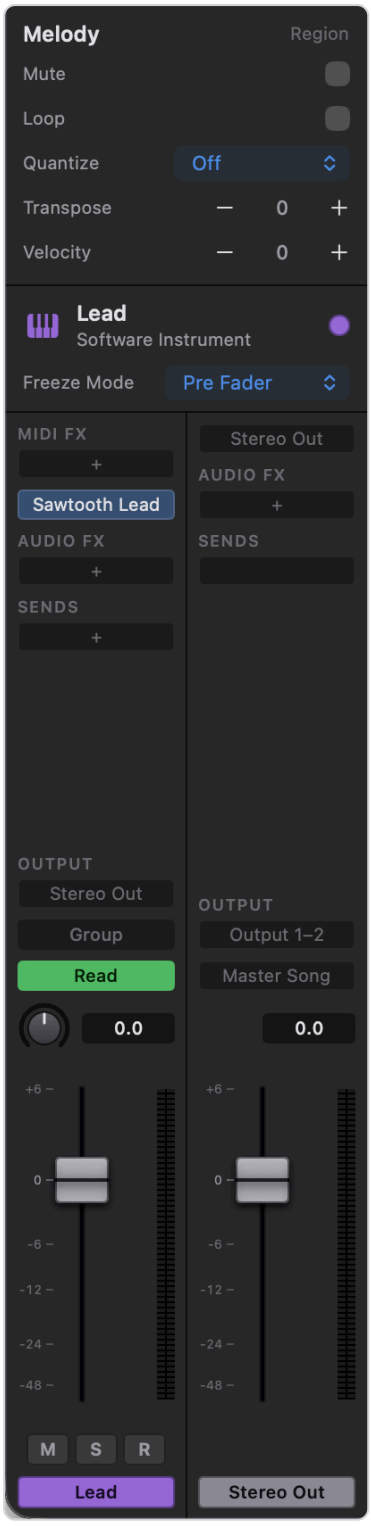
Gotcha: if a region crossing a locator can't be split (a take folder, or a flexed region), the whole command is refused, with a message naming the region.

The Region Inspector

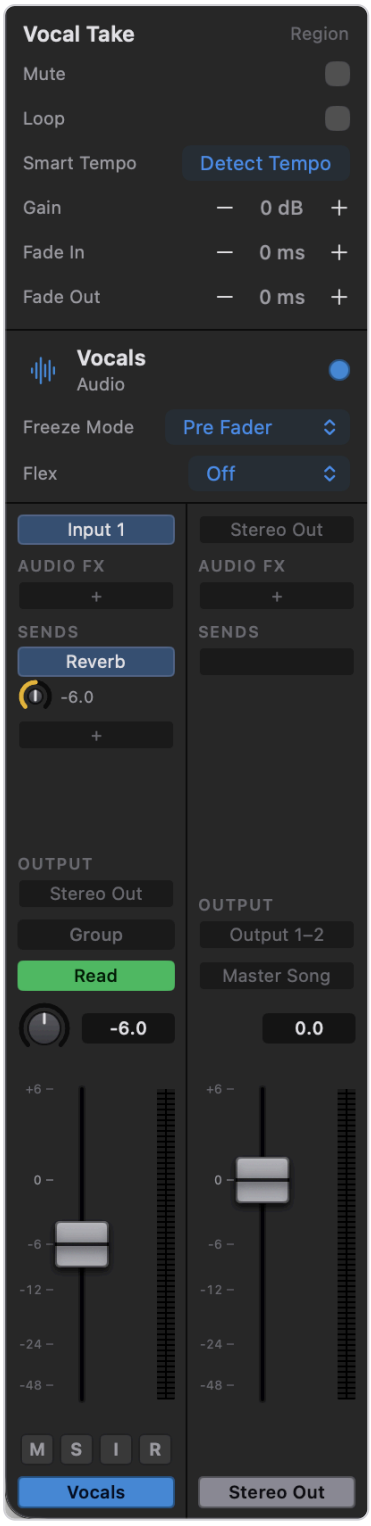
The **Region Inspector** appears at the top of the Inspector (press **I** if it's hidden) whenever regions are selected. It changes how the selected regions **play**, without changing the notes or audio inside them. That's what "non-destructive" means: set **Transpose** to +2 and the region plays a tone higher, but the notes in the Piano Roll are unchanged, and setting it back to 0 restores the original exactly.

The top line shows the region's name (or "3 Regions" if several are selected). Changes apply to every selected region; settings for MIDI only are skipped for audio regions in the selection, and the other way round.

Use the **–** and **+** buttons beside a value to change it. The **Fade In** and **Fade Out** lengths can also be typed: click the number, type a length, and press Return (Escape cancels).



The Region Inspector for a MIDI region



The Region Inspector for an audio region

Settings for all regions

Setting	What it does
Mute	Silences the region (see Muting regions)
Loop	Repeats the region until the next region on its track, or the end of the song (see Looping regions)

Settings for MIDI regions

Setting	Range	What it does
Quantize	**Off**, **1/4 Note**, **1/8 Note**, **1/16 Note**, **1/32 Note**, **1/4 Triplet**, **1/8 Triplet**, **1/16 Triplet**, **1/32 Triplet**, and the mixed grids **1/8 & 1/12**, **1/16 & 1/12**, **1/16 & 1/24**, **1/32 & 1/24**	Plays the notes moved onto the grid, to tighten up timing. Mixed grids move each note to the nearer of a straight line and a triplet line, which suits playing that mixes both
Transpose	–96 to +96 semitones	Plays every note higher or lower. 12 semitones is an octave
Velocity	–99 to +99	Adds to (or takes from) every note's velocity (how hard it was played), keeping the result between 1 and 127

Quantizing in the Region Inspector is different from quantizing in the Piano Roll: here the notes only play on the grid, and stay where you played them; in the Piano Roll (Chapter 4), the notes themselves are moved.

Settings for audio regions

Setting	Range	What it does
Flex	(only when Flex is on for the track)	**Quantize Audio**, see Chapter 6
Flex Markers	(only when a selected region has flex markers)	**Remove Flex Markers** removes them from every selected region, even while Flex is off for the track
Flex Pitch	(only when a selected region has retuned notes)	**Reset Retuned Notes** puts every retuned note back to the pitch it was sung at
Smart Tempo		Finds the recording's tempo, and matches it to the song or the song to it. **Remove Stretch** appears beside it whenever the region is stretched. See Chapter 6
Gain	–96 to +30 dB, in 1 dB steps	Makes the region louder or quieter before it reaches the track's fader. Useful for evening out takes recorded at different levels
Fade In	0 to 60,000 ms (60 s), in 10 ms steps, or typed	Length of a fade from silence at the start
Fade Out	0 to 60,000 ms (60 s), in 10 ms steps, or typed	Length of a fade to silence at the end

When you type a fade length, a plain number is in milliseconds. You can add a unit: "250", "250 ms" and "0.25 s" all mean a quarter of a second, and "1.5 s" means one and a half seconds. Lengths over 60 seconds are set to 60 seconds.

Fades and crossfades

A **fade** makes an audio region rise smoothly from silence at its start (a fade-in) or sink to silence at its end (a fade-out). Even a very short fade, a few milliseconds, removes the click you can hear when a region starts or stops in the middle of a sound.

A **crossfade** is a fade between two regions that touch or overlap: the first fades out while the second fades in. It smooths the join when you've cut between two takes.

Fades appear on the region as a darkened triangle under a white line. Fades are for audio regions only.

Making a fade with the Fade Tool

1. Press **T**, then **A**, to choose the Fade Tool.
2. To fade in, start dragging in the **first half** of an audio region and drag right; the fade runs from the region's start to where you let go.
3. To fade out, start dragging in the **second half** and drag left; the fade runs from where you let go to the region's end.

The fade appears as you drag. Clicking without dragging does nothing.

You can also set fade lengths in the Region Inspector's **Fade In** and **Fade Out**: step them with **–** and **+**, or click the number and type a length such as "250 ms" or "1.5 s" (see The Region Inspector). Changes there apply to every selected region at once.

Making a crossfade

There are two ways two audio regions on the same track can meet, and the Fade Tool crossfades both:

- **End to end**, so the first ends exactly where the second starts (splitting a region does this for you).
- **Overlapping**, so the second starts before the first ends (for example after you've dragged one region partly over another).

To crossfade them:

1. Choose the Fade Tool (**T** then **A**).
2. Start dragging where the regions meet:
    - for regions **end to end**, start right at the border between them, and drag left or right. The crossfade is twice as long as your drag, centred on the border. To make room for it, each region is lengthened into the other by half the crossfade, as far as its audio file allows. If neither region has any spare audio beyond its edge, Ongaku tells you there's nothing to crossfade with;
    - for **overlapping** regions, start anywhere over the part where they overlap (or just beside it), and drag a little. The crossfade covers the whole overlap: the first region fades out across it while the second fades in.

Gotchas:

- For regions end to end, you need to start the drag within a few pixels of the border; if you start further in, you get an ordinary fade instead. Zooming in helps.
- If there's a gap between the regions, there's nothing to crossfade. Move one of them until it snaps against the other, or overlaps it.

Looping regions

Looping makes a region repeat over and over without making copies. Change the original and every repeat changes with it, which is ideal for a drum pattern or a riff.

The repeats are drawn as fainter copies of the region after it. You can't click or select the repeats; edit the original region instead.

Turning looping on

1. Select the region.
2. Turn on **Loop** in the Region Inspector, or right-click the region and choose **Loop or Unloop**.

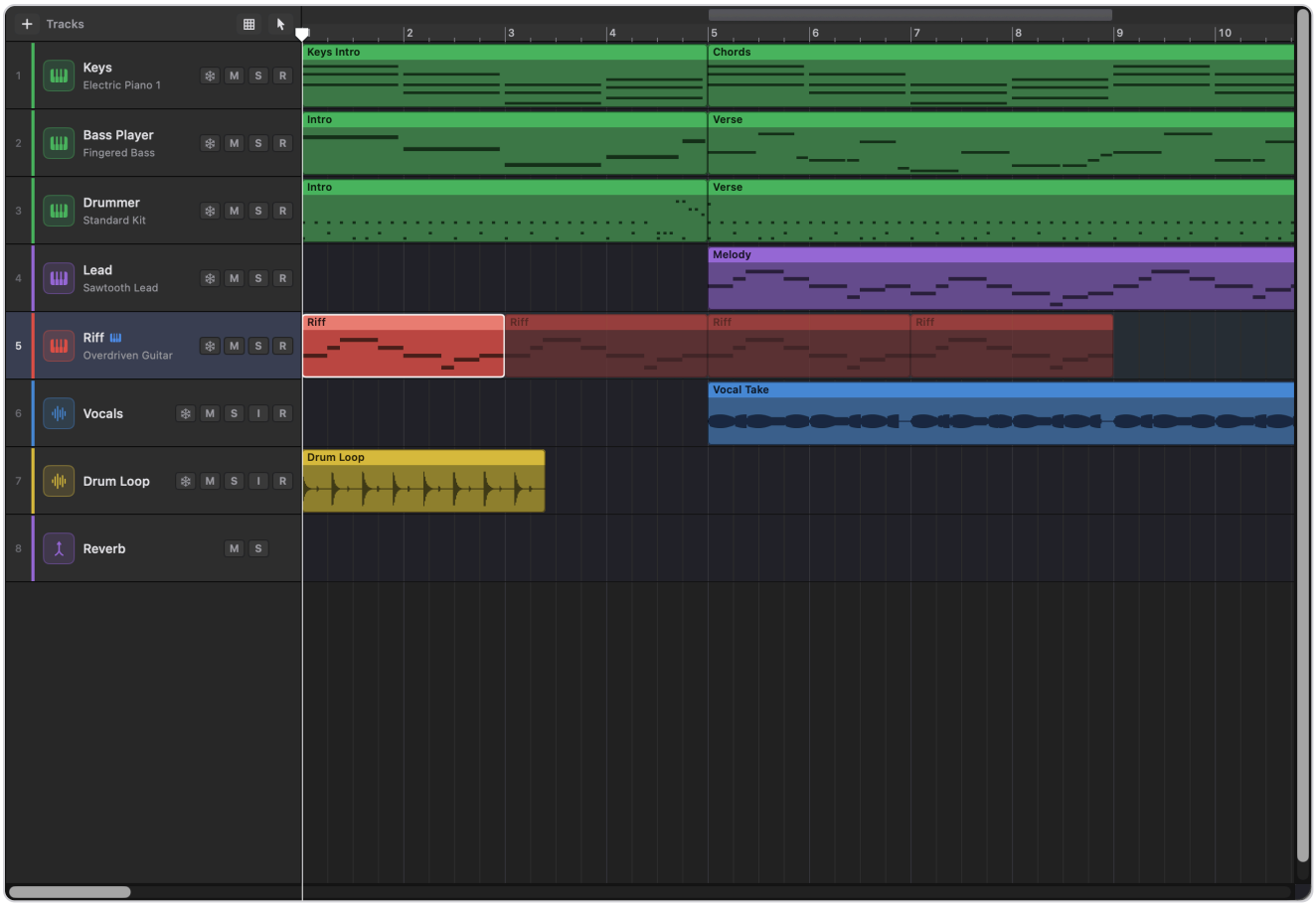
The region now repeats until the next region on its track, or the end of the song if nothing follows it.

Setting where the loop stops with the loop handle

To repeat a region for a set length instead:

1. Move the pointer to the **upper half** of the region's right edge. The pointer turns into a circular arrow: this is the **loop handle**.
2. Drag to the right. The repeats appear as you drag, and the end snaps to the grid. The last repeat can stop part way through.
3. To change it later, grab the end of the repeats (again, in the upper half of the track) and drag it.

Dragging the loop end back to the region's own end turns looping off.



A two-bar region on the Riff track, looped out to bar 9

Turning loops into regions

If you want to change one of the repeats on its own (say, add a fill to the last time round), right-click the region and choose **Convert Loops to Regions**. Each repeat becomes an ordinary region of its own, and the original stops looping.

Gotchas:

- Take folders don't have a loop handle.
- Trimming a looped region keeps the loop end where it was on the timeline.

Markers

Markers are named flags along the song, such as "Vocals in" or "Key change", that help you find your way around and jump straight to a spot. They don't change the sound.

Markers live on the **Marker** track, one of the global tracks under the ruler. Press **G** (or choose **Track › Show Global Tracks**) to see them.

To create a marker:

- choose **Transport › Create Marker** (⌥') to put one at the playhead (snapped to the grid), or
- click **+** beside **Marker** in the global track names, or
- double-click an empty spot on the Marker track.

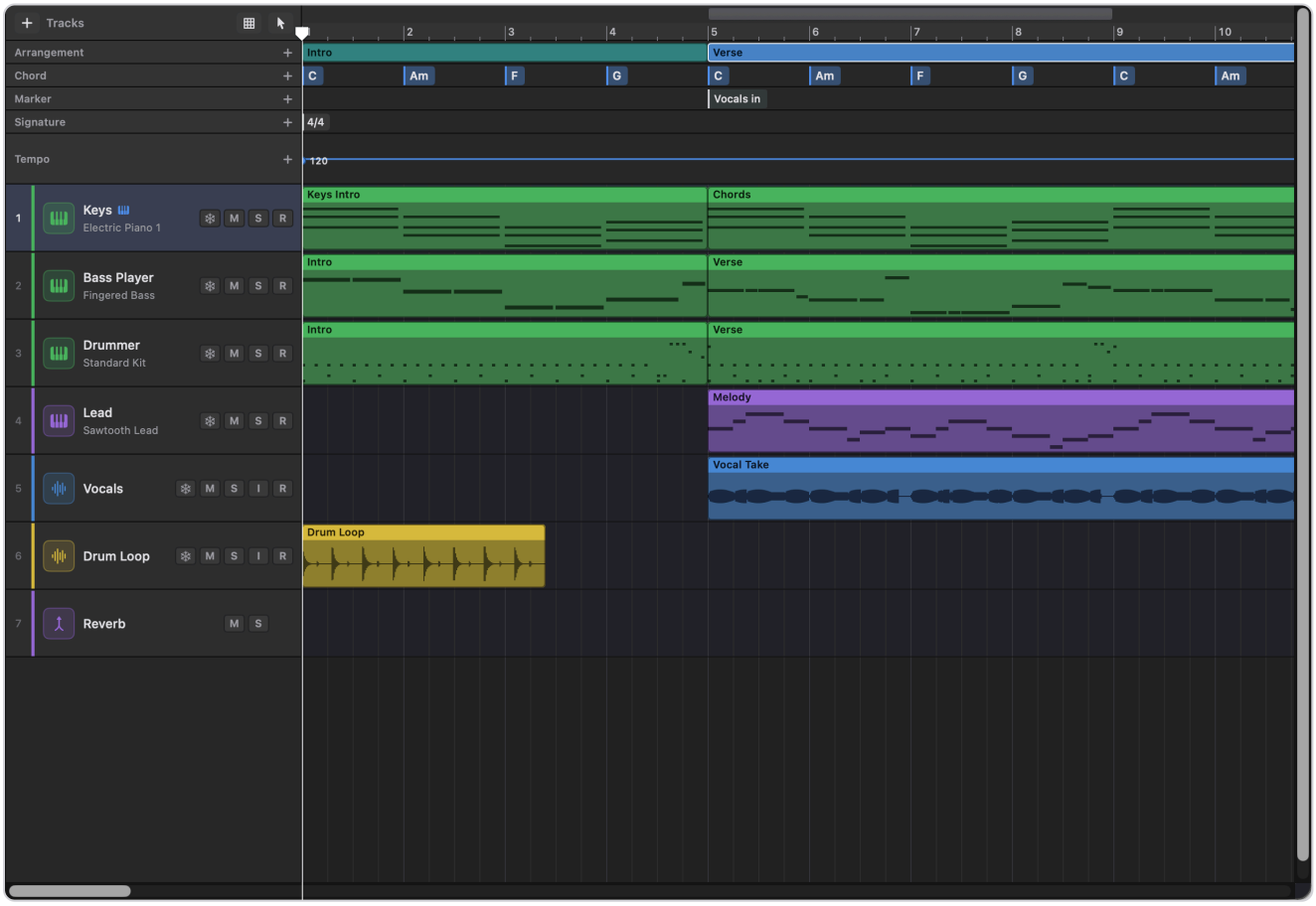
New markers are named "Marker 1", "Marker 2" and so on, each one number past the highest "Marker" number in the song, so deleting a marker never leaves two with the same name. Creating a marker also shows the global tracks. You can't put two markers in exactly the same place.

Working with markers:

To	Do this
Jump to a marker	Click it
Jump to the previous or next marker	**Transport › Go to Previous Marker** (⌥,) or **Go to Next Marker** (⌥.)
Move a marker	Drag it (it snaps to the grid)
Rename a marker	Double-click it, or right-click it and choose **Rename…**
Delete a marker	Right-click it and choose **Delete**

The Arrangement track

The **Arrangement** track (the top global track) divides the song into **sections**, such as Intro, Verse and Chorus. Once your song has sections, you can rearrange it by dragging whole sections around: drag the second chorus before the bridge, and every region, automation point and marker in it moves too. The menus call sections **arrangement markers**.



The Arrangement track with Intro, Verse, Chorus and Outro sections, Verse selected

Adding sections

Click **+** beside **Arrangement** in the global track names, or choose **Track › Add Arrangement Marker**. Each new section is eight bars long and goes straight after the last section (the first one starts at bar 1). They're named in turn Intro, Verse, Pre-Chorus, Chorus, Bridge, Solo, Breakdown and Outro.

Sections can't overlap.

Working with sections

To	Do this
Select a section and the regions that start in it	Click it
Rename it	Double-click it, or right-click and choose **Rename…**, or choose a name from right-click **Name**
Make it longer or shorter	Drag its right edge. Only the section's length changes; nothing in the song moves
Move it, with everything in it	Drag it left or right and drop it where it should go in the song order
Copy it, with everything in it	Right-click and choose **Duplicate**. The copy goes straight after it, and everything later moves along
Delete it, with everything in it	Right-click and choose **Delete with Contents**. Everything later moves back to close the gap

How moving works: the sections are laid end to end again in their new order, starting where the first section started, and each section's contents move with it. A region belongs to the section it **starts** in.

Gotchas:

- Gaps between sections close up when you move one.
- Clicking a section selects its regions, so pressing **Delete** afterwards deletes those regions (not the section). Use **Delete with Contents** to remove a section.
- To make a section that doesn't start after the last one, add it and then drag it into place, or resize its neighbours.

The Chord track

The **Chord** track, under the Arrangement track, holds the song's chords for the Bass and Keyboard players to follow. It's described in Chapter 5.

The Signature track

The **Signature** track shows the time signature and the key, and any changes of either part-way through the song: for example, one bar of 2/4 in a song in 4/4, or a last chorus that goes up a tone. Time signatures are white labels (**4/4**); keys are blue labels with a note (♪ **C maj**). Where a bar has both, they sit side by side.

Time signatures

To	Do this
Change the starting time signature	Click the first one (or right-click it) and choose a new one, or use **TIME** on the LCD
Add a change after the playhead	Click **+** beside **Signature** and choose **Time Signature at the Next Bar** (see below)
Add a change anywhere	Right-click an empty spot on the Signature track and choose from **Add Time Signature at Bar N**. It goes at the nearest bar line
Change an existing change	Click it and set it in the box that opens, or right-click it and choose another time signature
Move a change to another bar	Drag it (the first one can't be moved)
Delete a change	Click it and click **Delete** in the box, or right-click it and choose **Delete** (the first one can't be deleted)

Adding a change with +. Clicking **+** beside **Signature** and choosing **Time Signature at the Next Bar** adds a change at the start of the **next bar** after the playhead, in the time signature already in effect there. (If the playhead is exactly on a bar line, past bar 1, the change goes at that bar instead.) The global tracks are shown, and a box titled **Time Signature at Bar N** opens on the new change so you can set it straight away. If there's already a change at that bar, the box opens on it.

The time signature box. Clicking any change on the Signature track opens the same box. In it:

- the stepper sets the number of beats in a bar (VoiceOver calls it **Beats per Bar**);
- the menu after the **/** sets the beat unit (the note that counts as one beat, for example 4 for quarter notes or 8 for eighths);
- the buttons below set a common meter in one click: **2/4**, **3/4**, **4/4** and the rest of the list on the LCD;
- **Delete** removes the change (it isn't there for the first time signature, which can't be deleted).

Each change you make in the box is applied straight away and is its own undo step. Click outside the box to close it.

Key changes

A **key change** says the song is in a new key from that bar on. It doesn't transpose any of your regions: it's what the Chord track suggests chords and progressions from, and what the session players play where there are no chords (see Chapter 5). The key at bar 1 is the song's key.

To	Do this
Change the song's key	Click the key at bar 1 and choose one, or use **KEY** on the LCD with the playhead before the first key change
Add a key change after the playhead	Click **+** beside **Signature** and choose **Key Change at the Next Bar…**

To	Do this
Add a key change anywhere	Right-click an empty spot on the Signature track and choose **Add Key Change at Bar N…**. It goes at the nearest bar line
Change a key change	Click it and choose another key, or use **KEY** on the LCD with the playhead after it
Move a key change	Drag it to another bar
Delete a key change	Click it and click **Delete** in the box, or right-click it and choose **Delete**

The key box. Adding or clicking a key opens a box: choose **Major** or **Minor**, then click one of the twelve keys. When you're adding a change, the box says which key the song is in at that point; nothing is added until you choose a different one. A change to the key that's already playing does nothing, so it's left out (and a change that becomes the same as the key before it, say after you change the song's key, disappears).

Key changes move with the music when you cut a section, insert silence or repeat a section (see Editing a section between the locators), and they're written to MIDI files you export (Chapter 9).

The Tempo track

The **Tempo** track, at the bottom of the global tracks, draws the tempo as a line, with a round handle and its BPM at each tempo change. Use it when your song speeds up or slows down at a particular point.

To	Do this
Change the starting tempo	Drag the first handle up or down, or double-click **TEMPO** on the LCD and type
Add a tempo change	Double-click the Tempo track: the change goes at that spot (snapped), at the tempo level where you clicked. Or click **+** beside **Tempo** to add one at the playhead
Change a tempo	Drag its handle up (faster) or down (slower). Dragged tempos land on whole numbers
Move a tempo change	Drag its handle left or right (the first one can't be moved)
Delete a tempo change	Right-click its handle and choose **Delete**

Tempos can be from 5 to 990 BPM. Each change happens instantly at its position; there are no gradual speed-ups (ramps) yet.

Because regions are placed in bars and beats, they stay on the same bars when the tempo changes. MIDI regions simply play faster or slower. Audio isn't sped up or slowed down to match (use Smart Tempo for that, Chapter 6), so plan the tempo before recording or importing audio where you can.

Track stacks and track outputs

Where a track's sound goes

Every track has an **output**: where its sound goes after its fader. Normally that's **Stereo Out**, the main output you hear. You can send a track to an **aux track** instead, to treat several tracks as one: put all the drum tracks through one aux, and that aux's fader turns all the drums up and down together.

To change a track's output:

1. Select the track, and find its channel strip in the Inspector (or open the Mixer with **X**).
2. Click the **Output** slot (it says **Stereo Out** to begin with).
3. Choose **Stereo Out** or one of the aux tracks.

Ongaku won't let you make a loop (an aux that feeds back into itself), or output to an aux the track already sends to. Mixing, sends and auxes are covered in Chapter 7.

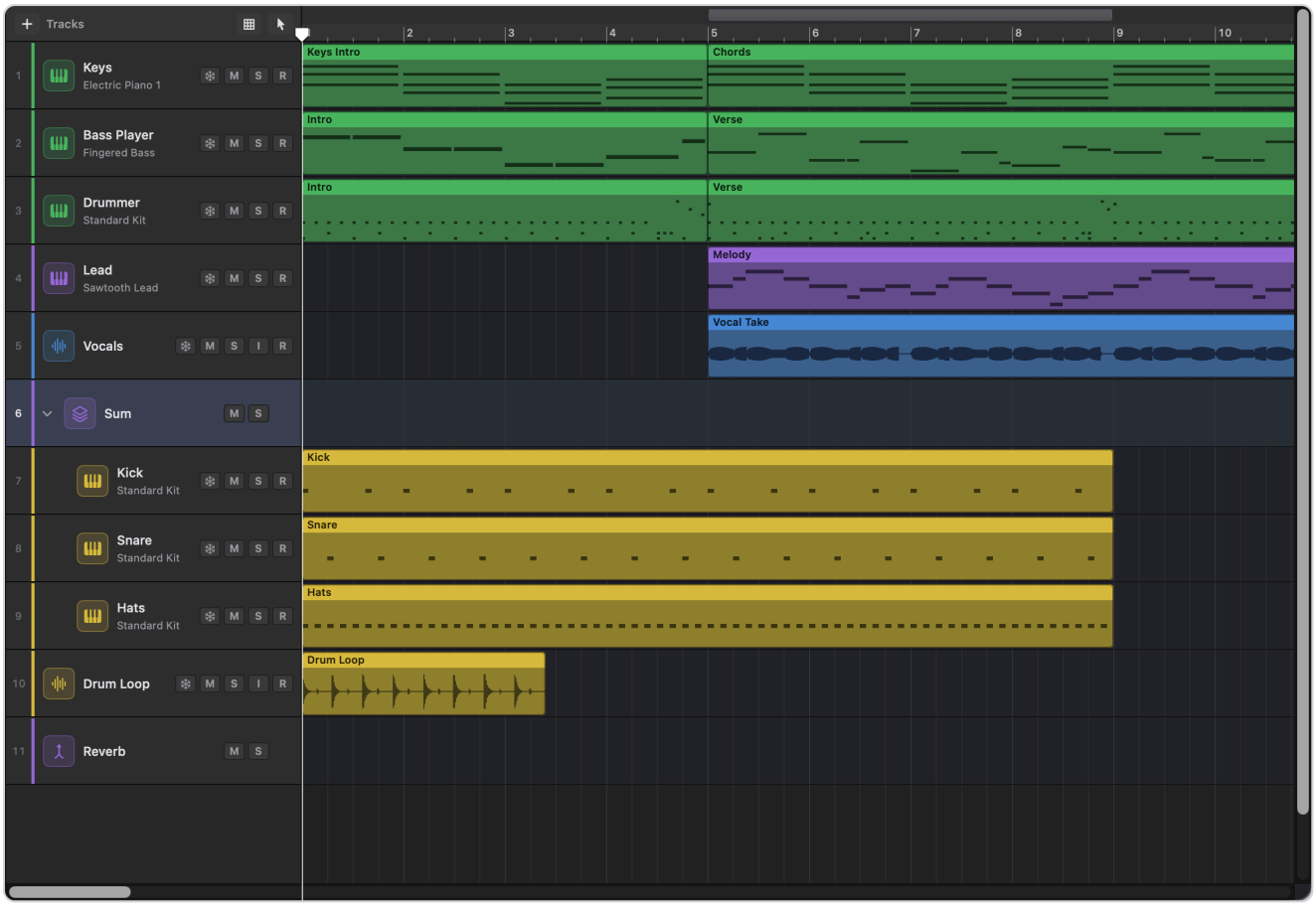
Summing stacks

A **summing stack** does the same job with less effort: it groups tracks under a new main track and sends all their outputs to it, so the main track's fader, effects, sends and mute work on all of them together. You can also fold the stack up to save space, which helps in a big song.

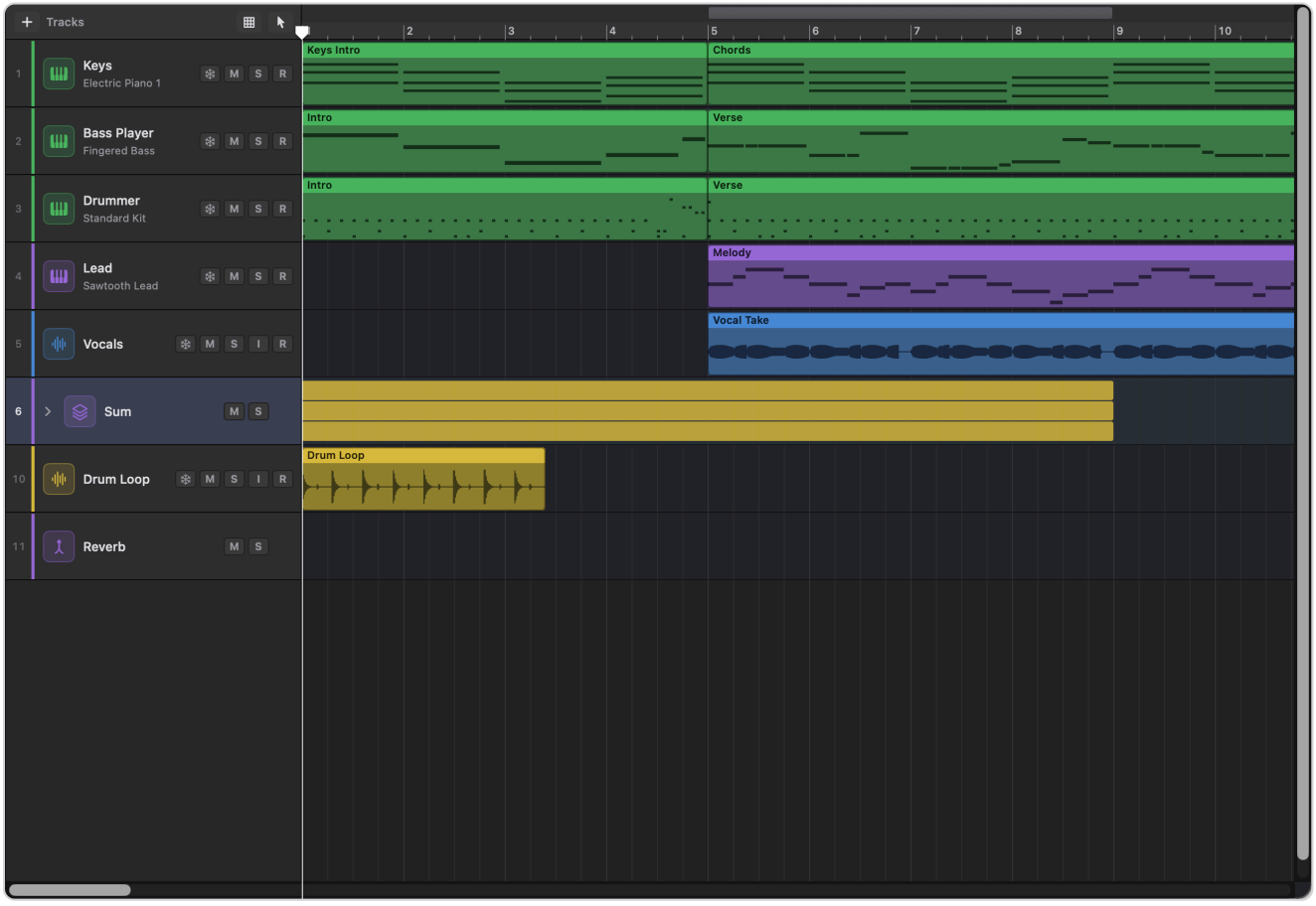
To create a summing stack:

1. Select the tracks: click the first track header, then Shift-click or ⌘-click the others. (None of them can already be in a stack.)
2. Choose **Track › Create Summing Stack** (⇧⌘D), or right-click one of the headers and choose **Create Summing Stack**.

A new main track called "Sum" (then "Sum 2" and so on) appears where the first selected track was, with the selected tracks indented below it as its **subtracks**. The main track is an aux track, so it has no regions of its own.



A summing stack of three drum tracks, expanded



The same stack, collapsed

Working with stacks:

To	Do this
Fold a stack up, or open it again	Click the triangle on the stack's header
Add a track to a stack	Select the stack or one of its subtracks, then add a new track: it joins the stack
Take a track out of a stack	Change the subtrack's **Output** to something other than the stack
Undo a stack	Select it and choose **Track › Flatten Stack**, or right-click and choose **Flatten Stack**: the subtracks become ordinary tracks again, sent wherever the stack was going, and the main track goes

When a stack is folded up, its row shows thin coloured strips where each subtrack has regions. These are just for looking at: open the stack to edit the regions.

Deleting the stack's main track (**Track › Delete Track**, ⌘⌫) deletes its subtracks too.

Importing audio files

You can bring in audio files (for example WAV, AIFF, CAF or MP3, anything your Mac can play) to use as loops, samples or backing tracks.

To import with the menu:

1. Put the playhead where the files should start. They start at the nearest grid line to the playhead, snapped as dropped files are.
2. To put them on an existing audio track, select that track. Otherwise, select any other kind of track (or none).
3. Choose **File › Import Audio Files…** (⇧⌘I), or click **Import Audio** in the control bar.
4. Choose one or more files and click **Open**.

To import by dragging:

1. Drag the files from the Finder onto the tracks area.
2. Drop them on an audio track to put them on that track, or anywhere else (another kind of track, or the empty space below the last track) to give each file a new track. They start where you drop them, snapped to the grid.

Where the files go:

- **Onto an audio track**: the files are placed one after another on that track.
- **Otherwise**: each file gets a new audio track, named after the file, and all of them start at the same position.

Each new region is named after its file and is exactly as long as the audio. If a long file runs past the end of the song, the song grows to fit it (see How long the song is). You can drop MIDI files the same way; they're described in Chapter 9.

What happens to the files

- Ongaku **copies** each file into the project (into its **Audio Files** folder, when you save). The original stays where it was, unchanged.
- If a file with the same name is already in the project, the copy gets a number ("Vocal 2.wav").
- Files are kept in their **own format and sample rate**: nothing is converted when you import. If a file's sample rate differs from your audio interface's (say, a 44.1 kHz file on a 48 kHz interface), Ongaku converts it as it plays, so it sounds at the right pitch and speed. The file itself isn't changed.
- Importing several files at once is a single step to undo.

If a file can't be read, Ongaku tells you which file and why, and imports the rest.

Limitations

- The song's length can't be set by hand. It grows by itself to 8 bars past the last region, marker or section, but never gets shorter by itself.
- There's only one tool at a time (no Command-click tool).
- Fades are straight lines; there's no control over their curve.
- Tempo changes are instant steps; there are no tempo ramps.
- Summing stacks can't be duplicated.
- Audio regions can't be pasted between projects.

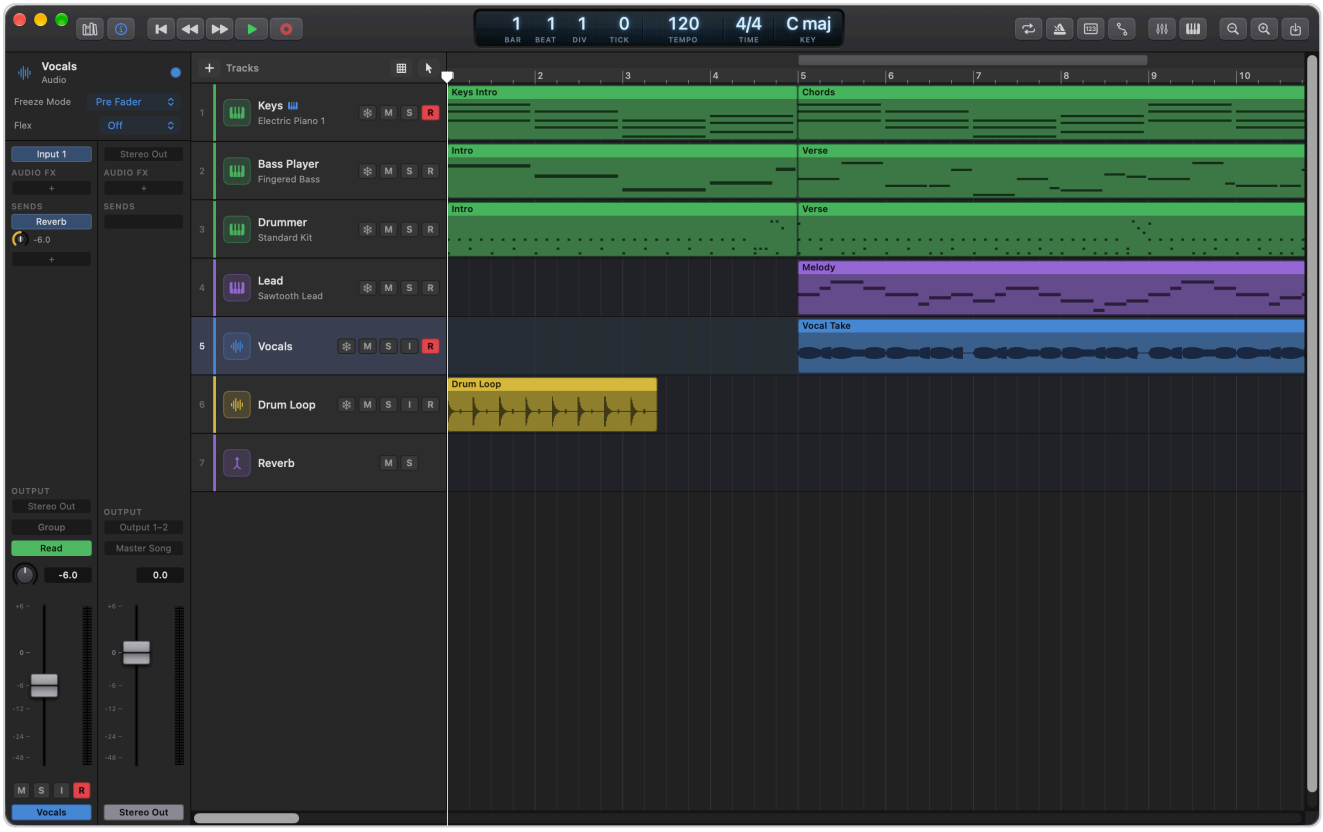
Quick reference

Action	How
Select, add to selection	Click, Shift-click
Select with a rectangle	Drag across empty space with the Pointer
Select all	⌘A
Move	Drag a region
Nudge a snap step earlier or later	⌥← or ⌥→
Copy by dragging	Drag, hold ⌥ as you release (the pointer shows a +)
Trim	Drag the left edge, or the lower half of the right edge
Loop to a length	Drag the upper half of the right edge (loop handle)
Loop on or off	Region Inspector **Loop**, or right-click **Loop or Unloop**
Open the Tool menu	T, then the tool's key
Pointer, Pencil, Eraser	T T, T P, T E
Scissors, Glue, Mute, Fade	T I, T G, T M, T A
New MIDI region	Double-click an empty instrument lane

Action	How
New pattern region at the playhead	**Track › New Pattern Region**
Fade in or out	Fade Tool: drag from the start or end of an audio region
Crossfade	Fade Tool: drag where two audio regions meet or overlap
Type a fade length	Click the **Fade In** or **Fade Out** number in the Region Inspector
Split at playhead	⌘T
Join regions	⌘J
Delete the selected regions	Delete
Choose what regions snap to	The **Snap** button above the track headers
Move freely, off the grid	Hold ^ after you start dragging
Delete the selected tracks	**Track › Delete Track** (⌘⌫)
Cut, copy, paste	⌘X, ⌘C, ⌘V
Repeat	⌘R (right-click **Repeat** for more)
Duplicate track	⌘D
Cut section between locators	^⌘X
Insert silence between locators	^⌘Z
Repeat section between locators	^⌘R
Show global tracks	G
Create marker	⌥'
Previous or next marker	⌥, or ⌥.
Add arrangement section	**Track › Add Arrangement Marker**
Add tempo change	Double-click the Tempo track
Add time signature change	**+** beside **Signature** › **Time Signature at the Next Bar**, or right-click the Signature track
Change a time signature	Click it on the Signature track
Add a key change	**+** beside **Signature** › **Key Change at the Next Bar…**, or right-click the Signature track
Change or delete a key change	Click it on the Signature track
Create summing stack	⇧⌘D
Import audio	⇧⌘I, or drag files from the Finder

3. Recording, Takes and Comping

This chapter covers getting sound into Ongaku Studio: setting up your audio interface and MIDI keyboard, recording audio and MIDI, playing along with the metronome, and recording several takes of a part so you can build one perfect performance out of the best bits. If you've never recorded on a computer before, read the sections in order. Each one builds on the one before.



The Keys and Vocals tracks record-enabled: their R buttons are lit

Before you record: audio devices

What it is

Your Mac can have more than one way to make and hear sound: the built-in speakers and microphone, headphones, or an audio interface (a box you plug microphones and guitars into). Ongaku Studio lets you choose where it **plays**. Where it **records from** comes from macOS.

Choosing the output device

To choose where Ongaku Studio plays:

1. Choose **Ongaku Studio › Settings…** and click the **Audio** tab.
2. Under **Output**, open the **Output device** menu. **System Default** follows whatever is set in macOS. Its name in brackets shows which device that is. Below it you'll find every connected device.
3. Choose an **I/O buffer size**, from 32 to 2048 samples. The default is 512.

Under the menus, Ongaku shows the device's **Sample rate** and the **Buffer latency** your buffer size gives you.

Setting	What it does
Output device	Where Ongaku Studio plays. If a device you chose is unplugged, it stays chosen and shows as **Not Connected**. Meanwhile Ongaku plays through the system default.
I/O buffer size	How much audio is worked out at a time. Small buffers make playing a keyboard feel immediate but work your Mac harder. If you hear crackles, choose a larger buffer. The menu only lists sizes the device accepts.
Sample rate (shown, not set)	The device's own rate. Recordings are made at this rate.
Buffer latency (shown)	How long one buffer lasts, in milliseconds.

Tip: Use a small buffer (128 or 256) while you record or play live, and a larger one (1024 or more) while you mix a big project.

Choosing the input device

Ongaku Studio records from the input device that's chosen in macOS.

To choose it:

1. In **Settings › Audio**, under **Input**, click **Open Sound Settings…**. This opens System Settings › Sound.
2. Choose your microphone or audio interface as the input there.

The first time you record audio (or turn on input monitoring), macOS asks whether Ongaku Studio may use the microphone. Click **Allow**. If you said no by mistake, recording stops with a message. Turn the permission on in **System Settings › Privacy & Security › Microphone**.

Choosing which inputs a track records

An audio interface often has several inputs, for example input 1 for a vocal mic and input 2 for a guitar. Each audio track records from its own input or pair of inputs. New audio tracks record from **Input 1** (mono).

To change a track's input:

1. Show the Mixer (press **X**).
2. At the top of the track's channel strip, click the input slot (it shows, for example, **Input 1**).
3. Choose a mono input (**Input 1** to **Input 8**) or a stereo pair (**Input 1–2**, **Input 3–4**, **Input 5–6**, **Input 7–8**).

If you choose an input your device doesn't have, Ongaku records from the nearest input it does have. For more on the Mixer, see Chapter 7.

Record-enabling tracks

What it is

Before you record, you tell Ongaku which tracks should record. This is called **record-enabling** or "arming" a track. Audio tracks record sound from their input. Software instrument tracks record what you play on a MIDI keyboard. You can arm several tracks and record them all at once, for example a drummer's three microphones.

To record-enable a track:

1. Find the **R** button at the right of the track's header. It shows **Record Enable** when you hover over it.
2. Click it. It lights red.
3. Click it again to disarm the track.

Gotchas:

- Aux tracks have no **R** button, because they can't record.
- A frozen track can't record. Its **R** button is greyed out and says **Frozen tracks can't record**. Unfreeze the track first (see Chapter 7).
- If you press Record with no track armed, Ongaku says: "Record-enable a track first (the R button on its header)."

Hearing yourself: input monitoring

What it is

Input monitoring plays what's coming into an audio track's input straight back to you, live, through the track's channel strip: its plug-ins, fader, pan and sends. Use it when you want to sing or play with the track's effects in your headphones (a little reverb on a vocal, an amp simulator on a guitar), or when your audio interface has no direct monitoring of its own.

It works whether or not you're playing or recording, so you can also use it to check your levels and sound before you start. Regions already on the track still play as usual.

To turn input monitoring on:

1. Find the **I** button on the audio track's header, between **S** and **R**. The same button is on the track's channel strip in the Mixer and the Inspector.
2. Click it. It lights orange, and you hear the track's input.
3. Click it again to turn it off.

The track listens to the input chosen in its input slot (see Choosing which inputs a track records). Input monitoring is off for new tracks. Turning it on or off is saved with the project and can be undone.

Gotchas:

- **Use headphones.** With speakers and a microphone in the same room, the mic picks up its own sound from the speakers and you get a loud howl (feedback).
- **There's a small delay.** What you hear is about one or two audio buffers behind what you play. A smaller **I/O buffer size** in **Settings › Audio** makes the delay shorter (see Choosing the output device). If it still bothers you, use your interface's direct monitoring instead.
- **The input and output devices must run at the same sample rate.** If they don't, Ongaku shows a message saying so. Open **Audio MIDI Setup** (in Applications › Utilities) and set both devices to the same rate.
- **Frozen tracks can't monitor.** The **I** button is greyed out until you unfreeze the track.
- Only audio tracks have an **I** button. Instrument tracks already play what you play on a MIDI keyboard.
- What gets recorded is your input as it arrives, without the track's plug-ins.

Recording audio and MIDI

What happens when you record

Recording starts at the playhead and runs until you stop. Everything already on other tracks plays while you record, so you can play along. Ongaku corrects for the small delay between hearing the music and your sound arriving, so your recording lines up with what you heard.

To record:

1. Arm one or more tracks (see above).
2. Put the playhead where you want to start. Click in the lower part of the ruler, or press **Return** to go to the beginning.
3. Start recording in one of these ways:
    - press **R**,
    - click the round **Record** button in the control bar,
    - or choose **Transport › Record**.
4. Play or sing.
5. Stop by pressing **R** again, pressing **Space**, clicking **Stop Recording**, or choosing **Transport › Stop Recording**.

While you record, the Record button pulses. Recording doesn't stop at the end of the song: the tracks area widens ahead of the playhead as you go, and the song grows to fit what you recorded (see How long the song is).

What you get

- **Audio tracks** get a new audio region that starts where the recorded sound starts. The region is named after the track. Ongaku saves the audio file inside the project.
- **Instrument tracks** get a new MIDI region. It begins at the start of the bar where recording began and ends at the end of the bar where you stopped (or where the last note ends). Any notes still held when you stop end there. Ongaku records the notes and their velocities, plus controllers such as the sustain pedal and mod wheel, pitch bend and aftertouch. Program changes are ignored.

- If you didn't play anything on an instrument track, no region is added to it.
- The whole recording, on every track, is **one undo step** named **Record**. Choose **Edit › Undo** to remove all of it at once.

Tip: New recordings are laid on top of whatever is already on the track, not in place of it. If you record over an old part, select the old and new regions and choose **Edit › Pack Take Folder** to keep both as takes (see "Packing regions into a take folder" below).

Audio file settings

To choose how audio is saved:

1. Choose **Ongaku Studio › Settings…** and click **Recording**.
2. Under **Audio Files**, choose a **File type** and **Bit depth**.

Setting	Choices	Notes
File type	**CAF** (default), **WAV**, **AIFF**	CAF is Apple's format and handles very long recordings. Choose WAV or AIFF if you want to open the files in other apps.
Bit depth	**16-bit**, **24-bit** (default), **32-bit float**	24-bit is the usual choice for recording.

Recordings are saved inside the project, at the output device's sample rate.

The metronome and count-in

What it is

The **metronome** clicks on every beat so you can keep time. It accents the first beat of each bar so you can tell where the bar starts. A **count-in** gives you one or two bars of clicks before recording starts, so you're ready on the downbeat.

To turn the metronome on or off, do one of these:

- press **K**,
- click the **Metronome** button in the control bar (it lights while on),
- or choose **Transport › Turn Metronome On** (or **Turn Metronome Off**).

The metronome follows the project's tempo and time signature. It clicks once for each beat of the time signature: four times in 4/4, and six times (once per eighth note) in 6/8.

Whether the metronome is on is saved with the project, so it's the way you left it when you open the song again. Switching it on or off can be undone, like any other change (the Edit menu calls it **Metronome Change**).

Metronome sound

To change how the metronome sounds:

1. Choose **Ongaku Studio › Settings…** and click **Metronome**.
2. Set the options:

Setting	What it does
Sound	**Click** (a short, bright blip), **Woodblock** (hollow and knocking), **Hi-Hat** (a burst like a closed hi-hat) or **Beep** (a longer, softer tone).
Volume	How loud the clicks are.
Accent the first beat of each bar	Makes beat 1 higher and a little louder.

Count-in

The quickest way to switch the count-in on or off is the **Count-in** button in the control bar, just right of **Metronome**. It lights while a count-in is on. Hold the pointer over it to see how many bars it counts in.

- Turning it **on** uses the number of bars you last chose in **Settings › Recording** (1 bar if you've never chosen).
- Turning it **off** sets the count-in to **None**.

The button and the setting are the same thing, so changing one changes the other. The count-in is part of the app's settings, not the song, so it applies to every project.

To choose how long the count-in is:

1. Choose **Ongaku Studio › Settings…** and click **Recording**.
2. Under **Metronome**, set **Count-in** to **None**, **1 bar** or **2 bars**. The **Count-in** button remembers the last length you chose here.
3. If you want clicks while you record even when the metronome is off, turn on **Click while recording, even with the metronome off**.

How these work together:

Metronome	Count-in	Click while recording	What you hear
On	any	any	Clicks during the count-in, recording and playback.
Off	1 or 2 bars	off	The count-in clicks, then silence once recording starts.
Off	any	on	Clicks while recording (and during any count-in), none during normal playback.
Off	None	off	No clicks.

Gotchas:

- The count-in is only used when you **record**. It isn't used when you just play.
- The **Count-in** button can only switch between your chosen length and none. To change between 1 and 2 bars, use **Settings › Recording**.

Playing and recording from a MIDI keyboard

What it is

A MIDI keyboard (or pad controller, or another app that sends MIDI) doesn't make sound by itself. It sends notes to an instrument on a software instrument track. Ongaku listens to every connected MIDI device automatically, and picks up keyboards you plug in while it's running.

Which tracks you hear

Your keyboard plays:

- **every record-enabled** instrument track, all together, if any are armed. This lets you layer sounds, for example a piano and a pad, and hear exactly what you'll record;
- otherwise, the **selected** track, if it's an instrument track.

A small keyboard icon appears after the name of each track your keyboard plays, so you can always see where your playing will go. Hold the pointer over it to see why that track is playing (it's record-enabled, or it's selected).

To play a different instrument, click its track (and disarm any armed instrument tracks). If a track has MIDI effects such as an arpeggiator, you hear them while you play, each track through its own effects (see Chapter 8). When you record, Ongaku records the notes you actually played, before the MIDI effects change them.

Turning inputs on and off

To stop Ongaku listening to a device, for example a controller that sends unwanted notes:

1. Choose **Ongaku Studio › Settings…** and click **MIDI**.
2. Under **Inputs**, turn off the device's switch.

Ongaku Studio plays and records only from inputs that are on. If nothing is connected, the tab says "No MIDI keyboards or controllers are connected."

Gotcha: If you arm **several** instrument tracks and record, every armed track gets the **same** recording of everything you played, just as you heard them all while playing. To record separate parts, arm one instrument track at a time.

Cycle recording and take folders

What it is

Often you want to try a part several times: sing the chorus five times, then keep the best one. **Cycle recording** makes this easy. Turn on the cycle (a looped section of the song) and record. Ongaku plays the section over and over, and each time round (each "pass") becomes a separate **take**. All the takes are

kept together in one **take folder**, a single region that holds them all. Afterwards you choose the best take, or build a **comp** (a compilation) from the best parts of several takes.

To cycle record:

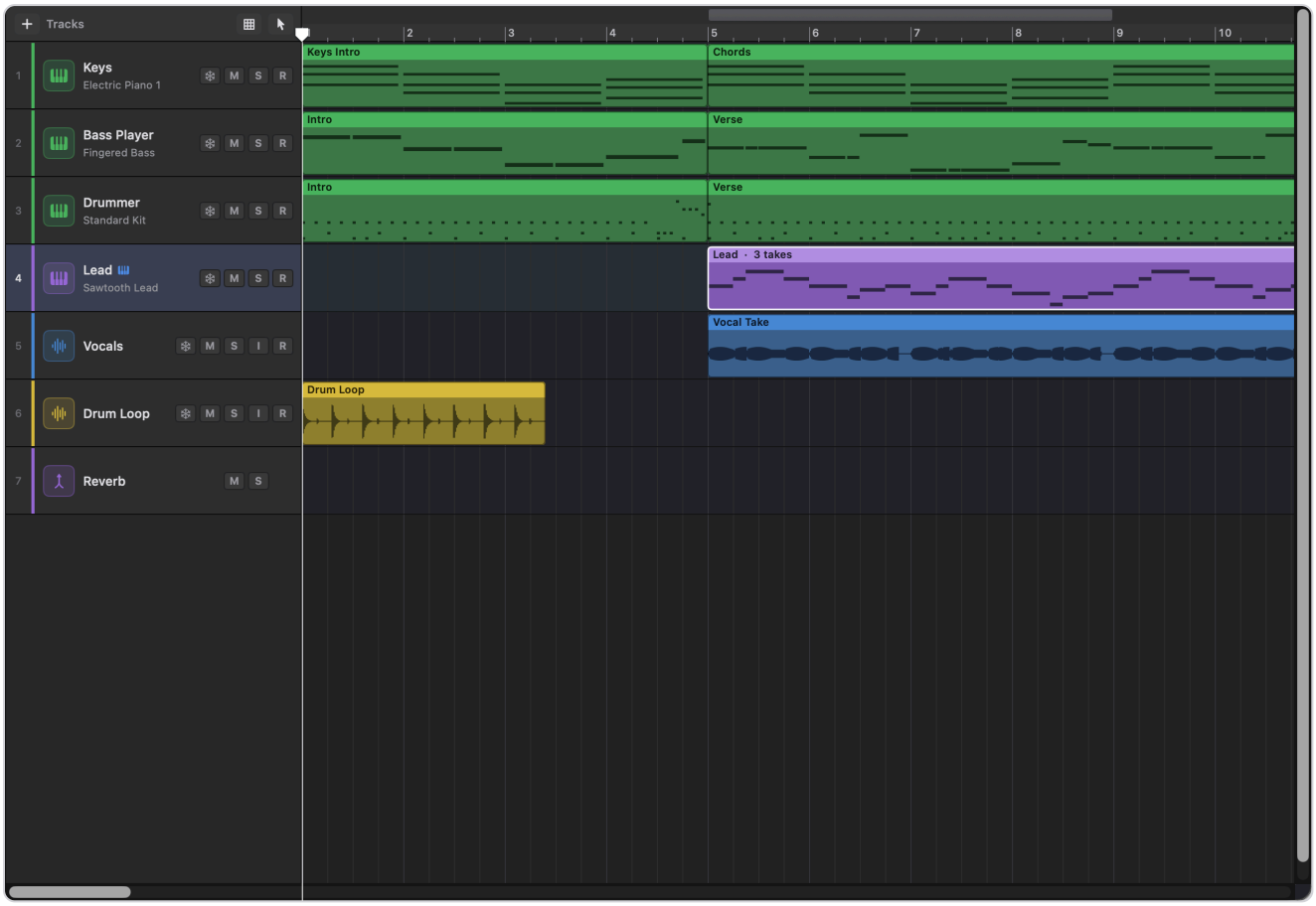
1. Set the cycle range by dragging in the upper strip of the ruler. Then turn the cycle on: press **C**, click the **Cycle** button, or click the cycle strip. See Chapter 2 for more about the cycle.
2. Arm the tracks you want to record.
3. Put the playhead at or a little before the cycle start, then press **R**.
4. Perform the part as many times as you like. Ongaku keeps looping.
5. Press **R** or **Space** to stop.

What you get:

- A take folder that covers exactly the cycle range. Its title shows how many takes it holds, for example "Vocal · 4 takes".
- The takes are named **Take 1**, **Take 2** and so on, oldest first.
- The **last** take plays, as in Logic. You can change this in the Takes editor.

How passes become takes:

- **Audio:** each pass that recorded at least one second becomes a take. If you recorded only one pass, you get a plain region instead of a take folder.
- **MIDI:** each pass in which you played something becomes a take. Passes where you played nothing are skipped. If only one pass had notes, you get a plain region. Notes still held at the end of a pass are cut off at the cycle end.



A take folder region on the Lead track, showing "· 3 takes" in its title

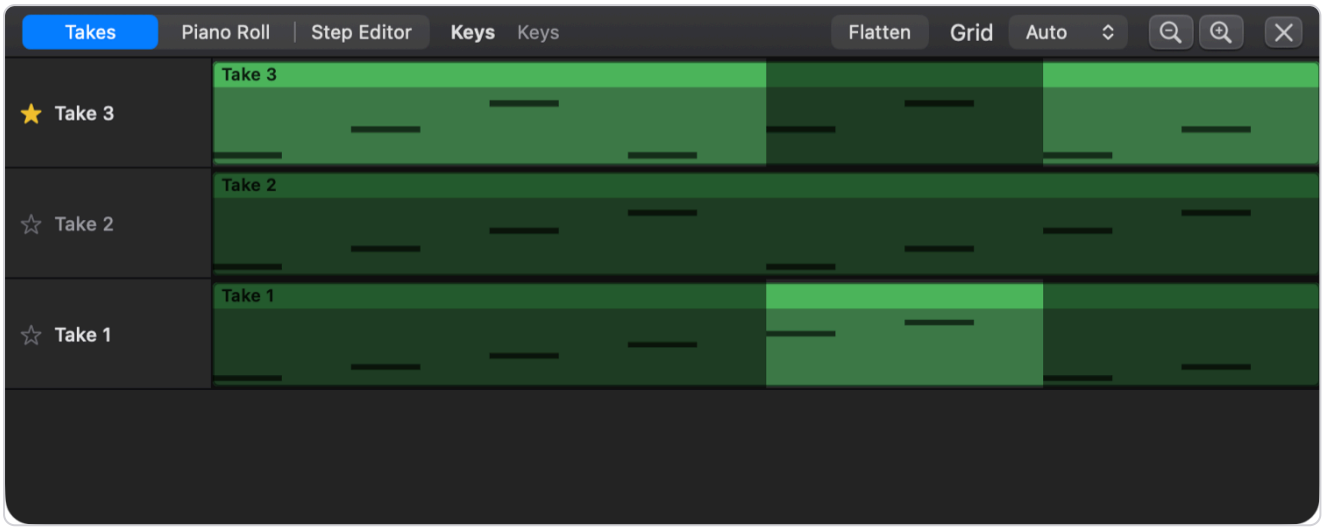
The Takes editor

What it is

The Takes editor shows every take in a folder as its own row, newest at the top. Each row's lit sections are the parts that play in the comp. Dimmed parts are not used. A take folder always plays its comp: for each moment, one chosen take.

To open the Takes editor, double-click a take folder in the tracks area, or select it and press **E**. Take folders always open in the Takes editor first.

A **MIDI** take folder can also be edited note by note: the editor pane shows a choice at the top left, **Takes**, **Piano Roll** and **Step Editor**. The Piano Roll and Step Editor edit one take, the one marked with the star (see below), and their header says which, for example ★ **Editing Take 3**. To edit a different take's notes, click that take's name in the Takes editor so it plays throughout, then switch to the Piano Roll. An audio take folder only has the Takes editor.



The Takes editor with three takes, parts of Take 1 and Take 3 lit

What you see in each row:

Part	Meaning
Star	A filled yellow star marks the take that **starts the comp**: the take that plays at the very start of the folder. For a MIDI folder, it's also the take the Piano Roll and Step Editor edit.
Take name	Bright if the take plays somewhere in the comp, grey if it's not used.
Take drawing	The take's waveform or notes. Parts that don't play are darkened.

With VoiceOver, each take row says how much of the comp it plays: **Not used**, **Partly used** or **Used throughout**, followed by "starts the comp" for the starred take.

Choosing a whole take

To play one take from start to end, click its name in the row header. Its tooltip says "Play Take *n* throughout the folder" (on the starred take, it says the take starts the comp).

Renaming and deleting takes

Right-click a take's row for its menu:

- **Rename…** opens a box where you type a new name, such as "Best verse". Click **Rename** (or press Return). A take's name can't be empty.
- **Delete Take** removes the take. The parts of the comp it played go silent, unless it was the only take used: then the first take left plays throughout.

When a folder has two takes and you delete one, the folder turns into an ordinary region playing the other take. A folder's only take can't be deleted. Renaming and deleting are ordinary edits, so **Edit › Undo** brings the name or the take back.

Quick-swipe comping

Quick-swipe comping is how you build a comp: you "swipe" across the bits of each take that you like.

To comp:

1. Open the folder in the Takes editor.
2. Drag across the part of a take that you want to hear. A white outline shows the range as you drag.
3. Let go. That take now plays across that range, replacing whatever played there before. The rest of the comp is unchanged.
4. Repeat with other takes. Neighbouring sections from the same take join up automatically.

Each swipe can be undone with **Edit › Undo**.

Tips:

- Swipes snap to the editor's **Grid** setting, in the editor's header (see Chapter 4). Choose a finer grid, such as **1/16**, for fine cuts, or **1/4** to comp beat by beat. On **Auto**, the grid gets finer as you zoom in. To make one cut off the grid, hold ^ (Control) after you start swiping.
- Use the zoom buttons in the editor's header to see more detail.
- Where two audio sections meet, Ongaku adds a very short crossfade so the switch doesn't click.

Flattening a take folder

When you're happy with the comp, you can **flatten** the folder. This replaces it with ordinary regions holding just what the comp plays.

To flatten:

- in the Takes editor, click **Flatten** in the header; or
- right-click the take folder in the tracks area and choose **Take Folder › Flatten Take Folder**.

The result:

- **Audio:** one region for each section of the comp, each playing its take, with short fades where they meet.
- **MIDI:** one MIDI region holding each section's notes.

Flattening throws away the takes the comp doesn't use. If you change your mind straight away, choose **Edit › Undo** to get the take folder back.

Packing regions into a take folder

You can also make a take folder out of regions you already have. For example, you can keep three separate guitar solos you recorded at different times and comp between them.

To pack regions:

1. Select two or more regions **on the same track**.
2. Choose **Edit › Pack Take Folder** (⌥⌘F), or right-click and choose **Take Folder › Pack Take Folder**.

Each region becomes a take. The folder spans from the earliest region's start to the latest region's end. At first the comp plays what you heard before. Where regions overlapped, the later one on the track plays, because it was the one on top.

You can't pack regions from different tracks, or pack a take folder into another one.

Limitations

- **Input monitoring has a little latency** (about one or two buffers), and needs the input and output devices at the same sample rate. For no delay at all, use your audio interface's direct monitoring.
- **Input device comes from macOS.** You choose the input device in System Settings › Sound, not in Ongaku.
- **No punch-in or punch-out.** Recording starts at the playhead and stops when you stop it.
- **New recordings layer on top** of existing regions. There's no "replace" mode.
- **A MIDI take folder's notes are edited one take at a time**: the take that starts the comp. To edit another take, make it play throughout first.
- **MIDI is played and recorded identically on every armed instrument track.** There's no per-track MIDI input or channel filtering.
- **MIDI clock, SysEx, program changes and MIDI 2.0 messages are ignored** when playing and recording.

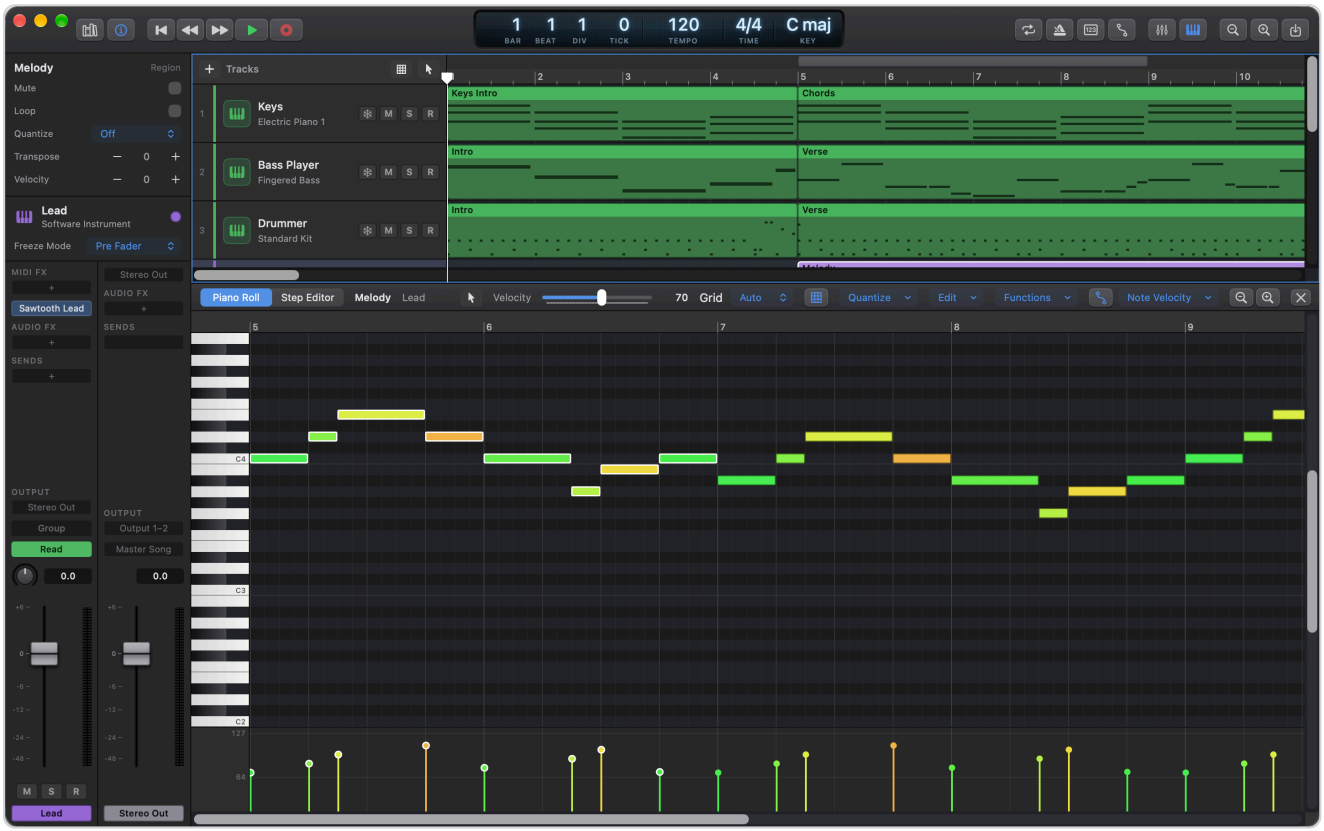
Quick reference

Action	How
Choose output device and buffer size	**Settings › Audio**
Choose input device	**Settings › Audio › Open Sound Settings…** (macOS Sound settings)
Choose a track's input channels	Mixer (**X**), input slot at the top of the channel strip
Record-enable a track	**R** button on the track header
Hear an audio track's input live	**I** button on the track header or channel strip (use headphones)
Record / stop recording	**R**, the **Record** button, or **Transport › Record**
Stop	**Space**
Go to beginning	**Return**
Metronome on/off	**K**, the **Metronome** button, or **Transport › Turn Metronome On**
Metronome sound and volume	**Settings › Metronome**
Count-in on/off	The **Count-in** button in the control bar
Count-in length, click while recording	**Settings › Recording › Metronome**
See which tracks a MIDI keyboard plays	The keyboard icon after the track name
Audio file type and bit depth	**Settings › Recording › Audio Files**
Turn MIDI inputs on/off	**Settings › MIDI › Inputs**

Action	How
Cycle on/off	**C**, or click the cycle strip in the ruler
Open the Takes editor	Double-click a take folder, or select it and press **E**
Play one take throughout	Click the take's name in the Takes editor
Comp part of a take	Drag across that part of the take's row
Rename or delete a take	Right-click its row › **Rename…** or **Delete Take**
Flatten a take folder	**Flatten** in the Takes editor, or right-click › **Take Folder** › **Flatten Take Folder**
Pack regions into a take folder	**Edit** › **Pack Take Folder** (⌥⌘F)
Undo a whole recording	**Edit** › **Undo** (⌘Z)

4. MIDI Editing: Piano Roll, Step Editor, Step Sequencer, MIDI Draw

A MIDI region doesn't hold sound. It holds instructions: which notes to play, when, for how long and how hard. Because of this, you can fix a wrong note, tighten the timing or change the instrument at any time without re-recording. This chapter shows the editors Ongaku Studio gives you for this work: the **Piano Roll** for notes, the **Step Editor** for drum hits and controllers, the **Step Sequencer** for pattern-based beats, and the **MIDI Draw** lane for velocity, controllers and pitch bend.



The Piano Roll open under the tracks area, with a melody and the MIDI Draw lane showing velocity

Instrument tracks and sounds

What it is

A **software instrument track** plays its MIDI regions through an instrument. New instrument tracks use **Acoustic Grand Piano**, a General MIDI sound built into macOS. You can switch to any of the 128 General MIDI sounds, a General MIDI drum kit, one of Ongaku's own instruments, or an Audio Unit instrument you've installed.

To add an instrument track, choose **Track › New Software Instrument Track** (⌥⌘S).

To choose a track's instrument:

1. Click the instrument name under the track's name in the track header, or right-click the header and choose **Instrument**. (In the Mixer, click the instrument slot at the top of the channel strip.)
2. Choose from the menu:

Menu section	What's in it
Ongaku	Ongaku's own instruments: **Retro Synth**, **Drum Machine**, **Sampler** and **Session Drummer**. See Chapter 8.
General MIDI	The 128 General MIDI sounds, grouped into families: **Piano**, **Chromatic Percussion**, **Organ**, **Guitar**, **Bass**, **Strings**, **Ensemble**, **Brass**, **Reed**, **Pipe**, **Synth Lead**, **Synth Pad**, **Synth Effects**, **Ethnic**, **Percussive** and **Sound Effects**.
Drum Kits	**Standard Kit**, **Power Kit**, **Electronic Kit**, **TR-808 Kit**, **Brush Kit**, **Orchestra Kit** and **Sound Effects Kit**.
Audio Units	Instrument plug-ins installed on your Mac, shown as "Maker: Name". If there are none, the menu says so.

For an Audio Unit or Ongaku instrument, the same menu has **Show Instrument Window…** at the top, which opens the instrument's own controls.

Tip: It matters whether a track plays drums. Choosing a drum kit (or the Drum Machine) gives you drum names in the Step Editor and a starter beat in new pattern regions.

Making a MIDI region

To make an empty MIDI region, do one of these:

- double-click an empty spot in an instrument track's lane;
- or right-click the lane and choose **New MIDI Region**.

Either way you get a one-bar region at that bar, and it opens in the Piano Roll. You can also record a region from a MIDI keyboard (see Chapter 3) or import a MIDI file (see Chapter 9).

Opening the editors

To open a region in the editor, double-click it, or select it and press **E**. You can also click the **Editor** button in the control bar, or choose **View › Show Editor**. Either way, the pane opens whichever editor suits the region: the Piano Roll, Step Editor, Step Sequencer, Drummer, Player, Takes or Flex Pitch editor. Press **E** again (or choose **View › Hide Editor**) to hide the editor, or click the **×** at the right of the editor's header.

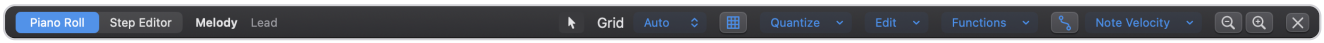
The editor pane holds several editors. When a region suits more than one, a switch at the top left of the editor's header lets you choose: **Piano Roll** or **Step Editor** for a MIDI region. A MIDI take folder offers **Takes**, **Piano Roll** and **Step Editor**, and opens in **Takes** first (Chapter 3). Pattern regions open in the **Step Sequencer**, and Drummer and session player regions open in their own editors (see Chapter 5).

The Piano Roll

What it is

The Piano Roll shows notes as bars on a grid. Higher notes are higher up, and a piano keyboard runs down the left so you can tell which pitch is which. Time runs left to right, with bar numbers in the ruler along the top. A note's **length** is how wide it is. Its **velocity** (how hard it was played) is its colour: soft notes are violet and blue, through green and yellow, to loud notes in red. Muted notes are grey.

Note names follow Logic: middle C (MIDI note 60) is **C3**.



The Piano Roll header, left to right: the editor switch, region and track name, Tool button, Velocity slider, Grid, Quantize, Edit, Functions, MIDI Draw and zoom

The header across the top of the Piano Roll holds, from left to right:

Control	What it does
Editor switch	Choose **Piano Roll** or **Step Editor** (only shown when there's a choice).
Region and track name	What you're editing. In a MIDI take folder, a yellow ★ **Editing Take *n*** follows them: the take whose notes you're editing (the one that starts the comp).
Tool button	The current tool's icon. Click it to choose another tool.
Velocity slider	Appears when notes are selected. Sets the velocity of all selected notes (1–127).
Grid menu	What notes snap to (see "The grid and snapping").
Snap button	Turns snapping to the grid on (lit blue) or off.
Quantize menu	Quantize options, strength and swing.
Edit menu	Clipboard and selection commands.
Functions menu	Note functions such as Transpose and Humanize.
MIDI Draw button and menu	Shows the MIDI Draw lane and chooses what it edits.
Zoom buttons	Zoom the Piano Roll in and out.

Clicking a note, or a key on the keyboard, plays it so you can hear what you've chosen.

Important: The Piano Roll's keys (Delete, Q, ⌥-arrows, ⌘C and so on) work on notes only when the Piano Roll was the **last place you clicked**. While the editor is open, a thin blue outline shows which area your keys act on: the tracks area or the editor. If the outline is round the tracks area, Delete, the clipboard keys and ⌥← / ⌥→ act on regions; click once in the Piano Roll to move the outline there.

The grid and snapping

Notes snap to the **grid** when you add, move, resize or split them, and quantizing moves them to it.

To set the grid, choose from the **Grid** menu in the header:

Choice	Meaning
Auto	The grid follows the zoom: the finest of 1/32, 1/16, 1/8 or a beat that's still comfortable to click. Zoom in for finer placement.
1/4, **1/8**, **1/16**, **1/32**	Straight notes.
1/4T, **1/8T**, **1/16T**, **1/32T**	Triplets: three notes in the time of two straight ones. Use these for shuffles, triplet fills and swing parts.

The grid is measured from each bar line, so it works in odd meters such as 5/8 or 7/8.

Turning snapping off. Click the **Snap** button, next to the **Grid** menu. It's lit blue while snapping is on, which is how every editor starts. With it off, notes go exactly where you draw, move or resize them. The grid lines still show, and **Quantize** still uses the grid, so you can play loose and tighten up later. Click **Snap** again to turn snapping back on.

Moving freely for one drag. To leave snapping on but place one note off the grid, hold **Control (^)** while you drag. Press it *after* you've started dragging: a Control-click is a right-click, so pressing it first opens the note menu instead.

The Piano Roll's Snap button is separate from the Snap button above the track headers, which sets what regions snap to (Chapter 2).

Tools

What they are

The Piano Roll has nine tools, as in Logic. The tool decides what a click or drag does. Most of the time you'll use the **Pointer**. The others are shortcuts for jobs you'd otherwise do with a menu.

To choose a tool, do one of these:

- click the Tool button in the header and choose from the list;
- or move the pointer over the Piano Roll, press **T** to open the Tool menu, then press the tool's key. Press **Escape** to close the menu without choosing.

While you hold ⌘ with the Pointer tool, it acts as the Pencil.

Key (after T)	Tool	What it does
T	**Pointer Tool**	Select, move and resize notes. Drag in empty space to select several. ⌥-drag to copy. Double-click empty space to add a note.
P	**Pencil Tool**	Click to add a note. Drag to set its length as you add it. It moves notes too, like the Pointer.
B	**Brush Tool**	Drag to paint notes along the grid at the pitch under the pointer.
E	**Eraser Tool**	Click a note to delete it (or the whole selection, if it's selected).
I	**Scissors Tool**	Click a note to split it there (snapped to the grid).

Key (after T)	Tool	What it does
G	**Glue Tool**	Select some notes, then click one of them to join the selected notes of each pitch into one long note.
M	**Mute Tool**	Click a note to mute or unmute it.
V	**Velocity Tool**	Drag a note up or down to change its velocity (and the selection's).
Q	**Quantize Tool**	Click a note to quantize it (or the selection) to the grid.

Tips:

- When you click a **selected** note with the Eraser, Mute, Velocity, Scissors or Quantize tool, the tool works on the whole selection.
- The Pencil remembers the length of the last note you drew or resized. A plain click gives a new note that length.
- New notes get the velocity of the last note you clicked (100 at first).
- The Brush skips grid cells that already have a note at that pitch.
- The Glue tool needs at least two selected notes. Otherwise it reminds you: "Select the notes to join, then click one of them with the Glue tool."

Adding notes

To add notes:

- **Pointer:** double-click empty space. You get a note one grid step long.
- **Pencil:** click to add a note, or drag right to set its length.
- **Brush:** drag across the grid to paint a run of notes, each one grid step long.

Selecting notes

To select notes:

To select…	Do this
One note	Click it.
Add or remove a note	⇧-click it.
Several notes	With the Pointer, drag a rubber band around them in empty space. ⇧-drag to toggle more notes in or out of the selection.
Every note of one pitch	Click that key on the keyboard at the left. ⇧-click more keys to add their notes.
All notes	⌘**A**, or **Edit › Select All** in the Piano Roll's header.
Every note of the pitches already selected	**Edit › Select Same Pitches**.
All muted notes	**Edit › Select Muted Notes**.
Everything not selected	**Edit › Invert Selection**.

To select…	Do this
Nothing	Click empty space, or **Edit › Deselect All**.

Moving, copying and resizing

To **move** notes, drag one of them. All selected notes move together, snapping to the grid (unless **Snap** is off, or you hold ^ while dragging). Drag up or down to change their pitch. You hear the new pitch as you go.

To **copy** notes, hold ⌥ as you start dragging. The originals stay put and the copies become the selection.

To **resize** notes, drag a note's left or right edge (the pointer changes to a resize arrow).

- The right edge changes the length.
- The left edge moves the start but keeps the end where it is.
- All selected notes change by the same amount.
- Notes can't get shorter than one grid step this way.

To **transpose and nudge** with the keyboard (selected notes only):

Keys	Action
⌥↑ / ⌥↓	Transpose up or down a semitone.
⇧⌥↑ / ⇧⌥↓	Transpose up or down an octave.
⌥← / ⌥→	Nudge earlier or later by one grid step.

Notes that would go above or below the MIDI range stay where they are. Notes can't be nudged before the start of the region.

Notes outside the region. The area to the right of the region's end is shaded. Notes there (dimmed) are kept but don't play. Make the region longer in the tracks area to hear them (see Chapter 2).

Cut, copy, paste and repeat

Use the Piano Roll's **Edit** menu, the right-click menu, or the usual shortcuts (when the Piano Roll was the last place you clicked):

Command	Shortcut	What it does
Cut	⌘X	Removes the selected notes and keeps them on the clipboard.
Copy	⌘C	Copies the selected notes.
Paste at Playhead (**Paste** in the right-click menu)	⌘V	Pastes at the playhead, or at the region's start if the playhead is before it. The pasted notes are selected.
Repeat (**Edit › Repeat Notes** in the menu bar)	⌘R	Copies the selected notes straight after themselves. A phrase of up to a bar repeats one bar later. A longer phrase repeats right after its own end.

Command	Shortcut	What it does
Delete	Delete	Deletes the selected notes. With no notes selected, it does nothing.

Copied notes can be pasted into another region or another project window.

Muted notes

A muted note stays in the region but doesn't play. It's drawn grey. Muting is a handy way to try a part without a note before you delete it.

To mute or unmute notes, do one of these:

- click a note with the **Mute Tool** (if it's selected, the whole selection is muted or unmuted);
- or select notes and choose **Functions › Mute / Unmute**. If every selected note is already muted, this unmutes them. With no notes selected, it mutes (or unmutes) every note in the region, like the other functions.

To find muted notes later, choose **Edit › Select Muted Notes**.

The right-click menu

Right-click a note, or empty space, for a menu of the most common commands: **Cut**, **Copy**, **Paste**, **Repeat**, **Delete**, **Select All**, **Select Same Pitches**, **Invert Selection**, then every note function (see below). If you right-click a note that isn't selected, it's selected first, so the command acts on it.

Note functions

The **Functions** menu (and the bottom of the right-click menu) changes notes in bulk. A function works on the **selected** notes, or on **every** note in the region if none are selected.

Function	What it does
Quantize	Moves notes to the grid (see "Quantize" below).
Transpose › Up a Semitone / Down a Semitone / Up an Octave / Down an Octave	Changes pitch.
Mute / Unmute	Mutes the notes, or unmutes them if they're all muted already.
Note Length › Force Legato	Makes each note last until the next one starts, for smooth, connected lines.
Note Length › Remove Overlaps	Shortens notes that run into the next note of the same pitch.
Note Length › Set to Grid	Makes every note one grid step long. The menu shows the current grid.
Velocity › Louder (+10 %) / Softer (–10 %)	Scales velocities up or down.
Velocity › Set to 127 / 100 / 80 / 64 / 40	Gives every note the same velocity.
Humanize	Moves notes and changes their velocities by small random amounts, so a programmed part sounds more played.

Function	What it does
Reverse	Plays the notes backwards in time, within the span they cover.
Invert Pitches	Mirrors the pitches: the highest note becomes the lowest.

Every function is a single undo step.

Quantize

What it is

Quantizing moves notes onto the grid, so a slightly loose performance plays exactly in time. Ongaku moves each note's **start**. Lengths are kept. Two settings let you keep some feel:

- **Strength** says how far notes move towards the grid. At 100 % they land exactly on it. At 50 % they move halfway, which tightens the timing but keeps some of your feel.
- **Swing** delays every second grid line, for a shuffle. At 100 % the off-beats land a third of a step late, a triplet feel.

To quantize notes:

1. Select the notes to quantize, or select none to quantize the whole region.
2. Set the **Grid** in the header to the smallest note value in your part, for example **1/16**.
3. If you want, open the **Quantize** menu and choose a **Strength** (**100 %**, **75 %**, **50 %**, **25 %**) and a **Swing** (**Off**, **25 %**, **50 %**, **75 %**, **100 %**). Ongaku remembers these until you change them.
4. Press **Q**, or choose **Quantize › Quantize to Grid (Q)**.

The **Quantize** menu also lets you quantize to a grid other than the current one:

Section	Choices
Straight	**1/4 Note**, **1/8 Note**, **1/16 Note**, **1/32 Note**
Triplets	**1/4 Triplet (1/6)**, **1/8 Triplet (1/12)**, **1/16 Triplet (1/24)**, **1/32 Triplet (1/48)**
Straight and Triplets	**1/8 & 1/12**, **1/16 & 1/12**, **1/16 & 1/24**, **1/32 & 1/24**

The names in brackets are Logic's names for triplet grids. They count how many fit in a 4/4 bar: twelve eighth-note triplets make a bar, so "1/12".

Mixed quantize (the **Straight and Triplets** choices) is for parts that mix straight notes with triplets, such as a drum fill. Each note goes to whichever grid line is nearer, the straight one or the triplet one, so neither kind gets pulled out of shape.

The **Quantize Tool** and **Functions › Quantize** use the current grid, strength and swing too.

Tip: To quantize how a region **plays** without moving its notes, use **Quantize** in the region's Inspector instead. It has the same straight, triplet and mixed grids, and you can turn it off again at any time (see Chapter 2).

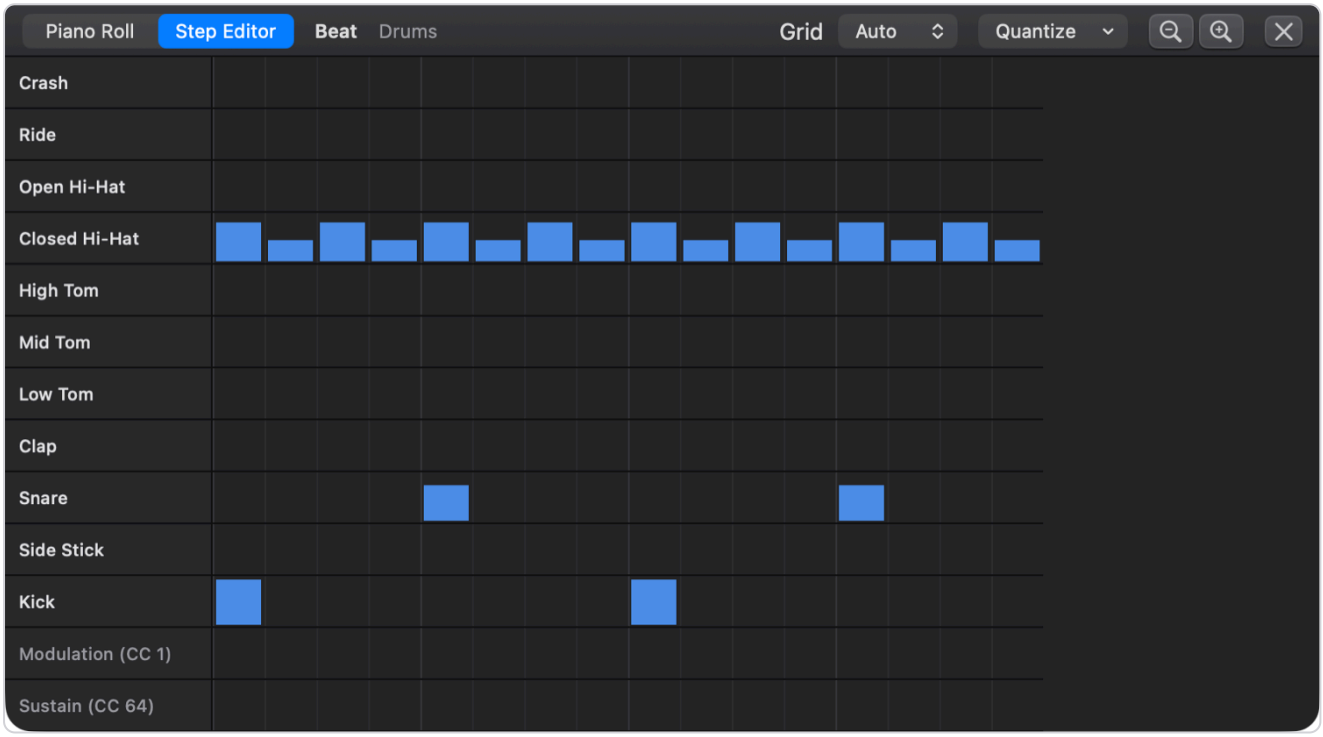
Gotcha: Q quantizes only in the Piano Roll. In the Step Editor, use the **Quantize** menu.

The Step Editor

What it is

The Step Editor shows a region as rows of steps, like a drum machine. Each row ("lane") is one note, for example a kick drum, or one controller. Each event is a bar as tall as its velocity or value. It's the quickest way to program a drum part, or to set the loudness of each hit at a glance.

To open it, open a MIDI region and click **Step Editor** at the top left of the editor's header.



The Step Editor on a drum track with Kick, Snare and Closed Hi-Hat lanes filled in

Which lanes you see:

- On a track playing a **General MIDI drum kit**: **Crash**, **Ride**, **Open Hi-Hat**, **Closed Hi-Hat**, **High Tom**, **Mid Tom**, **Low Tom**, **Clap**, **Snare**, **Side Stick** and **Kick**, then any other notes the region uses.
- On a track playing the Ongaku **Drum Machine**: its own pads, with its names: **Crash**, **Open Hat**, **Closed Hat**, **High Tom**, **Mid Tom**, **Low Tom**, **Shaker**, **Cowbell**, **Clap**, **Snare**, **Rim** and **Kick**, then any other notes the region uses.
- On other tracks: every pitch the region uses, highest first (or **C3** in an empty region).
- Then **Modulation (CC 1)** and **Sustain (CC 64)**, plus any other controller the region already uses.

The step size is the **Grid** setting in the header. On **Auto**, steps are sixteenth notes.

The Step Editor has no keys of its own. In particular, **Delete** does nothing while you're working here (remove events by double-clicking them), so it can't delete notes the Piano Roll had selected out of sight.

To edit steps:

To…	Do this
Add an event, or change one	Click the step. The bar's height is set by how high you click: near the top is loud, near the bottom is soft.
Fill several steps	Drag across them. Each step takes the height of the pointer as it passes.
Remove an event	Double-click its bar.

New notes are one step long. Each click or drag is one undo step.

Gotcha: The lanes on a drum track are only named for **General MIDI drum kits** and the **Drum Machine**. On a track playing another drum plug-in (including the Session Drummer), lanes are named by note, for example **C1**.

The Step Sequencer

What it is

The Step Sequencer is a pattern-based way to make beats and riffs, like the step sequencer in Logic or a hardware groovebox. A **pattern region** holds a pattern of steps rather than notes. The pattern loops through the region. Each row can have its own length, speed and direction, so rows of different lengths drift against each other for interesting polyrhythms. Each step can also have its own velocity, length, chance of playing and number of repeats (rolls).

Making a pattern region

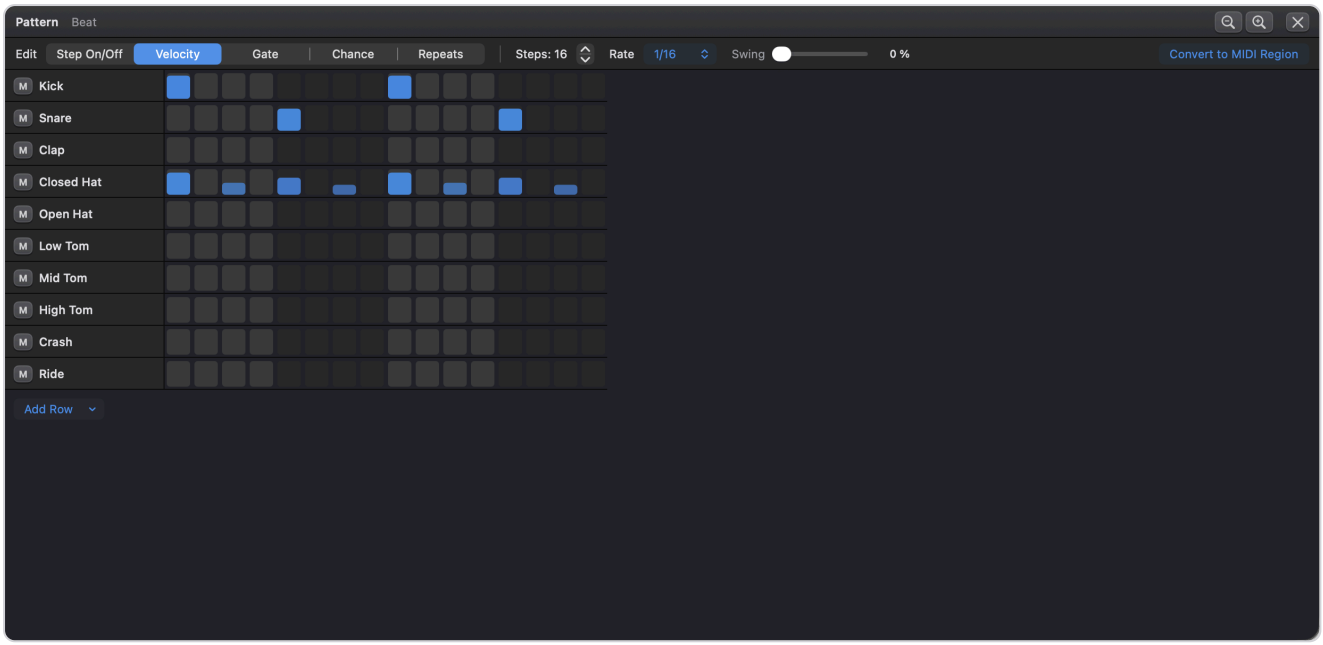
To make a pattern region:

1. Right-click an empty spot in an instrument track's lane.
2. Choose **New Pattern Region**.

Or select an instrument track and choose **Track › New Pattern Region**: the region goes in the bar containing the playhead.

You get a one-bar region named "Pattern", and it opens in the Step Sequencer.

- On a **drum track** (a General MIDI kit or the Drum Machine), it starts with rows for **Kick**, **Snare**, **Clap**, **Closed Hat**, **Open Hat**, **Low Tom**, **Mid Tom**, **High Tom**, **Crash** and **Ride**, with a simple beat to start from: kick on beats 1 and 3, snare on 2 and 4, eighth-note hats.
- On other tracks, it starts with empty rows for a C major scale from **C3** up to **C4**.



The Step Sequencer with the edit-mode switch set to Velocity

The pattern bar

The bar across the top of the Step Sequencer sets up the whole pattern:

Control	What it does
Edit switch	What clicking and dragging on steps changes: **Step On/Off**, **Velocity**, **Gate**, **Chance** or **Repeats**.
Steps	How many steps the pattern has before it starts again (1–64; 16 at first).
Rate	How long each step is: **1/4**, **1/4T**, **1/8**, **1/8T**, **1/16**, **1/16T**, **1/32** or **1/32T** (**T** means a triplet).
Swing	Delays every second step, for a shuffle.
Convert to MIDI Region	Turns the pattern into the notes it plays (see below).

Edit modes

Choose an edit mode, then click or drag across steps. In every mode except **Step On/Off**, a step's height shows its value, and clicking or dragging sets the value from how high you click. Setting a value also turns the step on.

Mode	What it changes	Range
Step On/Off	Turns steps on or off. Drag across steps to make them all match the first step you changed.	On or off
Velocity	How hard the step plays. Brighter steps are louder in every mode.	1–127
Gate	How long the note plays, as a part of the step. Short gates are staccato.	5 %–100 %, in 5 % steps

Mode	What it changes	Range
Chance	How likely the step is to play. A step with a chance below 100 % shows a **?**.	0 %–100 %, in 5 % steps
Repeats	Plays several quick notes in the step, for rolls and flams. A step with repeats shows **×2**, **×3** or **×4**.	1–4

Tip on Chance: Whether a "maybe" step plays is decided once for each place in the song, so the region plays the **same way every time** you play it. It varies from bar to bar, not from playback to playback.

While the song plays, a white outline shows the step each row is on.

Rows

Each row has an **M** button and a name. Click **M** to mute the row. Click the row's **name** for its menu:

Menu item	What it does
Length	**Pattern (*n*)** follows the pattern's step count. Or give the row its own length: **3**, **4**, **5**, **6**, **7**, **8**, **12**, **16**, **24** or **32 Steps**.
Rate	**Pattern (…)** follows the pattern's rate. Or give the row its own rate.
Direction	**Forward**, **Backward**, **Ping-Pong** (forwards then backwards) or **Random**.
Clear Row	Turns every step in the row off.
Remove Row	Deletes the row.

When a row has its own length, rate or direction, they're shown in grey next to its name.

Tip: For a polyrhythm, give the hi-hat row **3 Steps** while the kick plays 16. The hat pattern lines up with the kick only every three bars.

To add a row, click **Add Row** under the rows and choose:

- **Drums**: **Kick**, **Snare**, **Clap**, **Closed Hat**, **Open Hat**, **Low Tom**, **Mid Tom**, **High Tom**, **Crash**, **Ride**, **Pedal Hat**, **Side Stick**, **Cowbell** or **Shaker**.
- **Notes**: any pitch, grouped by octave (**C4–B4** down to **C0–B0**).

New rows appear at the top.

Converting a pattern to notes

A pattern region can only be edited in the Step Sequencer. To edit its notes in the Piano Roll (to add a fill by hand, say), convert it:

- click **Convert to MIDI Region** in the pattern bar; or
- right-click the region in the tracks area and choose **Convert to MIDI Region**.

The region becomes an ordinary MIDI region holding the notes the pattern played, and the editor switches to the Piano Roll. There's no command to turn notes back into a pattern. Only **Edit › Undo** takes you back.

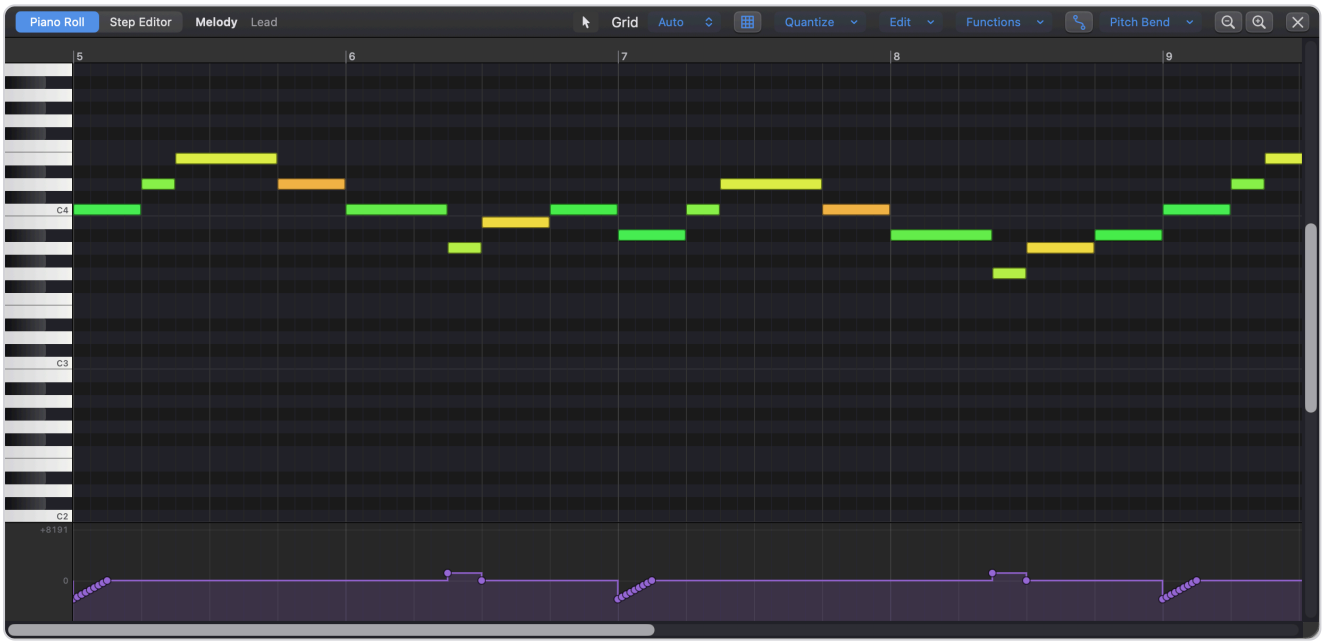
MIDI Draw

What it is

Besides notes, MIDI carries **controllers**: continuous movements such as the mod wheel, volume, pan, the sustain pedal, pitch bend and aftertouch (pressure on keys you're already holding). The **MIDI Draw** lane, under the Piano Roll's notes, lets you see and draw these over time. It also shows each note's velocity as a "stalk" you can drag. This is often easier than hunting for note colours.

To show the lane, click the **MIDI Draw** button in the Piano Roll's header (its tooltip says "Show MIDI Draw: velocity, controllers, pitch bend and aftertouch"). Click it again to hide it. Ongaku remembers whether it's shown.

To make the lane taller or shorter, drag its top edge.



The MIDI Draw lane showing a pitch bend curve (its parameter menu, at the right of the header, says Pitch Bend)

Choosing what to edit

When the lane is showing, a menu next to the MIDI Draw button names what it shows. Click it to choose:

- **Note Velocity**
- **Volume (CC 7)**, **Pan (CC 10)**, **Modulation (CC 1)**, **Expression (CC 11)**, **Sustain (CC 64)**
- **Pitch Bend**, **Channel Aftertouch**, **Poly Aftertouch**
- **Other Controllers**: every controller from 0 to 119, by name where it has one (for example **Brightness (CC 74)**).

A dot (•) after a name means the region already has data for it.

The scale at the left of the lane shows the range. Pan shows **L64** to **R63** around **C** (centre). Pitch bend shows **+8191** to **-8192** around **0**. Sustain shows **On** at 64 and above, **Off** below.

Editing velocity

With **Note Velocity** chosen, each note has a stalk at its start. The taller the stalk, the louder the note.

To…	Do this
Change one note	Drag its stalk up or down.
Change several notes together	Select them first, then drag one of their stalks. They all move by the same amount.
Sweep velocities	Drag across empty space in the lane. Each stalk you pass jumps to the pointer's height. If notes are selected, only selected notes change. Use this for quick crescendos.

A small label shows the value as you drag.

Editing controllers, pitch bend and aftertouch

Controller values are drawn as steps, the way MIDI plays them: each value holds until the next point. Before the first point, the line shows the controller's resting value (centre for pan and pitch bend, 100 for volume, full for expression).

To…	Do this
Add a point	Click an empty spot.
Move points	Drag a point (sideways for time, up and down for value). ⇧-click to select several, then drag one to move them all.
Draw freehand	Drag across empty space. The stroke replaces the points it passes over.
Delete points	Select them and press **Delete**.

Poly Aftertouch belongs to one pitch. Select exactly one note first, or the lane says "Select one note to edit its aftertouch."

Gotchas:

- New points are not snapped to the grid.
- If any points are selected in the lane, **Delete** removes those points, not your selected notes. Click a note to clear the point selection first.
- New points go on the MIDI channel of the selected note, or the channel most notes in the region use.

Compound and odd meters

What it is

In **compound meters**, the beat you feel is a group of three eighth notes (a dotted quarter). Examples are 6/8 (two felt beats in a bar), 9/8 (three) and 12/8 (four). Ongaku treats any meter with 6, 9, 12 or more beats (a multiple of three) over 8 or 16 this way.

You choose a time signature by clicking **TIME** on the control bar's display. The list has **2/4**, **3/4**, **4/4**, **5/4**, **6/4**, **7/4**, **2/2**, **3/8**, **5/8**, **6/8**, **7/8**, **9/8** and **12/8** (see Chapter 2).

How the editors handle them:

- In the Piano Roll and the tracks area, the stronger beat lines fall on the felt beats (every dotted quarter in 6/8). The eighth notes in between are drawn fainter once you zoom in far enough.
- On **Auto**, the coarsest Piano Roll grid is the dotted-quarter beat rather than a quarter note.
- Snapping and quantizing always count from the bar line, so odd meters like 5/8 and 7/8 line up correctly bar after bar.
- The metronome clicks every beat of the meter, for example six clicks per bar in 6/8, with the first one accented.

Tip: For a swung or shuffled part in 4/4, try the **1/8T** grid or a mixed quantize rather than switching to 12/8.

Limitations

- Keyboard shortcuts for notes (**Q**, ⌥-arrows, **Delete**) work only in the Piano Roll, not in the Step Editor.
- The Piano Roll's **Snap** setting isn't saved: each project window starts with snapping on.
- The Step Editor names drum lanes only for General MIDI drum kits and the Drum Machine.
- A pattern region can't be edited in the Piano Roll until it's converted, and only Undo turns a converted region back into a pattern.
- MIDI Draw points aren't snapped to the grid, and there's no line or curve tool. Draw freehand, or add points one by one.

Quick reference

Action	How
New instrument track	**Track › New Software Instrument Track** (⌥⌘S)
Choose an instrument	Click the instrument name in the track header
New MIDI region	Double-click an empty instrument lane, or right-click › **New MIDI Region**
New pattern region	Right-click an instrument lane › **New Pattern Region**, or **Track › New Pattern Region**
Open/close the editor	Double-click a region, or **E**

Action	How
Choose a tool	**T**, then **T P B E I G M V Q**, or the Tool button
Temporary Pencil	Hold ⌘ with the Pointer
Add a note	Double-click (Pointer), click (Pencil)
Select notes of one pitch	Click a key on the keyboard
Rubber-band select	Drag in empty space (Pointer)
Copy by dragging	⌥-drag notes
Transpose a semitone / an octave	⌥↑ ⌥↓ / ⇧⌥↑ ⇧⌥↓
Nudge a grid step	⌥← ⌥→
Snapping on or off	**Snap** button in the Piano Roll header
Move a note off the grid	Hold ^ after you start dragging
Quantize	**Q**, or **Quantize › Quantize to Grid (Q)**
Strength and swing	**Quantize › Strength**, **Quantize › Swing**
Cut / Copy / Paste	⌘X / ⌘C / ⌘V
Repeat notes	⌘R
Select all notes	⌘A
Delete notes (or MIDI Draw points)	Delete
Mute notes	**Mute Tool**, or **Functions › Mute / Unmute**
Step Editor: set / remove a step	Click / double-click
Step Sequencer: change edit mode	**Edit** switch in the pattern bar
Convert a pattern	**Convert to MIDI Region** (pattern bar, or right-click the region)
Show MIDI Draw	**MIDI Draw** button in the Piano Roll header
Zoom the editor	Zoom buttons in the editor header

5. Session Players: Drummer, Bass, Keys and the Chord Track

Session players are virtual bandmates. You tell them what kind of part you want and how hard to play, and Ongaku Studio writes the notes for you, section by section, to fit your song. This chapter covers the Drummer, the Chord track that the other players read from, and the Bass Player and Keyboard Player.

What a session player is

A session player region looks like any other MIDI region in the Tracks area, and it plays through the track's instrument like any other. The difference is where the notes come from. You don't draw them. The player works them out from a few settings: a style, how busy, how loud, and so on.

That has some useful results:

- **The part follows your song.** Make a drummer region longer and the drummer plays to the new end. Change the time signature and every drummer and player region is played again in the new meter. Change a chord on the Chord track and the bass and keys follow it straight away.
- **Each section gets its own part.** When your song has arrangement markers (Intro, Verse, Chorus and so on), a new player track gets one region per section. Each one starts with settings that suit its name: a quiet, simple intro, a loud and busy chorus, fills between sections.
- **You can always ask for another version. New Take** plays the same settings a different way. The playing is worked out from the settings, so a part sounds the same every time until you change something or ask for a new take.
- **You can turn it into ordinary notes.** If you want to edit single notes, convert the region to a normal MIDI region (see Converting a part to MIDI).

Every change you make in a player editor is a normal edit, so **Edit › Undo** (⌘Z) takes it back.

> Tip: The players work best when your song has arrangement markers. Add them first with **Track › Add Arrangement Marker**. Each new marker is eight bars long and is named after the next suggestion in the list (Intro, Verse, Pre-Chorus, Chorus, Bridge, Solo, Breakdown, Outro). See Chapter 2 for the Arrangement track.

How section names shape the part

When a player region is created for a section, it reads the section's name and picks a starting point. These names are recognised (capital letters don't matter):

Section name	How the Drummer starts	How the Bass and Keys start
Intro	Soft and fairly simple	Soft and simple
Verse	Medium	Medium

Section name	How the Drummer starts	How the Bass and Keys start
Pre-Chorus	A little louder and busier, more fills	A little louder and busier
Chorus	Loud and busy	Loud and busy
Bridge	Medium, with fills	Medium
Solo	Loud and busiest	Loud and busy
Breakdown	Soft and simple, half time, no toms	Softest and simplest
Outro	Medium-loud	Medium
Any other name	The default middle settings	The default middle settings

The Drummer also crashes into every section except the first, and fills into every section except the last. The Bass and Keys lead into every section except the last, and play an ending on the last one.

These are only starting points. You can change any of them in the editor afterwards.

The Drummer

The Drummer plays a full drum kit part: kick, snare, hi-hat or ride, toms, cymbals and, if you want, a shaker. Use it when you need drums quickly, for a demo, a practice track, or as a starting point you'll convert and edit later.

Adding a Drummer track

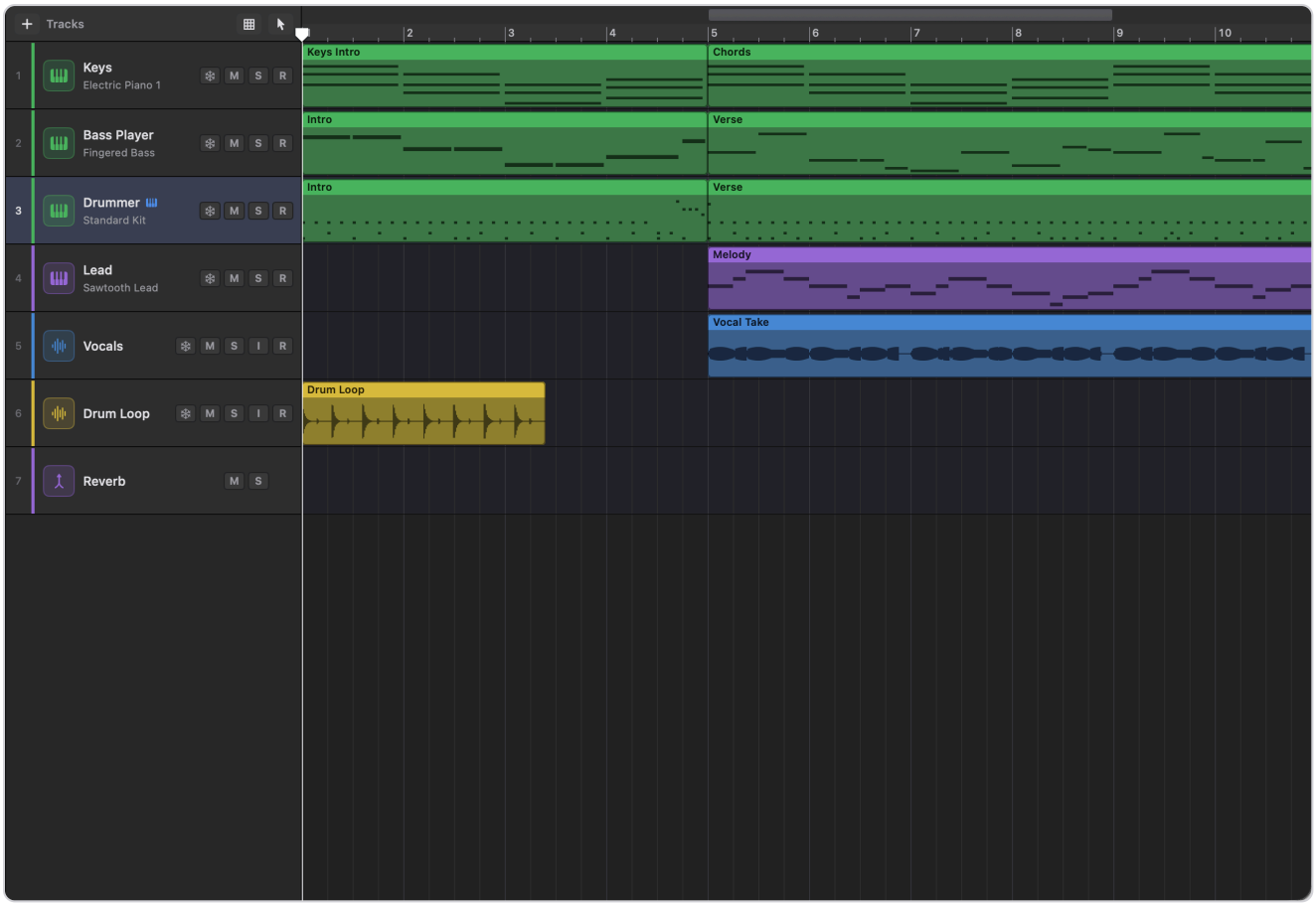
1. If you want the drummer to play each part of your song its own way, add arrangement markers first (see the tip above).
2. Choose **Track › New Drummer Track** (⌥⌘U).
3. A new track called **Drummer** appears below the selected track, using a standard drum kit sound. If there's already a track called Drummer, the new one is called **Drummer 2**, and so on.
4. The editor opens at the bottom of the window with the first drummer region selected, showing the Drummer editor.

The Drummer track gets:

- **One region per arrangement section**, named after the section, if your song has arrangement markers.
- **One region called "Drummer"** if it doesn't. It starts at bar 1 and runs to the end of the last region in the song, and is at least eight bars long. The drummer still adds fills along the way.

New Drummer tracks start in the **Rock** style.

For an electronic part, choose **Track › New Drummer Track (Drum Machine)** instead (or **Drummer (Drum Machine)** from the **+** button above the track headers). The track plays the Ongaku Drum Machine, and the drummer starts in the **Electronic** style.



The Drummer track, with one region per arrangement section

Adding more drummer regions

You can add a drummer region to any track that plays drums. That means a track using a General MIDI drum kit, or the Ongaku Drum Machine.

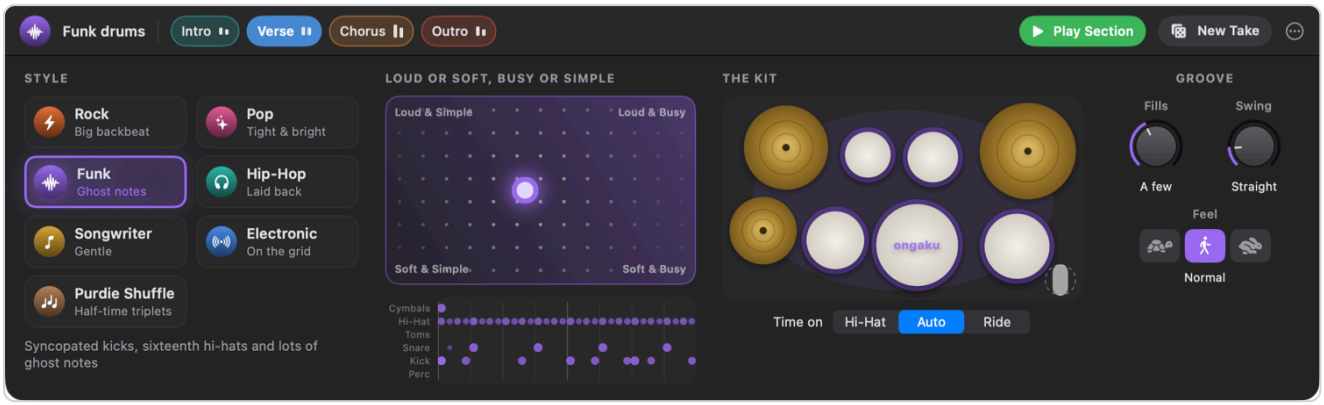
1. Right-click an empty spot in the drum track's lane.
2. Choose **New Drummer Region**.

If you click inside an arrangement section, the region fills that section and takes its name. If not, it's eight bars long from the bar you clicked. The new region uses the same style as the other drummer regions already on the track.

On tracks that don't play drums, this menu item says **New Drummer Region (Drum Tracks Only)** and is greyed out. That stops a drummer from playing its kit on a piano or a glockenspiel.

Opening the Drummer editor

Double-click a drummer region. The editor pane opens with the Drummer editor, because drummer regions can only be edited there. (The Piano Roll and Step Editor tabs don't appear for drummer regions until you convert them to MIDI.) You can also select the region and press **E** to show the editor.



The Drummer editor

The Drummer editor has a strip across the top, and four areas below it: **Style**, the pad and groove chart, **The Kit**, and **Groove**.

The song strip

The strip across the top shows the drummer's style (for example "Rock drums") and a row of chips, one for each drummer region on the track, in song order.

- **Click a chip** to edit that section. Each chip shows two small bars: how loud and how busy that section is played. Hover over a chip to see the numbers, for example "Chorus: 78 % loud, 60 % busy".
- Chips are coloured to match their arrangement sections. A region that isn't named after a song section shows its bars instead, for example **Bars 1–8**.
- **Play Section** turns on Cycle around the section you're editing and starts playback, so you can hear your changes as you make them. While it plays, the button changes to **Stop**.
- **New Take** (the dice) plays the same settings a different way.
- **More** (the **…** button) has these options:
  - **Fill into the Next Section**: play a fill at the end of the region, leading into whatever comes next.
  - **Crash as the Section Starts**: hit a crash cymbal on the first downbeat.
  - **Convert to MIDI Region**: turns the part into ordinary notes. See Converting a part to MIDI.

If your song has arrangement markers but the drummer's regions don't line up with them (perhaps you added the markers after the drummer, or moved a section), a **Match Song Sections** button appears. Click it to give each arrangement section exactly one drummer region:

- **Sections that already have a drummer region keep it, with your settings.** A region counts as a section's if it covers the same bars, or otherwise if it has the section's name. It's moved and resized to fit the section, renamed after it, and played again to its new length.
- **New sections** get a new drummer region, played to suit the section's name, in the style the track already uses.
- **Drummer regions left over**, for sections that are no longer in the song, are removed. Because their settings go with them, Ongaku asks first: "Remove the drummer regions for sections that are gone?", naming the regions. Click **Match and Remove 1 Region** (or **Match and Remove *n* Regions**) to go ahead, or **Cancel**.

The whole match is one undo step, so **Edit › Undo** puts everything back as it was.

If your song has no arrangement markers, the strip shows a tip instead: add markers with **Track › Add Arrangement Marker** and the drummer plays each section its own way.

Style

Pick who's playing. Click a style card to switch. The line of text under the cards describes the selected style, and hovering over a card shows its description too.

Style	Card says	How it plays
Rock	Big backbeat	Straight eighths on the hi-hat, a solid backbeat, tom fills
Pop	Tight & bright	Tight and steady, with a lighter touch and more kick variation
Funk	Ghost notes	Syncopated kicks, sixteenth hi-hats and lots of ghost notes
Hip-Hop	Laid back	Laid-back, swung sixteenths with off-beat kicks
Songwriter	Gentle	Sparse and gentle, cross-stick when quiet
Electronic	On the grid	Four on the floor, off-beat open hats, snare rolls
Purdie Shuffle	Half-time triplets	Half-time triplet shuffle: hi-hat on each beat's first and last triplet, ghost notes between, the backbeat on 3

Changing the style also resets **Swing** to that style's usual amount. Hip-Hop, Funk and Purdie Shuffle start with some swing; the others start straight.

Loud or Soft, Busy or Simple (the pad)

The big square pad is the main way to steer the drummer.

- **Left to right** goes from simple to busy. Busier parts add extra kicks, ghost notes on the snare, faster hi-hats and longer fills.
- **Bottom to top** goes from soft to loud.

The corners are labelled **Loud & Simple**, **Loud & Busy**, **Soft & Simple** and **Soft & Busy**.

1. Drag the dot around the pad.
2. While you drag, the groove chart underneath updates to show what the drummer would play.
3. Let go, and the region is played again with the new settings.

With VoiceOver, the pad has four actions: **Louder**, **Softer**, **Busier** and **Simpler**. Each moves the dot a tenth of the way. (Swiping up and down changes how loud it is.)

How loud the part is also changes what the drummer plays, not only the volume. In the quiet Songwriter and Pop styles the snare becomes a cross-stick. With **Time on** set to **Auto**, a loud part moves from the hi-hat to the ride.

The groove chart

Under the pad is a small drum chart of the region's first bars: two bars, or four if the pane is wide enough. It has a row each for **Cymbals**, **Hi-Hat**, **Toms**, **Snare**, **Kick** and **Perc**. Each dot is a hit, and bigger, brighter dots are harder hits. While the song plays, a line moves across the chart so you can follow along.

The Kit

The Kit is a drum kit seen from above. Click a drum to leave it out or bring it back in. Drums that are left out are greyed with a dashed outline. While the song plays, each drum lights up as it's hit.

Drums are switched in groups:

Click	Switches on or off
Crash, **Hi-Hat** or **Ride**	All the cymbals
High Tom, **Mid Tom** or **Floor Tom**	All the toms
Kick or **Snare**	Kick and snare together
Shaker	Percussion: a shaker, plus a cowbell in busy Funk parts

Percussion is off when a part starts. Click the shaker to add it.

If you turn off both the kick and snare and the toms, the drummer doesn't play fills, because there's nothing to fill with.

Under the kit, **Time on** chooses what keeps time:

Setting	What keeps time
Hi-Hat	Always the closed hi-hat
Auto	The hi-hat, opening up to the ride when the part gets loud (not in the Electronic style)
Ride	Always the ride

On the Ongaku Drum Machine, which has no ride cymbal, open hi-hats play instead of the ride.

Groove: fills, swing and feel

Fills sets how often and how long the drummer fills, from **None** through **A few** and **Some** to **Lots**. Fills come at the end of the region (if **Fill into the Next Section** is on) and now and then every four bars inside it. With lots of fills, some are longer.

Swing pushes the off-beats late, toward a triplet feel, from **Straight** through **A little** and **Swung** to **Shuffle**. Funk and Hip-Hop swing their sixteenth notes; the other styles swing eighths.

In the **Purdie Shuffle** style, the knob is called **Ghosts** instead. The triplets already are the swing, so this knob sets how loud the ghost notes on the snare are, from **Whisper** through **Soft** and **Present** to **Hard**.

To use a knob, drag up or down on it. Double-click it, or ⌥-click it, to set it back to zero.

Feel has three buttons:

Button	Feel	What it does
Tortoise	**Half Time**	The backbeat once a bar, for a big, slow feel
Walking figure	**Normal**	The usual backbeat
Hare	**Double Time**	Twice as driving

Editing several sections at once

Changes in the Drummer editor apply to the region you're editing. If you selected several drummer regions in the Tracks area first (⇧-click to add to the selection) and then opened one of them, each change applies to all the selected drummer regions. Only the setting you change is copied; everything else in each region stays as it was.

Clicking a chip in the song strip selects just that one section again.

Odd time signatures

The drummer plays in your song's meter, bar by bar, and plays again whenever the meter changes. Compound meters such as 6/8 are felt in dotted quarters, and uneven bars such as 7/8 are grouped 2 + 2 + 3.

The Session Drummer plug-in

Ongaku Studio also comes with a **Session Drummer** instrument plug-in that you can use in other music apps that support Audio Unit instruments. It has its own list of song sections, since other apps don't share their arrangement markers with plug-ins. See Chapter 8.

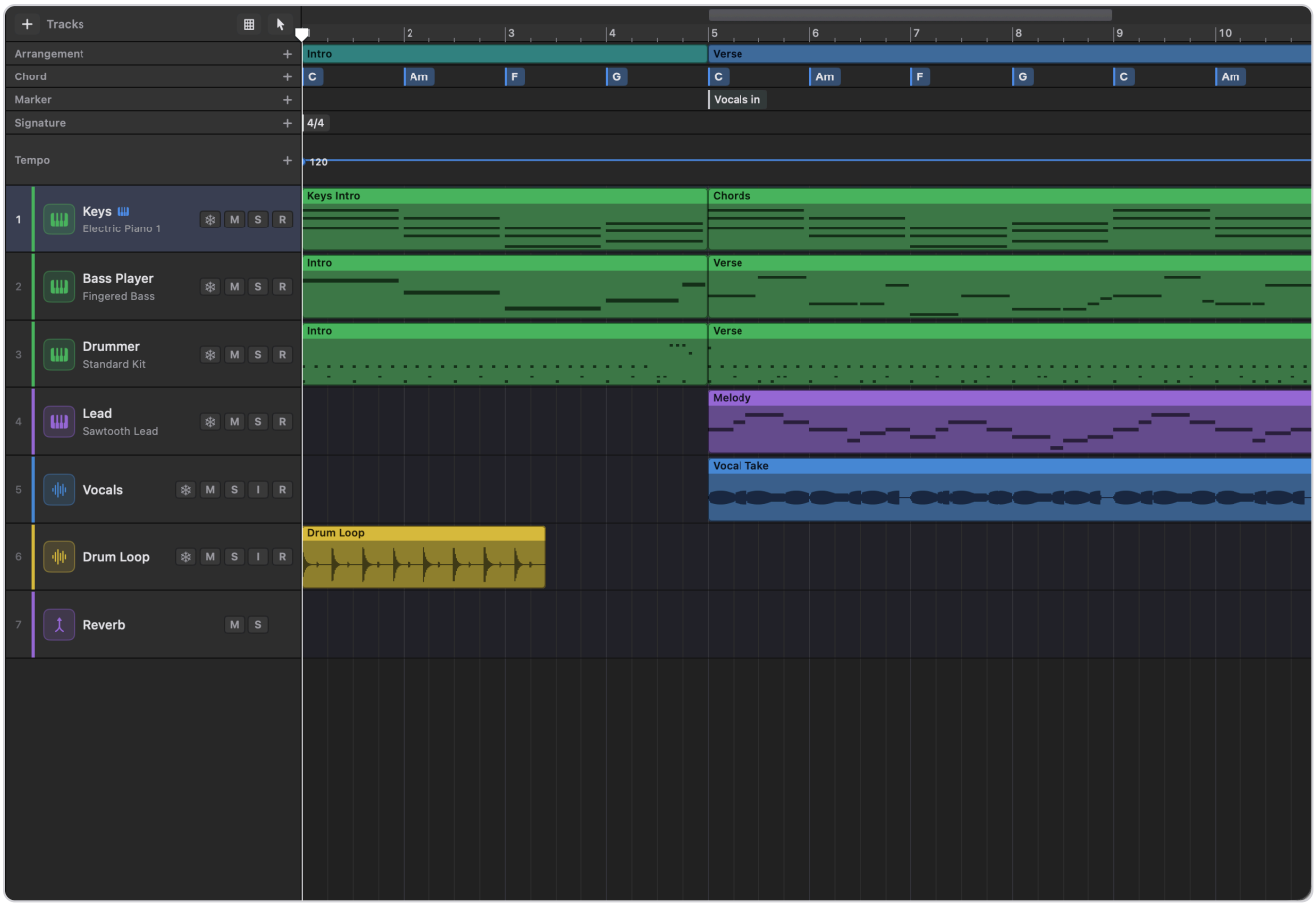
The Chord track

The Chord track is a lane of chord names that runs along the top of the song, with the other global tracks. It doesn't make any sound itself. It tells the Bass Player and Keyboard Player what to play. Whenever you add, change, move or delete a chord, every bass and keys player region is played again to match.

Use the Chord track when you know your song's chords, or when you want to try progressions quickly and hear a band play them.

Showing the Chord track

Choose **Track › Show Global Tracks**, or press **G**. The global tracks appear under the ruler: **Arrangement**, **Chord**, **Marker**, **Signature** and **Tempo**. The global tracks are also shown automatically whenever you add chords.



The Chord track, with chords under the Arrangement track

Adding a chord

There are several ways:

- **Double-click** an empty spot in the Chord lane.
- **Right-click** an empty spot and choose **Add Chord Here…**.
- Click the **+** at the Chord track's name to add a chord at the playhead.

Each of these opens a small **Add a Chord** box:

1. Type a chord name, for example `Am7`, `F#m` or `G/B`. While what you've typed isn't a chord yet, the box says **Not a chord name yet** and the **Add** button is greyed.
2. Press Return or click **Add**.

Or click one of the buttons under **In the key of…** to add one of your key's own chords with a single click.

To add a key chord without typing anything, right-click an empty spot and choose a chord from **Add a Chord of the Key**.

Chords land on the grid set with the **Snap** button above the track headers (Chapter 2), but never between sixteenth notes, because that's as finely as the players read the chords. With **Snap** set to **Off** or to a triplet grid, or while you hold ^ as you drag a chord, it lands on the nearest sixteenth note. If there's already a chord in that spot, the new one replaces it.

Chord names Ongaku Studio understands

A chord name is a root note, then the kind of chord, then (if you like) a slash and a different bass note.

Roots: A to G, followed by `#` or `♯` for sharp, or `b` or `♭` for flat. For example `C`, `F#`, `Bb`, `E♭`.

Kinds of chord:

Chord	You can type	Shown as
Major	nothing, `maj` or `M`	C
Minor	`m`, `min` or `−`	Cm
Dominant seventh	`7`	C7
Major seventh	`maj7`, `M7`, `ma7`, `Δ` or `Δ7`	Cmaj7
Minor seventh	`m7`, `min7` or `−7`	Cm7
Diminished	`dim`, `°` or `o`	Cdim
Half-diminished	`m7b5`, `m7♭5`, `ø` or `ø7`	Cm7♭5
Augmented	`aug` or `+`	Caug
Suspended second	`sus2`	Csus2
Suspended fourth	`sus4` or `sus`	Csus4
Sixth	`6`	C6
Minor sixth	`m6`	Cm6
Added ninth	`add9`	Cadd9
Ninth	`9`	C9
Minor ninth	`m9`	Cm9
Power chord (root and fifth)	`5`	C5

Slash chords: add `/` and a bass note, for example `C/E` (C major with E in the bass) or `Am7/G`.

Examples: `Am7`, `F♯m7♭5`, `C/E`, `Bbmaj7`, `Dsus4`, `E5`.

Ongaku Studio writes chord names its own way. Roots on black keys are shown as flats, except F♯ and C♯, the way most chord charts write them. So if you type `D#`, the chord shows as **E♭**.

If you type something it doesn't recognise, it shows a message like "“Xyz” isn't a chord name Ongaku Studio knows. Try something like C, Am7, F♯m or G/B."

Changing, moving and deleting chords

- **Change a chord:** double-click it, or right-click it and choose **Change Chord…**. Type the new name and click **Change**. To swap it for one of the key's chords, right-click it and choose from **Chords of the Key**.
- **Move a chord:** drag it left or right. It lands on the grid (see Adding a chord). If you drop it on another chord, it replaces that chord.

- **Select a chord and go to it:** click it, without dragging. The chord is highlighted, and the playhead moves there.
- **Delete a chord:** click it, then press **Delete** (or choose **Edit › Delete Selection**). Or right-click it and choose **Delete**.

A chord stays selected, and Delete keeps acting on it, until you click somewhere else. Clicking in the tracks area or an editor moves Delete back to regions or notes, as usual (Chapter 10).

A chord lasts until the next chord on the track.

Filling the track with a progression

You don't have to enter chords one at a time. Ongaku Studio has a library of common progressions, written so they play in any key. Each chord in a progression lasts one bar.

1. Right-click an empty spot in the Chord lane.
2. Choose **Fill *Section* with a Progression**. *Section* is the name of the arrangement section you clicked in, for example **Fill Chorus with a Progression**. If you didn't click inside a section, it says **Fill 8 Bars with a Progression** and fills eight bars from that bar.
3. Pick a progression. Each one is listed with its chords in your key, for example **Pop: C G Am F**.

The progression repeats to fill the space, and replaces any chords that were already there.

Progressions for major keys:

Name	Chords (in Roman numerals)	In C major
Pop	I V vi IV	C G Am F
Classic	I IV V I	C F G C
Fifties	I vi IV V	C Am F G
Sensitive	vi IV I V	Am F C G
Lift	IV I V vi	F C G Am
Build	ii IV V V	Dm F G G
Folk	I IV I V	C F C G
Turnaround	I vi ii V	C Am Dm G
Rock	I ♭VII IV I	C B♭ F C
Jazz	ii7 V7 Imaj7 Imaj7	Dm7 G7 Cmaj7 Cmaj7
Soul	Imaj7 iii7 IVmaj7 V7	Cmaj7 Em7 Fmaj7 G7
Canon	I V vi iii IV I IV V	C G Am Em F C F G
Blues	12-bar blues with sevenths	C7 ×4, F7 ×2, C7 ×2, G7 F7 C7 G7

Progressions for minor keys:

Name	Chords (in Roman numerals)	In A minor
Epic	i ♭VI ♭III ♭VII	Am F C G

Name	Chords (in Roman numerals)	In A minor
Andalusian	i ♭VII ♭VI V	Am G F E
Lift	♭VI ♭VII i i	F G Am Am
Build	iv ♭VI ♭VII ♭VII	Dm F G G
Classic	i iv V7 i	Am Dm E7 Am
Rock	i ♭VII ♭VI ♭VII	Am G F G
Ballad	i ♭III ♭VII iv	Am C G Dm
Jazz	iiø7 V7 i7 i7	Bm7♭5 E7 Am7 Am7
Blues	12-bar minor blues with sevenths	Am7 ×4, Dm7 ×2, Am7 ×2, F7 E7 Am7 E7

Which list you see depends on whether the key is major or minor where the progression goes, and its chords are in that key. Set the key by clicking **KEY** in the control bar, or add a key change on the Signature track (Chapter 2): after a key change, the suggestions, progressions and the players' fallback chord are all in the new key.

Writing chords for the whole song

If you just want to hear something, right-click the Chord lane and choose **Write Chords for the Whole Song**. Ongaku Studio clears the Chord track and writes a progression for each arrangement section. Verses get one progression, pre-choruses another, choruses and outros another, and bridges another, so the song hangs together. The last section ends on the key's home chord. Each time you choose it, you get a different combination.

If your song has no arrangement markers, the whole song is treated as one section.

Detecting chords from a MIDI region

If you've already played chords into a MIDI region, for example a piano part, Ongaku Studio can work out what they are and put them on the Chord track.

1. Right-click the MIDI region in the Tracks area.
2. Choose **Detect Chords**.

Or select one or more MIDI regions and choose **Track › Detect Chords**. Every selected region that can be read is used, and the whole detection is one undo step.

It listens every half bar, and gives longer and harder notes more weight. When the same chord carries on, it's written once. The detected chords replace any chords that were on the Chord track over the length of each region.

Things to know:

- **Detect Chords** works on ordinary MIDI regions only. It's greyed out for drum tracks, and for drummer, player and Step Sequencer pattern regions. **Track › Detect Chords** skips those and is greyed out when no selected region can be read.

- It recognises major, minor, seventh, major seventh, minor seventh, diminished and suspended fourth chords. It adds a slash bass when the lowest note is a chord tone other than the root. Richer chords (ninths, sixths and so on) come out as the nearest simpler chord, which you can then change by hand.
- If there are too few notes to tell, you'll see a message such as "There aren't enough notes in “Piano” to tell its chords."

When there are no chords

The players always need a chord to play:

- **If the Chord track is empty**, they play your key's home chord all the way through: C major in C major, A minor in A minor, and so on.
- **Before the first chord** on the track, they play that first chord.

> Note: When you create a Bass Player or Keyboard Player track while the Chord track is empty, Ongaku Studio writes chords for the whole song for you, the same as **Write Chords for the Whole Song**, and shows the Chord track. A player stuck on one chord isn't much use. Undo once to remove the new track and the new chords together.

The Bass Player and Keyboard Player

The Bass Player plays a bass line and the Keyboard Player plays chords, both following the Chord track. Use them together with the Drummer to hear a whole band playing your progression, or on their own to give you a part to sing or play along with.

Adding a Bass or Keyboard Player track

1. (Optional) Add arrangement markers and chords first.
2. Choose **Track › New Bass Player Track** or **Track › New Keyboard Player Track**.
3. A new track appears below the selected track, called **Bass Player** or **Keyboard Player** (numbered if that name is taken). The Bass Player starts with a fingered electric bass sound and the **Root Notes** style. The Keyboard Player starts with an electric piano sound and the **Pad** style.
4. The track gets one region per arrangement section (or one region for the whole song if there are none), and the Player editor opens on the first one.

You can change the track's sound to any instrument you like; the notes stay the same. See Chapter 8.

Adding more player regions

On any instrument track that doesn't play drums:

1. Right-click an empty spot in the lane.
2. Choose **New Player Region**.

As with the drummer, the region fills the arrangement section you clicked in, or eight bars from that bar. It continues in the style the track's other player regions use. On a track with no player regions yet, it plays bass (**Root Notes**) if the track has a General MIDI bass sound, and keys (**Pad**) otherwise.

Bass styles

Style	Card says	How it plays
Root Notes	Space and roots	Long roots with room to breathe, and a pickup into each phrase
Driving Eighths	Steady eighths	Eighths that drive the song, with octave pops and walk-ups
Walking	Quarter-note line	A line through the chords a beat at a time, leaning into each one
Funk	Syncopated	The one, then syncopated sixteenths, octave pops and dead notes
Motown	Melodic pickups	Melodic sixths and chromatic climbs, pushing ahead of the beat

Keyboard styles

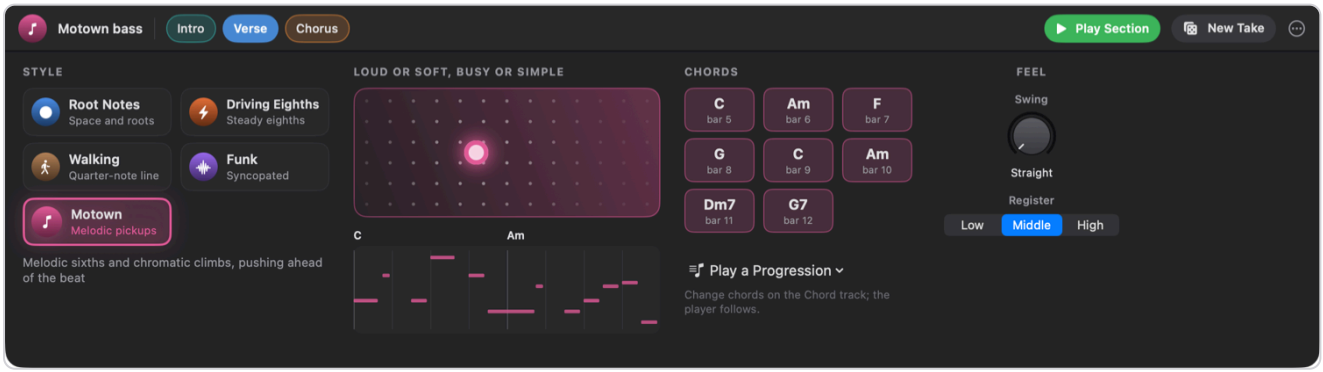
Style	Card says	How it plays
Pad	Held chords	Held chords with ninths, re-voiced so the top line moves
Block Chords	Every beat	Piano chords on the beat over octaves, pushing into changes
Pop Pulse	Eighth-note chords	Driving eighths and 3-3-2 pushes over a left-hand root
Ballad Arpeggio	Broken chords	A rolling left hand under held and broken chords
Funk Stabs	Short stabs	Syncopated ninth and minor-seventh stabs and chanks

A bass region only offers bass styles, and a keys region only keys styles.

The players think in four-bar phrases. They play the groove, a variation on the second bar of a phrase (once the part is busy enough), and a fill or turnaround on the fourth bar and on the region's last bar. When a chord change is coming, the bass often leans into it with a passing note.

The Player editor

Double-click a player region to open the Player editor.



The Player editor for a Bass Player region

The **song strip** across the top works like the Drummer's. It shows the style (for example "Root Notes bass"), a chip for each player region on the track (click one to edit it), **Play Section**, **New Take**, and a **…** menu with:

- **Lead into the Next Section**: end the region with an approach note or pickup into what comes next. Turn it off on the last section to play an ending instead.
- **Convert to MIDI Region**.

While the Chord track is empty, the strip shows a tip saying so: the player plays the key's chord until you add chords on the Chord track or play a progression.

Below the strip, from left to right:

Area	What it does
Style	Click a card to choose the style. Changing style resets Swing to that style's usual amount (Walking starts swung; the others start straight).
Loud or Soft, Busy or Simple	Drag the dot: right for a busier part, up for a louder one. Let go to hear the new part. With VoiceOver, use the **Louder**, **Softer**, **Busier** and **Simpler** actions.
The note chart	The section's first two bars as a small piano roll, with the chord names above. It follows the dot while you drag.
Chords	The chords this section follows, each with its bar number. Click one to move the playhead there.
Play a Progression	A menu of progressions in your key. Choosing one puts it on the Chord track over this section, replacing the chords there. **Write Chords for the Whole Song** is here too.
Swing	Drag the knob from **Straight** through **A little** and **Swung** to **Shuffle**. Double-click or ⌥-click to reset it. Funk bass and Funk Stabs swing sixteenths; the others swing eighths.
Register	**Low**, **Middle** or **High**: play an octave lower or higher than usual.

Under the chord tiles, a line of text tells you what the player is following:

- "Change chords on the Chord track; the player follows." when the section has chords.
- "No chords yet: playing C, the key's chord. Choose a progression." (with your key's chord) when the Chord track is empty.
- "No chords here yet: playing Am, the Chord track's first chord. Choose a progression." (with that chord) when the Chord track has chords, but none before the end of this section.

As with the Drummer, if you select several player regions of the same kind (all bass or all keys) before opening one, your changes apply to all of them.

> Tip: To change what the player plays harmonically, change the chords, not the player. Edit the Chord track and the part follows straight away, even while the song is playing.

Converting a part to MIDI

When you want to change single notes, for example to move one kick drum or fix one bass note, turn the region into an ordinary MIDI region:

- In the Drummer or Player editor, open the **…** menu and choose **Convert to MIDI Region**, or
- Right-click the region in the Tracks area and choose **Convert to MIDI Region**.

The region keeps exactly the notes it was playing, and the editor switches to the Piano Roll. See Chapter 4.

After converting, the region no longer follows the song. It won't fill a new length, follow chord changes or respond to the player's controls. Converting can only be reversed with **Edit › Undo**. If you selected several drummer or player regions, they're all converted.

Limitations

- **Joining differently set drummer or player regions converts them to MIDI.** Splitting a drummer or player region keeps both halves "live", and joining regions with the same settings (such as two halves you split) gives one live region again. But **Join Regions** (⌘J) on regions with different settings (a verse and a chorus, say) can't keep both, so the result is an ordinary MIDI region with the notes they were playing, which you can no longer edit in the Drummer or Player editor. The Edit menu calls this step **Join and Convert to MIDI**. See Chapter 2.
- **Renaming an arrangement section doesn't restyle its region.** The section-name presets are only used when a region is created. A region keeps its settings if you rename or move the section afterwards. On a drummer track, **Match Song Sections** fits the regions to the sections again, keeping each section's settings, and only new sections get the presets.
- **The drum kit isn't chosen in the Drummer editor.** The drummer plays the track's instrument. Its notes are standard General MIDI drum notes, and if the track uses the Ongaku Drum Machine it plays to suit that machine's pads.
- **Kit pieces are switched in groups**, not one drum at a time (for example, all cymbals together).
- **Chords can only be selected and deleted one at a time.**
- **Chords can't go between sixteenth notes**, even with Snap off.
- **Match Song Sections is for Drummer tracks only.** Bass and keys tracks have no button to line their regions up with the sections.

Quick reference

Action	How
New Drummer track	**Track › New Drummer Track** (⌥⌘U)
New Drummer track on the Drum Machine	**Track › New Drummer Track (Drum Machine)**
New Bass Player track	**Track › New Bass Player Track**
New Keyboard Player track	**Track › New Keyboard Player Track**

Action	How
Add arrangement markers (sections)	**Track › Add Arrangement Marker**
Add a drummer or player region	Right-click a lane › **New Drummer Region** or **New Player Region**
Open the Drummer or Player editor	Double-click the region (or select it and press **E**)
Steer loudness and busyness	Drag the dot on the **Loud or Soft, Busy or Simple** pad
Hear just this section, looped	**Play Section**
Another version, same settings	**New Take**
Line the drummer up with sections	**Match Song Sections** (keeps each section's settings)
Reset a knob	Double-click or ⌥-click it
Show the Chord track	**Track › Show Global Tracks** (**G**)
Add a chord	Double-click the Chord lane, or click **+** for the playhead
Change a chord	Double-click it
Move a chord	Drag it (it snaps to the grid, no finer than sixteenths)
Select a chord and go to it	Click it
Delete a chord	Click it and press Delete, or right-click it › **Delete**
Fill a section with a progression	Right-click the Chord lane › **Fill *Section* with a Progression**
Write chords for the whole song	Right-click the Chord lane › **Write Chords for the Whole Song**
Detect chords in a MIDI region	Right-click the region › **Detect Chords**, or select regions › **Track › Detect Chords**
Convert to ordinary MIDI	**…** › **Convert to MIDI Region**, or right-click the region › **Convert to MIDI Region**
Undo any change	**Edit › Undo** (⌘Z)

6. Flex Time, Flex Pitch and Smart Tempo

Audio recordings are harder to fix than MIDI. A drum hit or a sung note is baked into the sound wave, so you can't just drag it. The tools in this chapter let you do that anyway: **Flex Time** moves the timing of things inside a recording, **Flex Pitch** retunes single sung or played notes, and **Smart Tempo** finds a recording's tempo so the song and the recording can agree.

How flexing works

All three tools leave your original recording untouched. Ongaku Studio remembers what you asked for (move this hit here, raise this note a semitone, play this loop at 120 BPM), then makes a new, processed copy of the region's audio in the background. Playback switches to the processed copy once it's ready.

From your point of view this means:

- **Your edits are safe and reversible.** Every flex edit can be undone with **Edit › Undo** (⌘Z), and you can remove flex markers, pitch changes or stretching at any time to get the recording back exactly as it was.
- **There can be a short wait.** After an edit, the region keeps playing as it was recorded until the processed copy is ready. While it's being made, the region shows a small spinner and **Processing…** at its bottom left; when the label goes, you're hearing the new version. For a short region that's usually almost instant. For a long one, give it a moment before you judge the result. If something goes wrong, you'll see a message such as "“Vocal” couldn’t be flexed: …".
- **Processed copies aren't saved in your project.** They're made again when you open the project, so flexed regions may show **Processing…** and play unflexed for a moment right after opening.
- **Bouncing and freezing wait for them.** If you bounce the song or freeze a track while regions are still processing, Ongaku finishes processing them first, so what you get includes every flex edit. The Bounce dialog says **Finishing processing flexed audio…** while it waits. See Chapter 9 for bouncing and Chapter 7 for freezing.

Flex Time

Flex Time lets you move a moment inside an audio recording, such as a snare hit that's a little late or a word that's sung early, without cutting the region up. You grab the moment and drag it to where it should be, and the audio around it is stretched or squeezed to fit.

Use it to tighten a drum or bass recording, line up a double-tracked part, or fix one late note in an otherwise great take.

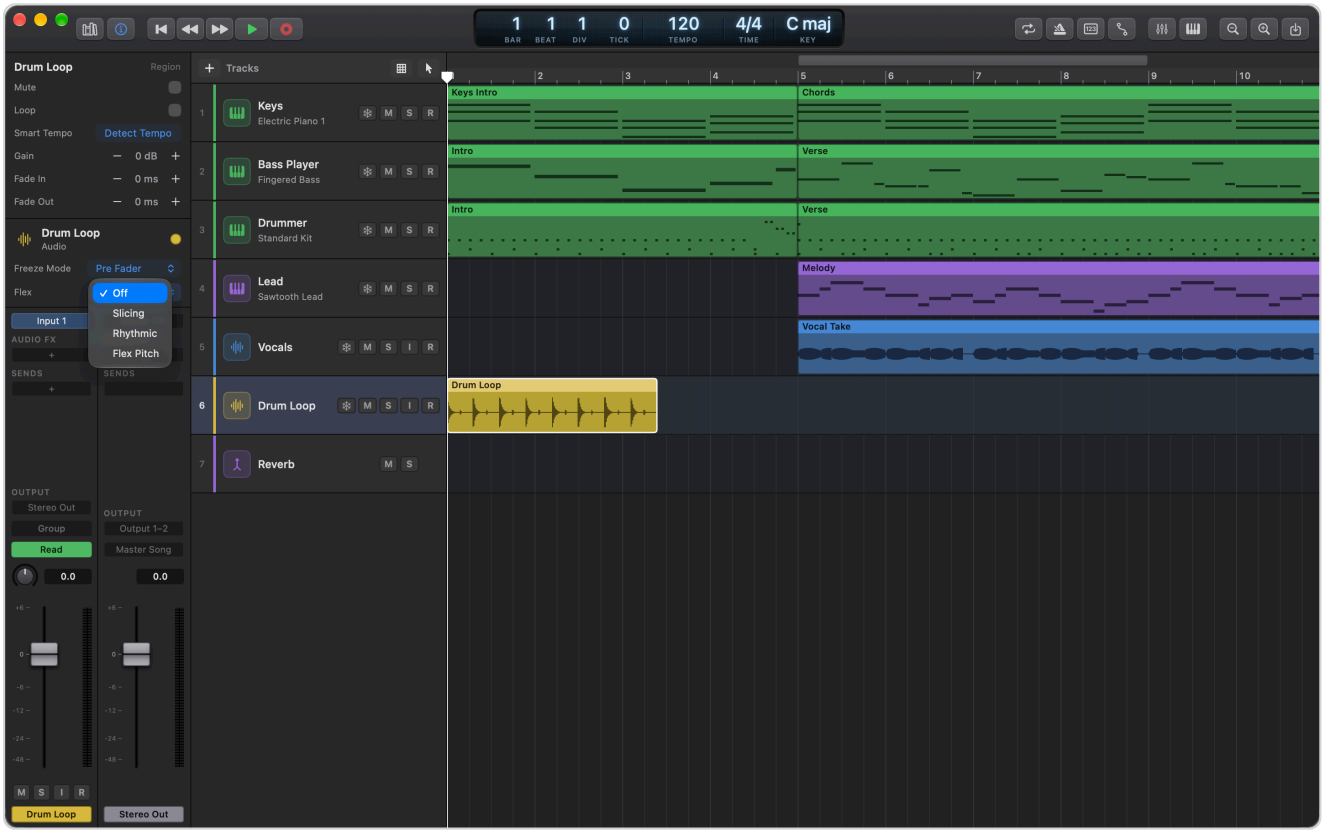
Turning on Flex for a track

Flex is set per audio track, in the Inspector.

1. Select the audio track.
2. Show the Inspector if it's hidden: press **I**, or choose **View › Show Inspector**.

3. Under the track's name, find **Flex** and choose a mode.

Mode	How it moves the audio	Best for
Off	Flex is off. Regions play as recorded.	Everything you don't want to flex
Slicing	Cuts the audio at each flex marker and moves the slices without stretching them. Hits stay crisp, but a slice moved later leaves a short silence before it, and a slice moved earlier cuts off the end of the slice before it.	Drums and percussion
Rhythmic	Stretches or squeezes the audio between markers, keeping its pitch.	Rhythm guitar, bass, keys, loops, anything with sustained or overlapping notes
Flex Pitch	Lets you retune single notes (see Flex Pitch). Flex markers stretch the audio as in Rhythmic.	A voice or a solo instrument playing one note at a time



The Inspector with the Flex mode menu open

Once Flex is on, the track's audio regions show their flex markers, and you can make new ones.

Flex is only available on audio tracks. Software instrument tracks don't need it, because you can move MIDI notes directly (see Chapter 4).

Flex markers: the idea

A flex marker pins a moment of the recording to a place on the timeline. Think of the region as a strip of elastic that's pinned at its start and its end. Each flex marker is another pin. When you drag a pin, the elastic either side of it stretches or squeezes, but everything stays in order, and the region's start and

end stay where they are.

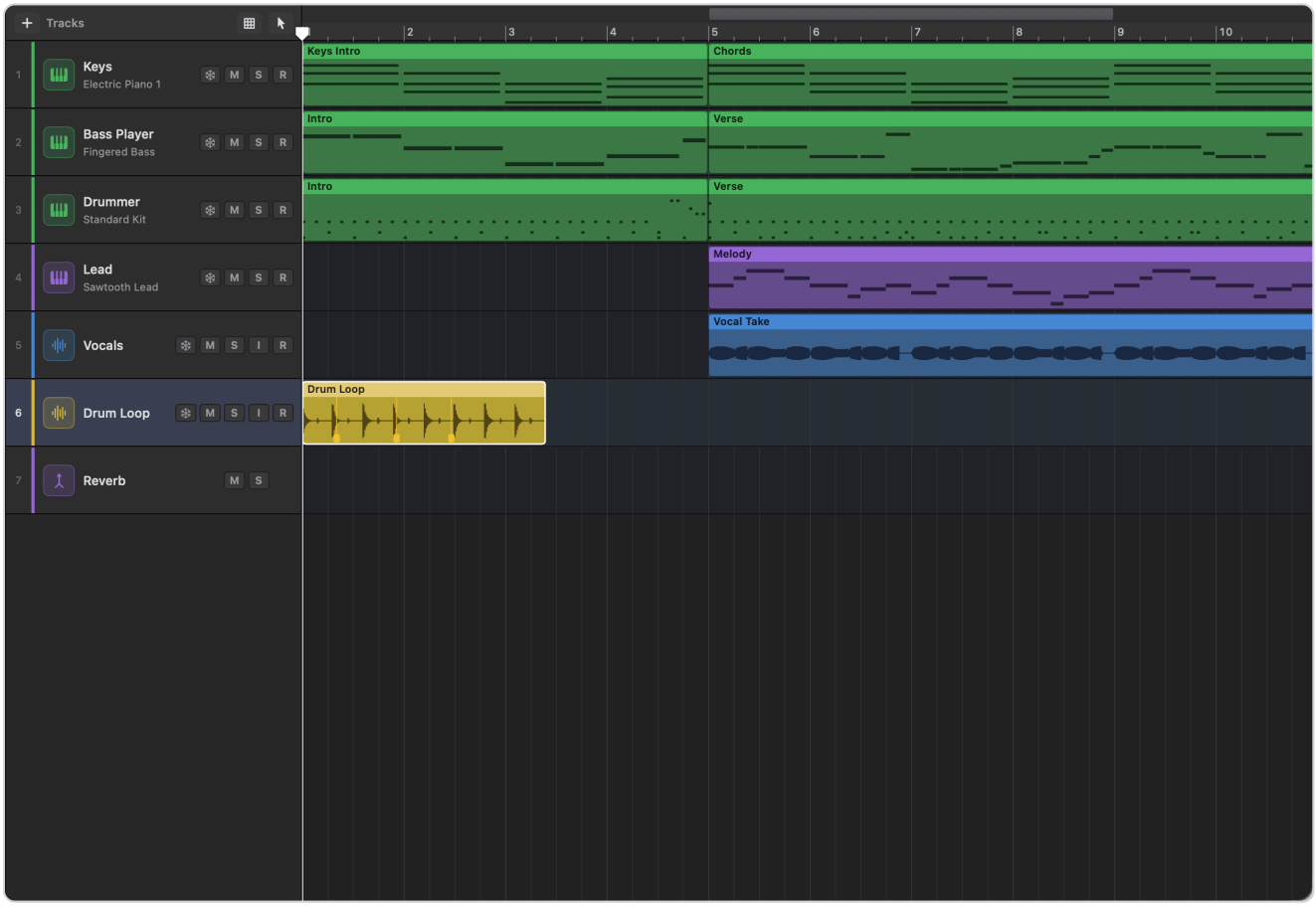
Flex markers show as thin coloured lines through the region, each with a small handle at the bottom.

Moving audio in time

1. Make sure the track has a Flex mode, and that the **Pointer** or **Pencil** tool is selected.
2. Find the moment you want to move, such as a hit you can see in the waveform.
3. Press in the **lower half** of the region at that moment, and drag left or right. (Dragging in the upper half moves the whole region, as usual.)
4. Let go where it should play. It lands on the nearest grid line, according to your Snap setting (see Chapter 2).

If you pressed on an existing flex marker, you move that marker. If not, a new marker is made at the point you pressed, and you drag it.

While you drag, the region's drawing updates to show the result. When you let go, the new version is processed and plays as soon as it's ready.



Flex markers (the small handles at the bottom) in the Drum Loop region, with hits moved in time

> Tip: The first marker you drag stretches everything between the region's start and that marker, and between the marker and the region's end. To move one hit without nudging its neighbours, first drag the hits on either side onto their grid lines. That pins them, so only the audio between them changes.

Deleting flex markers

- **One marker:** double-click it, in the lower half of the region. This only works while the track has a Flex mode, since that's when the markers are shown.
- **All markers in a region:** select the region, and in the Region Inspector click **Remove Flex Markers** (on the **Flex Markers** row). It removes the markers of every selected region at once, as one undo step.

The **Remove Flex Markers** button is there whenever a selected region has flex markers, even while the track's Flex is **Off** (see Turning Flex off).

Quantizing audio

Quantizing audio does for a recording what quantizing does for MIDI. It finds the hits in the region and moves each one to the nearest grid line, using flex markers.

1. Make sure the track has a Flex mode.
2. Select the audio region.
3. In the Region Inspector, next to **Flex**, open **Quantize Audio** and choose a grid: **1/4 Note**, **1/8 Note**, **1/16 Note**, **1/32 Note**, or one of the triplet grids (**1/4 Triplet** to **1/32 Triplet**).

Ongaku Studio looks for the hits first, which may take a moment, then places a marker on each one and moves it onto the grid. Any markers already in the region are replaced. A hit that's too close to the region's edges, or that would have to cross its neighbour, is left where it is.

If it can't find any hits, you'll see "No hits were found in “…” to quantize."

> Tip: Choose the grid that matches the part. If you pick 1/4 Note for a part full of sixteenths, the in-between notes will be pulled onto beats, and it'll sound wrong. Undo and try a finer grid.

Turning Flex off

Set the track's **Flex** menu back to **Off**. The track's regions play as recorded again, but their flex markers are kept (hidden). Turn the same mode back on and they come back.

Hidden markers still stop you resizing, splitting or joining the region. To get rid of them without turning Flex on again, select the region and click **Remove Flex Markers** in the Region Inspector.

Flex Pitch

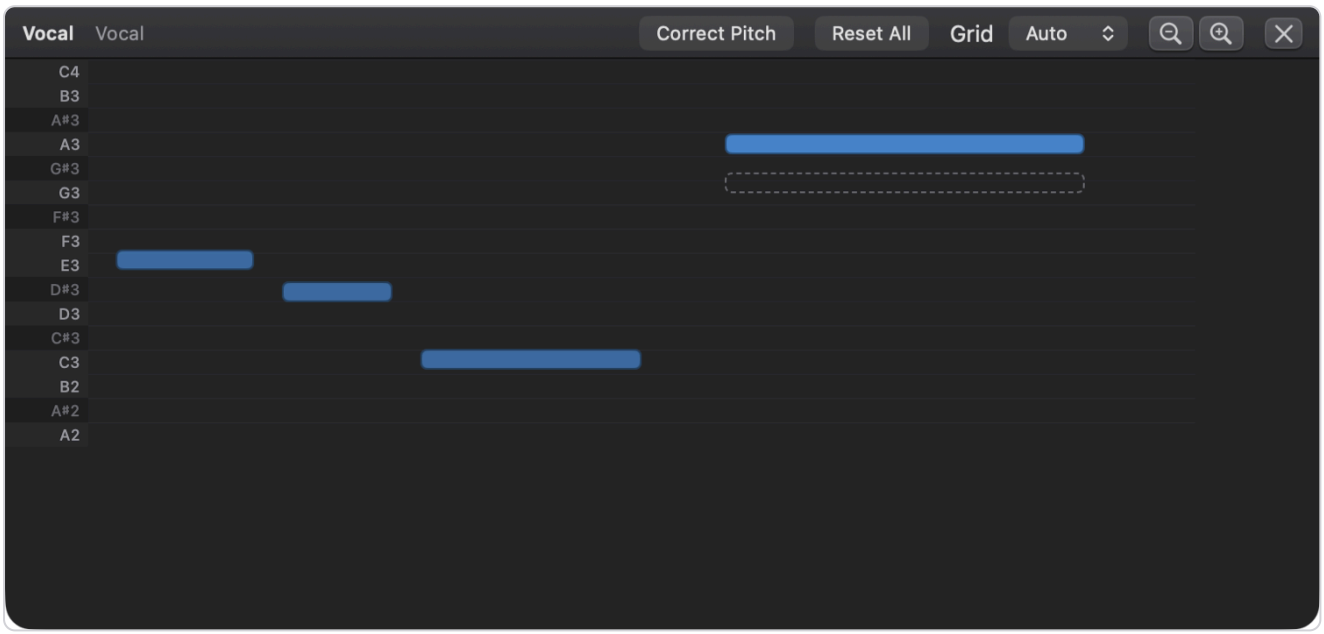
Flex Pitch finds the notes in a recording of one voice or instrument and shows them like notes in a piano roll. You can drag a note up or down to retune it, or snap every note to the nearest in-tune pitch in one go.

Use it to fix a sharp or flat note in a vocal, or to try a different melody note without singing it again.

Opening the Flex Pitch editor

1. Select the audio track and set its **Flex** mode to **Flex Pitch** in the Inspector.
2. Double-click an audio region on that track, or select it and press **E**.

The editor pane opens with the **Flex Pitch** editor. It shows **Finding notes…** while Ongaku Studio listens to the recording, which happens once for each region.



The Flex Pitch editor with one note dragged up

Each note found is drawn as a bar, at the pitch it was sung, against note names down the left side. A note sung in tune sits in the middle of its row; one that's a little sharp or flat sits a little above or below.

If no notes are found, you'll see **No Notes Found**. Flex Pitch works on monophonic audio: one voice or instrument playing one note at a time. Chords, full mixes and drums won't work.

Retuning a note

- **Drag a note up or down.** It moves in steps and lands exactly on a semitone, so dragging a slightly flat note into the row above or below also puts it in tune. While a note is moved, a dashed outline shows where it was sung.
- **Double-click a note** to put it back where it was sung.
- **Hover over a note** to see its name and where it's been moved to, for example "E3 → F3".

You can move a note by up to two octaves either way. The timing stays the same; only the pitch changes.

Correcting every note

Click **Correct Pitch** at the top of the Flex Pitch editor. Every note that's more than a tiny amount out of tune is moved to the nearest semitone.

If you click it while the notes are still being found, you'll see "Ongaku Studio is still finding the notes. Try again in a moment."

> Gotcha: **Correct Pitch** replaces any notes you've already retuned by hand. It moves each note to its nearest semitone, measured from where it was sung. So if you'd dragged a note up a whole tone, it goes back near its original pitch. Use Correct Pitch first, then make your own changes.

Correct Pitch moves notes to the nearest semitone of the full chromatic scale. It doesn't know your song's key, so a badly out-of-tune note can land on a semitone that's in tune but not in the key. Drag that note by hand.

Resetting pitch

Click **Reset All** at the top of the Flex Pitch editor to put every note back where it was sung. To reset just one note, double-click it.

You can also reset without opening the editor: select one or more regions and click **Reset Retuned Notes** in the Region Inspector (on the **Flex Pitch** row). The button is there whenever a selected region has retuned notes, whatever the track's Flex mode, and resets every selected region as one undo step.

Flex markers on a Flex Pitch track

You can use flex markers on a Flex Pitch track too, the same way as in Moving audio in time. They stretch the audio as in Rhythmic mode.

> Note: Retuned notes only play while the track's mode is **Flex Pitch**. If you change the track to Slicing, Rhythmic or Off, the region plays at its sung pitch again, but your changes are kept and come back when you return to Flex Pitch.

Smart Tempo

Smart Tempo listens to an audio recording and works out its tempo, in beats per minute. Then it can do one of two things:

- **Adapt Project Tempo**: change your song's tempo to match the recording. Use this when the recording came first. For example, you recorded a guitar idea without a click and now want to build a song around it.
- **Match Project Tempo**: stretch the recording to fit your song's tempo, without changing its pitch. Use this when the song came first. For example, you've imported a drum loop at 95 BPM into a 110 BPM song.

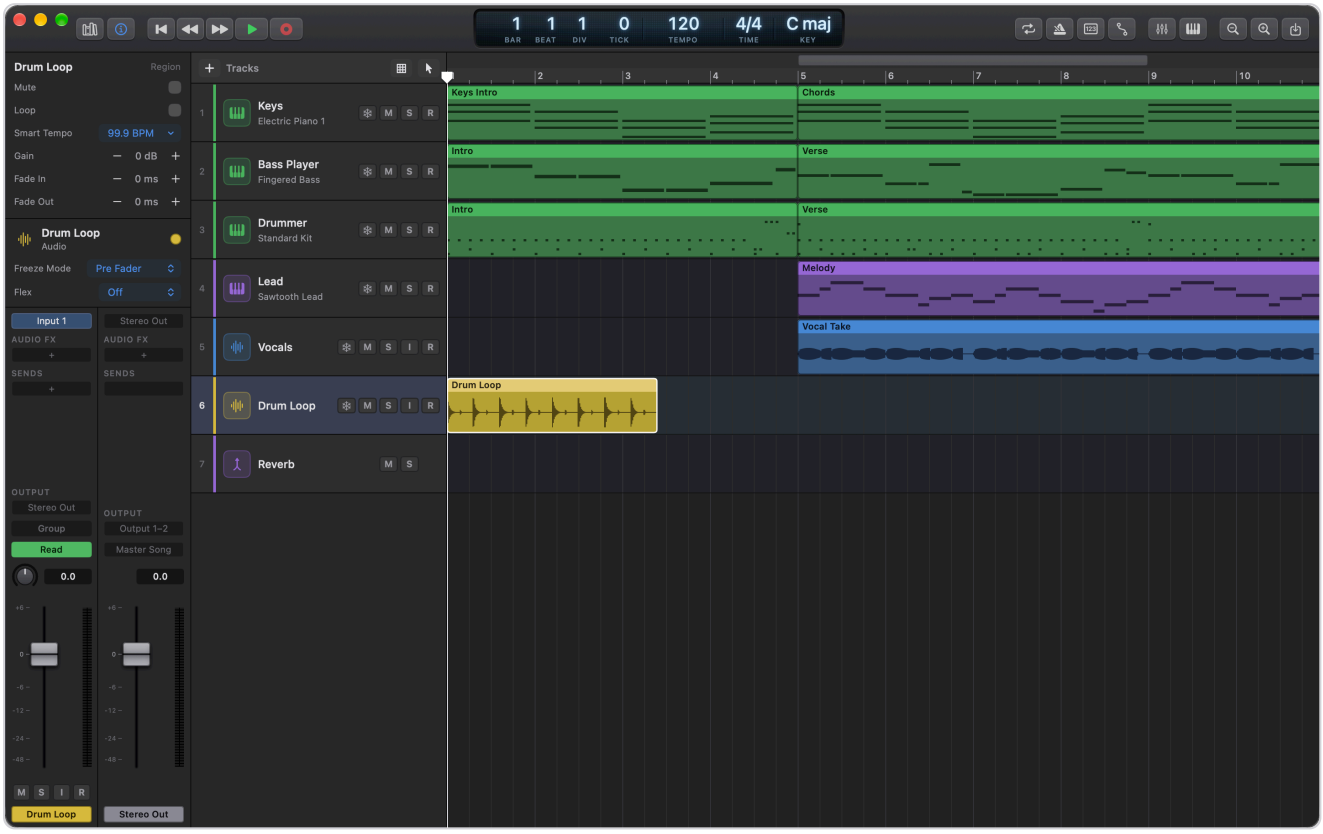
Either way, Ongaku Studio also moves the region slightly so its first beat lands on a beat line.

Smart Tempo works on any audio region. The track doesn't need a Flex mode.

Detecting a recording's tempo

1. Select the audio region.

2. In the Region Inspector, next to **Smart Tempo**, click **Detect Tempo**.
3. A small spinner shows while it listens. When it's done, the button shows the tempo, for example **96.0 BPM**.



The Region Inspector showing the Drum Loop's detected tempo, 99.9 BPM; click it for the Smart Tempo menu

If the recording doesn't have a clear, steady beat, you'll see "“…” doesn’t have a steady enough beat to find its tempo." Recordings with strong, regular hits (drums, strummed guitar, piano rhythm parts) work best. Free-time playing, pads and spoken word won't.

Smart Tempo looks for tempos between 60 and 200 BPM and leans toward tempos around 120. A song you think of as 70 BPM may therefore be detected as 140, or the other way round. Both describe the same beat. If the number is twice or half what you expected, the result is usually still usable; when you choose **Adapt Project Tempo**, your bars will just be twice or half as long as you might expect.

Making the song follow the recording

1. Detect the tempo (see above).
2. Click the tempo (for example **96.0 BPM**) and choose **Adapt Project Tempo**.

The song's tempo changes to the recording's. If your song has tempo changes, the one that's in effect where the region starts is the one that changes. If there's only one tempo, that's the tempo for the whole song. The region is moved slightly so its first beat lands on a beat, and it still plays all of its audio.

> Gotcha: Changing the song's tempo affects everything in it. MIDI regions and the session players follow the new tempo; other audio regions don't speed up or slow down with it. Use Adapt Project Tempo early, before you've recorded much else.

Making the recording follow the song

1. Detect the tempo.
2. Click the tempo and choose **Match Project Tempo**.

The region is stretched (or squeezed) so it plays at the song's tempo where it starts, keeping its pitch, and it's moved so its first beat lands on a beat. The region gets longer or shorter to match. The stretched version is processed in the background, as described in How flexing works.

Match Project Tempo can't be used on a region that has flex markers. You'll see "Remove the region’s flex markers first." It also doesn't work on take folders; flatten the take folder first (see Chapter 3).

Removing the stretch

1. Select the stretched region.
2. Next to **Smart Tempo** in the Region Inspector, click **Remove Stretch**. (If the tempo is showing instead, click it and choose **Remove Stretch** from its menu.)

The region plays its recording at its original speed again, starting at the same place.

Remove Stretch appears beside **Detect Tempo** whenever the region is stretched, so you don't have to detect the tempo again first, for example after reopening the project. The tempo menu also has **Detect Again**, which listens to the recording again.

Limitations

- **Flexed, retuned and stretched regions can't be resized, split, joined or crossfaded.** If you try, the message names what's in the way and the button that removes it, for example "To resize, split, join or crossfade “Vocal”, first remove its flex markers (Remove Flex Markers): select it and use the button in the Region Inspector." Do your trimming and splitting first, then flex. Or remove the markers (**Remove Flex Markers**), the pitch changes (**Reset Retuned Notes**) or the stretch (**Remove Stretch**), make your edit, and flex again.
- **Turning Flex off doesn't remove markers.** A region on a track with Flex set to **Off** can still have hidden flex markers or retuned notes, and those still stop you resizing or splitting it. The Region Inspector's **Remove Flex Markers** and **Reset Retuned Notes** buttons appear for such regions, so you can remove them without turning Flex back on.
- **No formant preservation.** Flex Pitch changes pitch by speeding up or slowing down the sound itself. Small corrections sound natural, but moving a voice several semitones also changes its character: higher sounds thinner and smaller, lower sounds darker and bigger.
- **Flex Pitch is for one note at a time.** It finds pitches between about 60 Hz and 1.2 kHz (roughly the range of a bass voice up to a high soprano). Very short sounds, under about 60 milliseconds, are skipped, so consonants, quick slides and grace notes aren't shown as notes.
- **No hit markers to start from.** Ongaku Studio doesn't show where the hits are before you flex; you place markers by eye from the waveform. Only **Quantize Audio** finds hits for you.
- **Quantize Audio always moves hits all the way** to the grid. There's no strength setting.
- **Big stretches soften hits.** In Rhythmic and Flex Pitch modes, stretching a passage a long way can slightly smear sharp attacks. Use Slicing for drums if that happens.

- **Smart Tempo needs a few seconds of audio** with a steady beat, and finds one tempo per region. It doesn't follow a recording that speeds up or slows down.
- **Smart Tempo works on one region at a time**: the first selected region, shown in the Region Inspector.
- **Detected tempos aren't saved with the project.** After you reopen a project, click **Detect Tempo** again to see the tempo. (**Remove Stretch** is shown either way.)
- **There's no progress bar for processing. Processing…** shows that a region is being processed, but not how far it has got.

Quick reference

Action	How
Show the Inspector	**I**, or **View › Show Inspector**
Turn on Flex for an audio track	Inspector › **Flex** › **Slicing**, **Rhythmic** or **Flex Pitch**
Turn Flex off	Inspector › **Flex** › **Off** (markers are kept)
Move audio in time	Drag in the lower half of the region (Pointer or Pencil tool)
Move a flex marker	Drag its handle in the lower half
Delete a flex marker	Double-click it in the lower half (with Flex on)
Quantize audio to the grid	Region Inspector › **Quantize Audio** › choose a grid
Remove all flex markers	Region Inspector › **Remove Flex Markers**
Open the Flex Pitch editor	Set Flex to **Flex Pitch**, then double-click the region (or select it and press **E**)
Retune a note	Drag it up or down in the Flex Pitch editor
Reset one note	Double-click it
Correct every note	**Correct Pitch**
Reset every note	**Reset All**, or Region Inspector › **Reset Retuned Notes**
Find a recording's tempo	Region Inspector › **Smart Tempo** › **Detect Tempo**
Song follows the recording	Tempo menu › **Adapt Project Tempo**
Recording follows the song	Tempo menu › **Match Project Tempo**
Remove Smart Tempo stretch	Region Inspector › **Remove Stretch** (or the tempo menu)
Undo any of these	**Edit › Undo** (⌘Z)

7. Mixing and Automation

Mixing is where your tracks become one song: you balance their levels, place them left and right, add effects, and send them to shared reverbs and buses. This chapter walks through the Mixer and its channel strips, plug-ins, sends and outputs, mixer groups, freezing tracks to save processing power, and automation, which lets any of those settings change as the song plays.

The Mixer and the Inspector

Every track in your song has a **channel strip**: a column of controls that decides how that track sounds in the mix. Think of it as one channel on a mixing desk. Sound enters at the top (from a recording, an instrument or other tracks), passes down through effects, and comes out of the fader at the bottom.

You can see channel strips in two places:

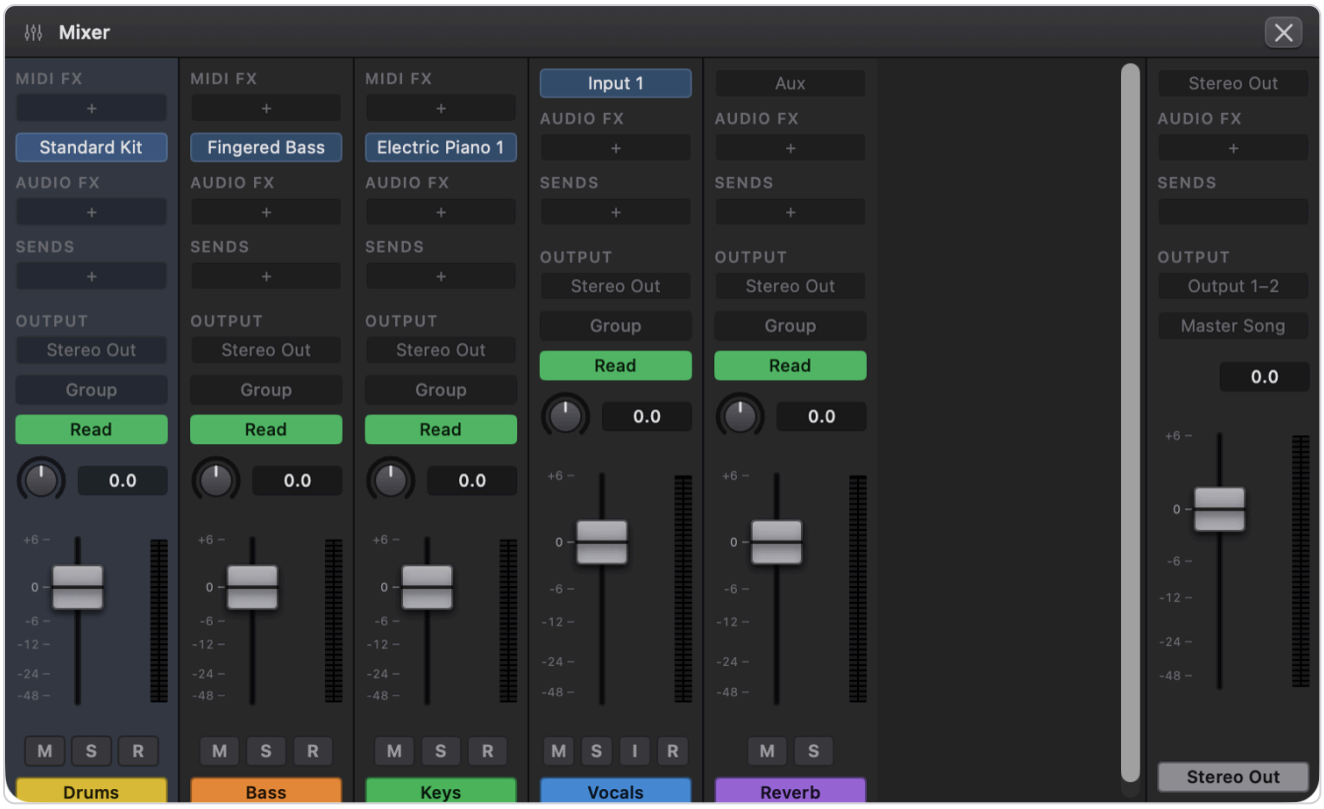
- **The Mixer** shows a strip for every track side by side, with **Stereo Out** (the final mix) on the right. Use it when you're balancing the whole song.
- **The Inspector** shows the selected track's strip next to Stereo Out. Use it when you're working on one track and don't want to leave the Tracks area.

Opening the Mixer

1. Press **X**, click **Mixer** in the control bar, or choose **View › Show Mixer**.
2. The Mixer opens in the bottom half of the window. If there are more strips than fit, scroll sideways; if a strip has many plug-ins and sends, the Mixer scrolls up and down too.
3. To close it, press **X** again, choose **View › Hide Mixer**, or click the **×** (**Close Mixer**) at the right of its title bar.

To show the Inspector instead, press **I**, click **Inspector** in the control bar, or choose **View › Show Inspector**.

Click anywhere on a strip's background to select that track. As on the track headers, **Shift-click** another strip to select every track in between, and **⌘-click** to add or remove one track at a time. The selected tracks' strips are tinted.



The Mixer with several channel strips and Stereo Out on the right

A tour of the channel strip

From top to bottom, a channel strip has these parts. Not every track has all of them.

Part	What it does
MIDI FX slots	Software instrument tracks only. Effects that change the notes before the instrument plays them, such as the Arpeggiator. See Chapter 8.
Input or instrument slot	On an audio track, the input you record from (for example **Input 1**, or **Input 1–2** for stereo). On a software instrument track, the instrument (or **No Instrument**). On an aux track it just says **Aux**.
Audio FX slots	Insert effects, which process the track's sound in order from top to bottom. Up to eight.
Sends	Copies of the track's sound sent to aux tracks, each with its own level knob.
Output	Where the track's sound goes: **Stereo Out**, or an aux track.
Group	Which mixer group the track belongs to, if any.
Automation mode	**Off**, **Read**, **Touch** or **Latch** (coloured grey, green, yellow and purple). See Automation.
Pan knob and level readout	Where the track sits from left to right, and the fader's level in decibels.
Fader and meter	The track's volume, and how loud it is right now.
M, **S**, **I**, **R**	Mute, Solo, Input Monitoring and Record Enable. Only audio tracks have **I**; aux tracks have no **R**.
Name	The track's name on its colour.

An empty slot shows **+**. Click it to add something there.

Stereo Out

Stereo Out is the strip at the far right of the Mixer (and of the Inspector). Everything you hear passes through it on the way to your speakers. It has **Audio FX** slots, a fader and a meter, and its output is labelled **Output 1–2**. Its send slots are empty placeholders that keep the faders lined up. Under its output is **Master Song**, which opens Mastering (see below).

Plug-ins on Stereo Out process the whole mix, after every track and aux and before the Stereo Out fader. This is the place for mastering. The easiest way is **Mastering**: click **Master Song** at the bottom of the Stereo Out strip, and Ongaku's Mastering plug-in is put on Stereo Out and opened. It listens to your song and sets up the EQ, compression and loudness for it (see Mastering in Chapter 8). If you'd rather build your own chain, use a gentle EQ, a bus compressor and, last of all, the **Maximizer** (its **Loud Master** preset is made for this). You add, open, bypass, reorder and remove them exactly as on a track (see Plug-ins on a channel strip), with the same **Add Insert** menu and up to eight slots.

Things to know about Stereo Out's plug-ins:

- They're included when you bounce (Chapter 9), and their delay is compensated like any other plug-in's.
- They aren't part of any track's freeze file. Freezing a track renders that track only, so Stereo Out's plug-ins keep running.
- Their settings can't be automated: there's no automation lane for Stereo Out.
- The metronome click doesn't pass through them, or through the Stereo Out fader, so a limiter on the mix won't squash the click.

Fader, pan, mute and solo

- **Fader:** drag the cap up or down. 0 dB is the track at its natural level; the fader goes up to +6 dB, and all the way down is silence (shown as **-∞**). The marks beside it show +6, 0, –6, –12, –24 and –48 dB. Option-click the fader to put it back to 0 dB (its tooltip reminds you).
- **Pan:** drag the small knob up or down. The readout shows **C** at the centre and **L 64** to **R 64** at the extremes. Double-click or Option-click it to centre it again.

Every knob and fader on a strip resets with Option-click, and each reset is one undo step.

- **Meter:** the stereo meter beside the fader shows the level after the fader, from –60 to +6 dB. It's green, turns yellow above –12 dB and red above 0 dB. Red means the track is very hot; pull the fader down.
- **M** (Mute) silences the track. **S** (Solo) plays only soloed tracks. **R** arms the track for recording (see Chapter 3).
- **I** (Input Monitoring, audio tracks only) lets you hear the track's input live through this strip: its plug-ins, fader, pan and sends. It lights orange while it's on. Use headphones, or the speakers can feed back into the microphone. See Chapter 3.

Every change you make on a strip is an ordinary edit: **Edit › Undo** (⌘Z) takes it back.

> Tip: If a fader keeps jumping back while the song plays, the track has volume automation in **Read** mode. The automation is in charge. Change the mode to **Off**, or edit the automation instead (see Automation).

Plug-ins on a channel strip

A **plug-in** is a piece of software that processes sound. An **insert** is a plug-in placed directly in a track's signal path, in one of its **Audio FX** slots. The whole sound passes through it, top slot first, so order matters: an EQ before a compressor sounds different from a compressor before an EQ.

Adding an insert

1. On the track's strip (or the Stereo Out strip), click the **+** slot under **Audio FX**.
2. The **Add Insert** menu opens with three groups:
    - **Ongaku**: Ongaku Studio's own effects, in submenus by kind: **EQ**, **Dynamics**, **Reverb & Delay**, **Distortion**, **Imaging** and **Vocal**. These are described in Chapter 8.
    - **Apple (built into macOS)**: Apple's own Audio Unit effects that come with your Mac. Each is listed with Apple's name for the Audio Unit underneath, so you can tell them from Ongaku's: | Menu item | Apple's Audio Unit | |---|---| | **Channel EQ** | Apple AUNBandEQ | | **Compressor** | Apple AUDynamicsProcessor | | **Multiband Compressor** | Apple AUMultibandCompressor | | **Limiter** | Apple AUPeakLimiter | | **Reverb** | Apple AUReverb2 | | **Delay** | Apple AUDelay | | **Distortion** | Apple AUDistortion | | **High Pass Filter** | Apple AUHipass | | **Low Pass Filter** | Apple AULowpass |
    - **Audio Units**: other Audio Unit effects installed on your Mac, listed as "Maker: Name". If you have none, it says **No Audio Unit effects are installed**.
3. Choose an effect. It appears in the slot, and a new **+** slot appears below it until all eight slots are used.

Note that **Channel EQ**, **Compressor**, **Reverb** and **Delay** appear under both **Ongaku** and **Apple (built into macOS)**. They're different plug-ins: the Ongaku versions have the visual editors described in Chapter 8.

Opening a plug-in window

Click the insert's slot. The plug-in's own window opens, titled with the track and plug-in name (for example "Vocals — Compressor"). This works on Stereo Out's slots too.

If the window is already open, this brings it to the front. If the plug-in is still loading you'll see a message saying so; try again in a moment. The slot's menu (see below) also has **Show Plug-in Window…**.

For an instrument plug-in, click the instrument slot (or the instrument name on the track header) and choose **Show Instrument Window…** at the top of the menu.

> Important: A plug-in's settings are stored in your song when you close its window (and whenever you save). Closing the window is one undoable step, called **Plug-in Settings**. Changes inside the window have their own undo in Ongaku's plug-ins (see Chapter 8).

Bypassing, reordering and removing inserts

Click and **hold** an insert's slot, or **Control-click** (right-click) it, for its menu:

- **Bypass**: switches the plug-in off without removing it, so you can hear the track without it. A bypassed slot is dimmed. Choose **Bypass** again to switch it back on.
- **Move Up** / **Move Down**: changes the order of the inserts.
- **Remove**: takes the plug-in off the strip and closes its window. Any automation of that plug-in's settings goes with it (undo brings both back).

Instruments and MIDI FX

On a software instrument track, the slot under **MIDI FX** shows the instrument. Click it to choose a different one (the same menu as on the track header). MIDI FX slots are above it, because MIDI effects change the notes before the instrument plays them. Like insert slots, a MIDI FX slot opens its effect when you click it, and shows its menu when you hold or Control-click it. Both are covered in Chapter 8.

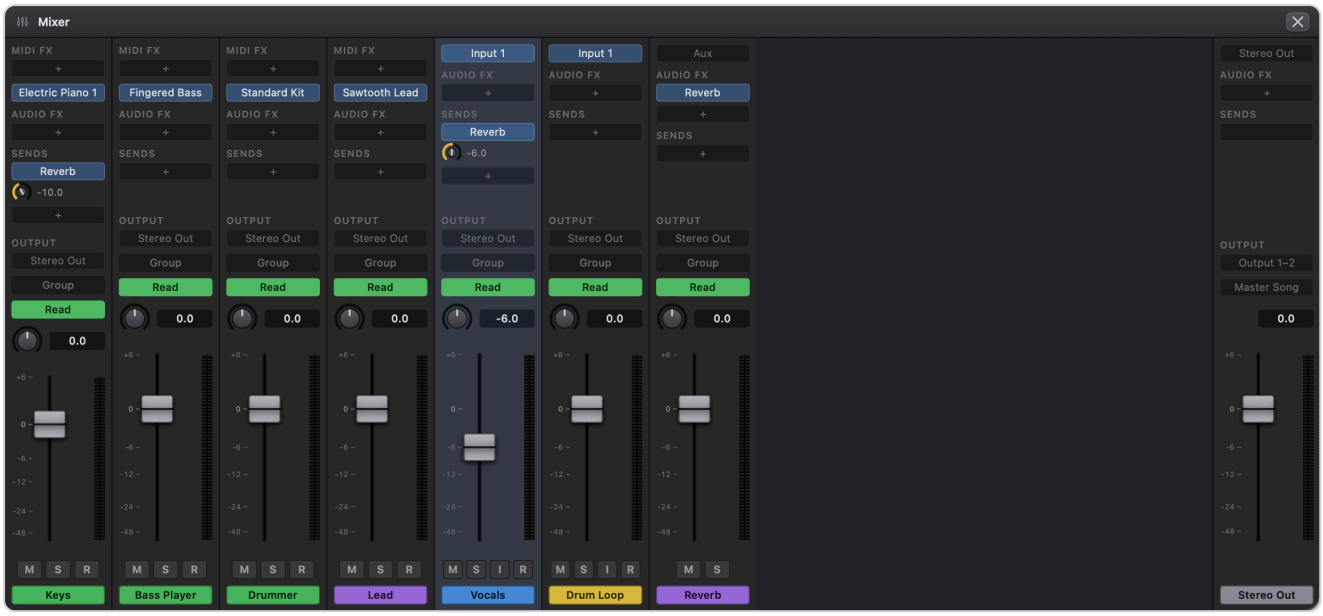
Sends, auxes and buses

What sends and auxes are

Suppose you want the vocals, the guitar and the snare to share one reverb, so they sound as if they're in the same room. You could put a reverb insert on each track, but that's three reverbs to set up and three times the processing. Instead:

- An **aux track** is a track with no recordings of its own. Its sound comes from other tracks. You put the reverb on the aux.
- A **send** is a knob on each track that sends a copy of that track's sound to the aux. Turn the knob up for more reverb on that track.

The original sound still goes through the track's fader as usual, so you hear the dry track plus the shared reverb. That's why the Ongaku Reverb and Delay say to set their **Mix** to 100 % when they're on an aux: the aux should be all effect.



Keys and Vocals sending to the Reverb aux, which has a Reverb insert

Adding a send

1. On the track's strip, click the **+** slot under **Sends**.
2. Choose an aux from the list, or choose **New Aux** to make a new aux track and send to it in one step.
3. The send appears, named after the aux, with a level knob underneath. New sends start at –6 dB.

You can also make an aux with **Track › New Aux Track**.

Setting a send

- **Level:** drag the send's knob up or down. The number beside it shows the level in dB. The knob turns like the fader: most of its travel is the useful range around 0 dB, it goes up to +6 dB, and all the way down is off (**-∞**). Double-click or Option-click the knob to put it back to –6 dB, where new sends start.
- **Automating it:** the knob follows the send's automation as the song plays, and in **Touch** or **Latch** mode, turning it during playback records send automation, just as the fader does (see Recording automation by moving the fader).
- **Pre-Fader:** click the send's name and choose **Pre-Fader**. A pre-fader send is taken before the track's fader and mute, so it stays at the same level when you move the fader. Its slot then reads " (pre)" after the name. Use this for things like a headphone mix; for reverbs and delays you usually want the normal (post-fader) send, so the effect follows the fader.
- **Remove Send:** click the send's name and choose **Remove Send**.

A send whose aux has been deleted is labelled **Missing**.

Track outputs and buses

The **Output** slot decides where the track's fader sends its sound. Normally that's **Stereo Out**. Click it to choose an aux instead. Sending several tracks' outputs to one aux makes a **bus** (sometimes called a submix): for example, route all your drum tracks to an aux called "Drums" so one fader controls the

whole kit, and one compressor on the aux squeezes them together.

1. Make an aux (**Track › New Aux Track**) and rename it if you like.
2. On each track you want on the bus, click **Output** and choose the aux.
3. The Output slot shows the aux's name and is highlighted, so you can tell at a glance that the track isn't going straight to Stereo Out.

Tracks inside a summing stack output to their stack automatically; hovering over their Output slot says "This subtrack plays through its summing stack". Summing stacks are covered in Chapter 2.

> Note: Ongaku Studio won't let you make a loop, where a track ends up feeding itself. Auxes that would cause a loop simply aren't offered in the send and output menus, and a track can't both send to and output to the same aux.

Mixer groups

A **mixer group** ties several tracks together so that moving one moves them all. It's handy for a set of backing vocals or a multi-miked drum kit: move one fader and the whole group goes up or down by the same amount, keeping their balance.

Making a group

1. Select the tracks you want to group: click the first track header (or channel strip in the Mixer), then Shift-click or Command-click the others.
2. On one of their channel strips, click the **Group** slot (just below **Output**).
3. Choose **New Group with Selected Tracks**. The tracks join the lowest free group number, and the slot shows the group's name, **Group 1** to begin with (or 2, 3 and so on, up to 32).

To add one more track to an existing group, click its **Group** slot and choose the group from the list (shown as its number and name, for example "1: Group 1"). Choose **No Group** to take it out.

Naming a group

To give a group a name you'll recognise, such as "Backing Vocals":

1. Click the **Group** slot of any member.
2. Choose **Rename Group *n*…**.
3. Type the name and click **Rename**.

Every member's Group slot then shows the new name. Renaming can be undone with ⌘Z.

What a group links

Click the **Group** slot of any member to see the group's settings, under **Group N Settings**:

Setting	Default	What it does
On	On	Switch the group off to adjust one member on its own, then on again.
Volume	On	Moving one member's fader moves the others by the same number of dB.
Pan	Off	Moving one member's pan moves the others by the same amount.
Mute	On	Muting one mutes them all.
Solo	On	Soloing one solos them all.

Rename Group N… is described above. **Delete Group N** removes the group (the tracks stay, just ungrouped). When the group is on, its slot has an accent-coloured outline.

Plug-in delay compensation

Some plug-ins need a moment to think. A look-ahead limiter such as the **Maximizer** listens slightly ahead so it can catch peaks before they happen, and the **Convolution Reverb** works in blocks. Both hand their sound back a little late. If nothing were done about it, a track with one of these on it would play slightly behind the others, and drums would flam against each other.

Ongaku Studio fixes this automatically. It knows how late each plug-in is, and delays every other path through the mixer by just enough that everything arrives at Stereo Out (and at each aux) together. This is called **plug-in delay compensation**. There's nothing to switch on.

What it means for you:

- **The mix stays in time**, however many plug-ins you use, and bounces line up the same way.
- **Recording is lined up too.** What you record is placed where you played it against what you heard.
- **Adding or removing a delaying plug-in during playback restarts playback** from where you are, so everything lines up again. That brief restart is expected.
- **Playing live through a delaying plug-in feels slightly late**, because the whole mix is delayed to match it. If you're recording and it bothers you, bypass the Maximizer or Convolution Reverb until you've finished recording.

Freezing tracks

A big software instrument with several plug-ins can use a lot of your Mac's processing power. **Freezing** a track renders it to an audio file in the background; from then on, the track plays that file instead of running its instrument and plug-ins. The track sounds the same, but costs almost nothing to play. When you want to change it again, you unfreeze it.

Audio and software instrument tracks can be frozen. Aux tracks and summing stacks can't, and a track with no regions has nothing to freeze.

Choosing what freezing includes

Before freezing, choose a **Freeze Mode** in the Inspector (press **I**; the row is just under the track's name), or right-click the **Freeze** button (the snowflake) on the track header and choose from **Freeze Mode** there:

Freeze Mode	What's rendered	What you can still change while frozen
Pre Fader (the default)	The instrument (or audio regions) **and** the inserts	Fader, pan, sends, output, mute, solo and automation
Source Only	Only the instrument (or audio regions)	All of the above, and the inserts, which keep running

Pre Fader saves the most processing. Source Only is useful when the instrument is the heavy part but you're still tweaking the effects.

Freezing and unfreezing

1. Click the **Freeze** button (the snowflake) on the track header. Or Control-click the header and choose **Freeze Track**, or select several tracks and choose **Track › Freeze Tracks**.
2. While it renders, the button shows a progress circle. Click it again (or choose **Stop Freezing**) to cancel.
3. When it's done, the snowflake lights up blue and the track's lane gets an icy tint.

If the track has flexed audio regions that are still being processed (Chapter 6), the freeze waits for them first, so the frozen file includes every flex edit.

To unfreeze, click the lit snowflake, choose **Unfreeze Track** from the header's menu, or choose **Track › Unfreeze Tracks** (this menu item says Unfreeze when every selected track is frozen). Unfreezing is instant: the track goes back to playing live, with all its settings as they were.

What you can't do while a track is frozen

A frozen track's file was made from its regions and plug-ins, so Ongaku Studio stops you changing those until you unfreeze:

- You can't edit, move or add regions on it, or move other regions onto it.
- You can't change its instrument, MIDI FX, or input, record-enable it, or monitor its input (the **R** and **I** buttons are disabled).
- In **Pre Fader** mode you can't add, remove, open or reorder its inserts.
- You can't change its Freeze Mode.

If you try, you'll see a message such as "“Synth” is frozen. Unfreeze it to change its regions or plug-ins." If you edit the track while it's still rendering, the freeze is abandoned and you're asked to freeze it again.

> Tip: Freeze files are saved inside your song, so a frozen song opens frozen. They can always be made again, so unfreezing and re-freezing is safe.

Automation

What automation is

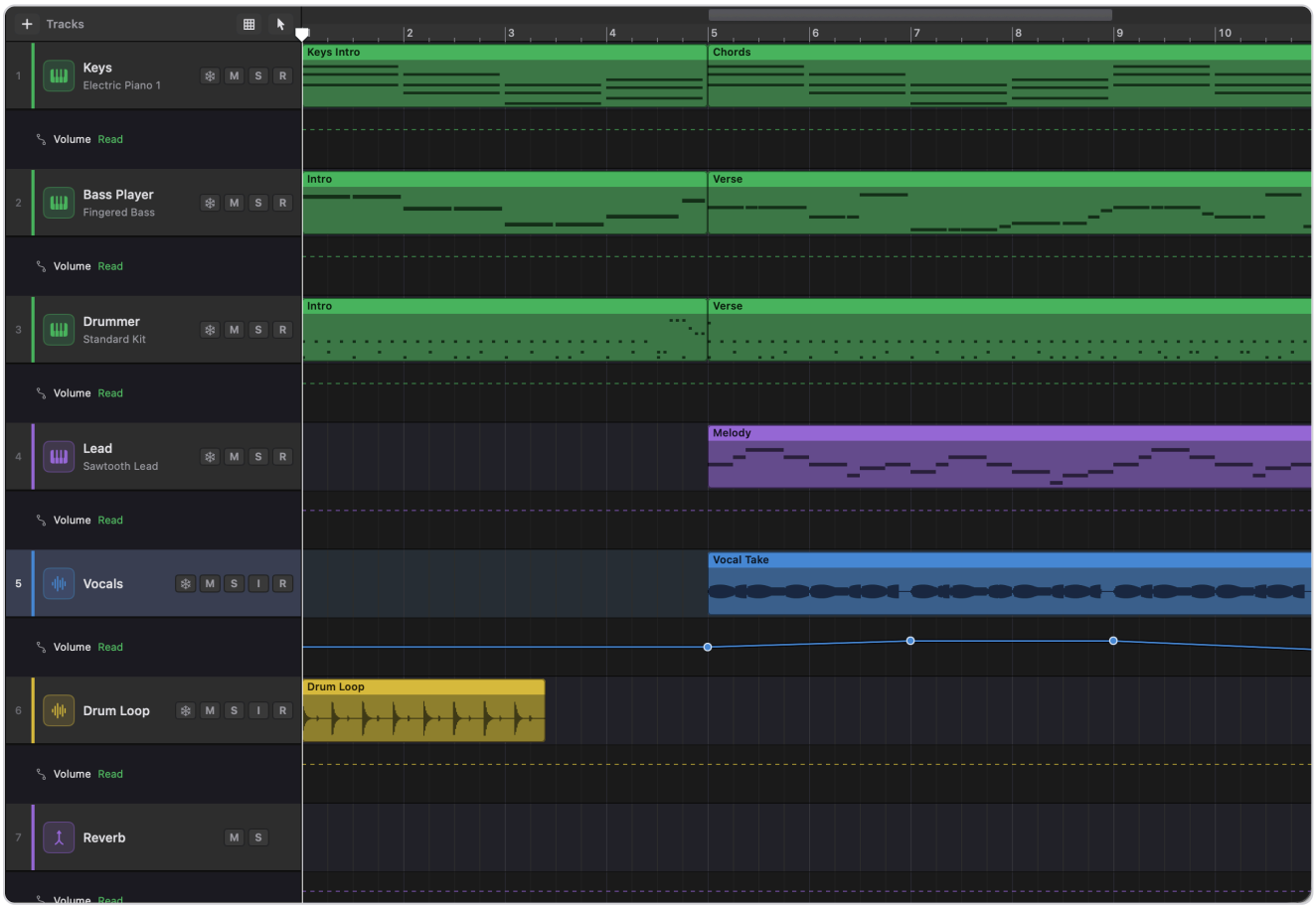
Automation is a recording of a control moving over time. Instead of setting a track's volume once, you draw a line that fades it up for the chorus and down for the outro, and the fader follows that line every time the song plays. You can automate a track's **volume**, **pan**, each **send level**, and the settings of its **insert plug-ins**.

Each track shows one automation line at a time, in its own **automation strip** directly below the track's lane in the Tracks area. The line is made of **points**; between two points the value moves in a straight line. Before the first point the value is the first point's, and after the last point it stays at the last point's.

Showing automation

Press **A**, click **Automation** in the control bar, or choose **View › Show Automation**. Every track gets an automation strip just below its lane, showing its automation line. To the left of each strip, under the track header, is the strip's own header with two small menus: the parameter being shown (for example **Volume**) and the automation mode.

The regions stay where they were, above the strips, and you can go on selecting, moving and editing them while automation is showing. Press **A** again to hide the strips.



Automation strips below each track, with a volume curve on Vocals; each strip's header has its parameter and mode menus

Choosing what to automate

Click the parameter menu in the automation strip's header (it starts as **Volume**). It lists:

- **Volume** and **Pan**.
- Each send, as "Send to *aux name*".
- Each insert, as a submenu of that plug-in's settings (for example **Compressor › Threshold**). A plug-in with nothing to automate says **No automatable parameters**.

Parameters that already have automation are marked with a dot (•). Choosing one shows its line in the strip. Until you add points, the strip shows the current setting as a dashed line.

Drawing and editing points

- **Add a point:** click in the track's automation strip. The point goes where you click: across is the time, up and down is the value.
- **Move a point:** drag it. While you drag, its value is shown above it (for example "−6.0 dB" or "L 40"). Plug-in settings show in their own units, such as "1.50 kHz", "+3.0 dB", "120 ms" or a choice's name like "Saw", while the plug-in is loaded; otherwise they show as a percentage of their range, such as "75%".
- **Delete a point:** double-click it.
- **Delete the whole line:** open the parameter menu and choose **Delete *Volume* Automation** (named after the parameter shown).

Volume and send lanes use the same scale as the fader, so the curve looks like the fader moving. Pan runs from left at the bottom to right at the top. Plug-in settings run from the bottom of each setting's range to the top.

Every point you add, move or delete can be undone with ⌘Z.

Points snap across to the same grid as regions, set with the **Snap** button above the track headers (Chapter 2). Their value (up and down) isn't snapped.

> Tip: To place a point between grid lines, hold ^ (Control) while you drag it. Press Control after you've started dragging, since a Control-click is a right-click. Or set **Snap** to **Off**.

Automation modes

The mode decides what a track does with its automation. Set it from the mode menu in the automation strip's header (shown while automation is visible) or from the coloured slot on the channel strip, just above the pan knob.

Mode	Colour	What happens
Off	Grey	Automation is ignored. Controls stay where you set them. The line is drawn grey.
Read	Green	Controls follow the automation as the song plays. This is the usual mode.

Mode	Colour	What happens
Touch	Yellow	Like Read, but while the song plays, moving the fader, pan knob or a send knob records over the automation. When you let go, the control goes back to following the existing line.
Latch	Purple	Like Touch, but when you let go, the control stays where you left it and that value keeps being written until you stop playback.

New tracks start in **Read** mode. You can change that in **Settings › Automation** (see below).

Recording automation by moving the fader

Drawing points is precise, but sometimes it's easier to ride the fader by hand while you listen:

1. Set the track's mode to **Touch** or **Latch**.
2. Start playback.
3. Move the track's fader, pan knob or a send knob in the Mixer or Inspector. Your moves are recorded as you make them.
4. Let go. In **Touch**, the control returns to the existing line straight away. In **Latch**, it holds its last value until you stop.
5. Stop playback. What you recorded becomes one undoable step, named for example **Write Volume Automation**.

In cycle mode, when the playhead jumps back to the start of the cycle, what you've written so far is kept and the next pass records over it from there.

When you've finished, switch the track back to **Read** so you don't record over it by accident.

The **fader**, the **pan** knob and the **send** knobs record automation this way. Plug-in settings are automated by drawing points.

Grouped tracks. If the track is in a mixer group that links volume (or pan), moving its fader moves the other members too. Members that are also in **Touch** or **Latch** record their own automation of the move, all in the same undo step. Members in **Read** or **Off** move along while you hold the control, but nothing is written on them.

Automation settings

Choose **Ongaku Studio › Settings…** (⌘,) and click **Automation**:

- **Mode for new tracks**: the automation mode new tracks start in (**Read** unless you change it).
- **Thin recorded automation**: moving a fader records a great many points. Thinning keeps only the ones that change the curve's shape, so the line is easy to edit afterwards. Choose **Off (keep every point)**, **Light**, **Normal** (the default) or **Heavy**.

See Chapter 9 for the other settings.

Limitations

- **Stereo Out can't be automated.** Its fader and its plug-ins' settings stay where you set them.
- **Stereo Out's plug-ins aren't in freeze files**, and the metronome doesn't pass through them.
- **Only volume, pan and send levels can be recorded by moving controls.** Plug-in settings are automated by drawing points, and instrument plug-in settings can't be automated at all (only inserts appear in the parameter menu).
- **Plug-in values show in their own units only while the plug-in is loaded.** Otherwise the automation strip shows them as a percentage of their range.
- **Each track shows one automation line at a time.** Switch between parameters with the parameter menu.
- **Inserts are reordered with Move Up and Move Down**; you can't drag them between slots.
- **Grouped tracks only record automation if they're in Touch or Latch themselves.** Groups link volume and pan; send knobs aren't linked.

Quick reference

Action	How
Show or hide the Mixer	**X**, **Mixer** button, or **View › Show Mixer**
Show or hide the Inspector	**I**, or **View › Show Inspector**
Select several strips	Shift-click or ⌘-click them
Reset a fader to 0 dB	Option-click the fader
Centre the pan	Double-click or Option-click the pan knob
Reset a send to –6 dB	Double-click or Option-click its knob
Add an insert	Click **+** under **Audio FX**
Add a mastering effect to the whole mix	Click **+** under **Audio FX** on the Stereo Out strip
Master the song	Click **Master Song** on the Stereo Out strip, then **Listen** (or **Analyze Whole Song**)
Hear an audio track's input live	**I** button on the strip or track header
Open a plug-in's window	Click the insert slot
Bypass, reorder or remove an insert	Hold or Control-click the insert slot › **Bypass**, **Move Up**, **Move Down**, **Remove**
Add a send	Click **+** under **Sends**, choose an aux or **New Aux**
Make a send pre-fader	Click the send's name › **Pre-Fader**
Route a track to a bus	Click **Output**, choose an aux
Group tracks	Select tracks › **Group** slot › **New Group with Selected Tracks**
Name a group	**Group** slot › **Rename Group *n*…**
Freeze or unfreeze a track	Snowflake button on the track header, or **Track › Freeze Tracks**

Action	How
Choose what freezing renders	Inspector › **Freeze Mode**, or right-click the snowflake
Show or hide automation	**A**, **Automation** button, or **View › Show Automation**
Add, move or delete a point	Click the automation strip, drag a point, double-click a point
Move a point off the grid	Hold ^ after you start dragging
Choose what to automate	The automation strip header's parameter menu
Set the automation mode	The automation strip header's mode menu, or the channel strip's mode slot
Record automation by hand	**Touch** or **Latch**, play, and move the fader, pan or a send knob
Undo a mixer or automation change	⌘Z

8. Ongaku Instruments, Effects and MIDI FX

Ongaku Studio comes with its own set of effects, instruments and MIDI effects. The effects and instruments are full plug-ins with visual editors that show what they're doing to your sound; the MIDI effects reshape the notes you play before they reach an instrument. This chapter explains how the plug-in windows work, then goes through every effect, instrument and MIDI effect with its settings.

About the Ongaku plug-ins

Ongaku's effects and instruments are **Audio Units**, the standard plug-in format on the Mac. They live inside the Ongaku Studio app, and macOS makes them available to every app on your Mac once Ongaku Studio has been opened at least once. So you can also use them in Logic Pro, GarageBand and other Mac apps that host Audio Units. There they're listed under the maker **Ongaku**, with names such as **Ongaku: Compressor** and **Ongaku: Retro Synth**.

In Ongaku Studio you'll find them here:

- **Effects:** on a channel strip, click **+** under **Audio FX** and look in the **Ongaku** section of the menu. They're grouped as **EQ**, **Dynamics**, **Reverb & Delay**, **Distortion**, **Imaging**, **Vocal** and **Mastering**. See Chapter 7 for inserts in general.
- **Instruments:** click a software instrument track's instrument name (on the track header or the channel strip) and look in the **Ongaku** section: **Retro Synth**, **Drum Machine**, **Sampler** and **Session Drummer**. They're also in the Library (see Chapter 9).
- **MIDI FX:** click **+** under **MIDI FX** on a software instrument track's channel strip. MIDI FX are part of Ongaku Studio itself rather than plug-ins, so they only work inside Ongaku Studio.

To open a plug-in's window, click its insert slot, or for an instrument click the instrument name and choose **Show Instrument Window…**.

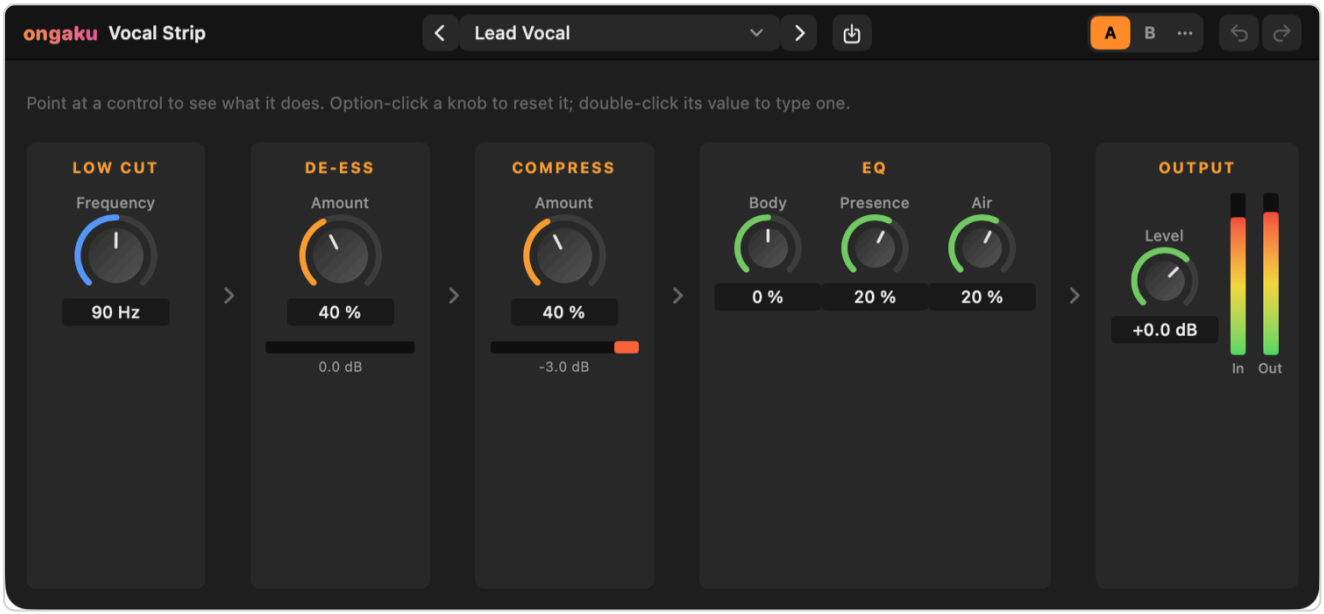
The Ongaku effects aren't the same as Apple's effects of the same name in the **Apple (built into macOS)** section of the **Add Insert** menu (for example Apple's **Compressor**, listed with "Apple AUDynamicsProcessor" under it). See Chapter 7.

Effect	Menu group	Factory presets
Channel EQ	EQ	none
Compressor	Dynamics	none
Noise Gate	Dynamics	5
Transient Shaper	Dynamics	5
Maximizer	Dynamics	5
De-Esser	Dynamics	5
Reverb	Reverb & Delay	5 rooms
Delay	Reverb & Delay	none
Convolution Reverb	Reverb & Delay	7

Effect	Menu group	Factory presets
Saturation	Distortion	5
Doubler	Imaging	5
Vocal Strip	Vocal	5
Mastering	Mastering	4 destinations

Working in a plug-in window

Every Ongaku plug-in window has the same header across the top, then the plug-in's own panel. Most panels also have a **hint line**: point at any control and a sentence appears explaining what it does. It's worth leaving your mouse over a knob you don't understand.



An Ongaku plug-in window with the header, hint line and knobs

Turning knobs and typing values

- **Drag up or down** on a knob (or its value) to turn it. A full sweep is about 200 points of mouse travel. Frequency and time knobs move evenly by ratio, so a frequency knob sweeps by octaves.
- **Option-click** any knob or value, in any Ongaku plug-in, to reset it to its default. The reset is one step in the window's undo (see below).
- **In the effects**, double-click a knob's value to type a number, then press Return (Escape cancels). You can type units: "2k" or "2 kHz" for 2,000 Hz, "250 ms", "-6 dB".
- **In the instruments** (Retro Synth, Drum Machine, Sampler), double-click a knob to reset it to its default, as Option-click does.

The plug-in header

From left to right, the header has the plug-in's name, the preset controls, the save button, the A/B switch and undo and redo.

Presets. The wide menu in the middle shows the current preset, or **No Preset**. Click it to choose one. This is the only preset menu in the window: the Reverb's rooms, the Retro Synth's patches, the Drum Machine's kits and the Sampler's presets are all chosen here. **Factory** presets come with the plug-in; **My Presets** are the ones you've saved. The **<** and **>** arrows either side step through all of them in turn. An orange dot after the name means you've changed the settings since choosing that preset.

Saving your own preset:

1. Set the plug-in up the way you like.
2. Click the save button (the arrow pointing into a tray) next to the preset menu.
3. Type a name and click **Save** (or press Return). Saving with the name of an existing preset of yours replaces it.

Your presets are kept by macOS for that plug-in, so they're also offered when you use the plug-in in other apps.

Renaming and deleting: open the preset menu and choose **Rename** or **Delete**, then the preset. These appear once you've saved at least one preset. Factory presets can't be renamed or deleted.

A/B compare. The **A** and **B** buttons hold two separate versions of the settings. You start in A. Click **B**, and B starts as a copy of A; change anything you like, then click **A** and **B** to switch back and forth and hear which is better. Changes in each are kept separately. The ⋯ menu beside them has **Copy A to B** (or **Copy B to A**) to make the other one match the one you're hearing.

Undo and Redo. The two arrows undo and redo changes made in this window (⌘Z and ⇧⌘Z work too while the window is in front). A whole knob turn counts as one step. Choosing a preset can be undone too. Changes made by automation aren't added to this history.

> Note: The window's undo history and A/B versions last while the window is open. When you close it, its settings are stored in your song as one step called **Plug-in Settings**, which **Edit › Undo** in the main window can take back.

Effects

Each effect section gives what the effect does, when you'd reach for it, and a table of its settings. Ranges and defaults are exactly what the knobs offer. Names in the tables are the knob labels; they're also the names you'll see in the automation menu (see Chapter 7).

Channel EQ

What it is. An equalizer (EQ) turns particular frequency ranges up or down: more bass, less boxiness, more sparkle. The Channel EQ has eight bands, from a low cut on the left to a high cut on the right, shown as one curve over a live picture of your sound (the **Analyzer**).

When to use it. On almost any track: to remove rumble under vocals and guitars, to cut a muddy frequency, or to brighten something that sounds dull.

How to use it.

1. Play the track and watch the analyzer (the blue area) to see where its energy is.
2. Drag a band's coloured handle on the graph: left and right change its frequency, up and down change its gain. Dragging a band that's off turns it on.
3. Click a band's name button under the graph to switch it on or off. Bands that are off are dimmed.
4. Fine-tune each band's **Frequency**, **Gain** and **Q** with the values in its column.

Q is how wide a band is: low Q affects a broad range, high Q a narrow one. The **Low Cut** and **High Cut** bands have a **Slope** instead of a gain: how steeply they cut, from **6 dB/Oct** to **48 dB/Oct** (default **12 dB/Oct**). Click **Analyzer** to hide the spectrum.

Band	On at first	Frequency	Q
Low Cut	No	30 Hz	0.71
Low Shelf	Yes	80 Hz	0.71
Low Mid	Yes	200 Hz	1.1
Mid	Yes	500 Hz	1.1
High Mid	Yes	1.5 kHz	1.1
Presence	Yes	4 kHz	1.1
High Shelf	Yes	10 kHz	0.71
High Cut	No	17 kHz	0.71

Setting	Range	Default
Band **Frequency**	20 Hz – 20 kHz	see above
Band **Gain** (all but the cuts)	–24 to +24 dB	0 dB
Band **Q**	0.1 – 12	see above
Cut **Slope**	6, 12, 18, 24, 36 or 48 dB/Oct	12 dB/Oct
Gain (output, left of the bands)	–24 to +24 dB	0 dB

The bands start at 0 dB, so a new Channel EQ changes nothing until you move something.

> Tip: A good first move on vocals, guitars and keys is to switch on **Low Cut** and raise its frequency until the sound starts to thin out, then back it off a little.

Compressor

What it is. A compressor turns down the loud parts of a sound, so the difference between loud and quiet is smaller. The result sounds more even and more solid.

When to use it. On vocals that jump between whispers and shouts, on bass to keep every note the same weight, on drums for punch.

How it works. When the sound goes above the **Threshold**, it's turned down according to the **Ratio**: at 4 : 1, a sound 8 dB over the threshold comes out only 2 dB over. **Attack** is how quickly it reacts, **Release** how quickly it lets go. The panel shows the **Gain Reduction** meter across the top (how much it's turning down right now), the compressor's curve with a dot riding on it at the current level, and **In** and **Out** meters.

Setting	Range	Default	What it does
Threshold	–60 to 0 dB	–20 dB	The level above which it starts turning down. Lower means more compression.
Ratio	1 : 1 to 30 : 1	4 : 1	How hard it turns down. 2 : 1 is gentle; 10 : 1 and above acts like a limiter.
Knee	0 – 24 dB	6 dB	How gradually it starts. A soft (higher) knee sounds smoother.
Make Up	–20 to +40 dB	0 dB	Turns the result back up to make up for what was taken off.
Attack	0.1 – 200 ms	10 ms	How quickly it clamps down. Longer lets the start of each note through, for punch.
Release	5 – 2,000 ms	150 ms	How quickly it recovers.
Mix	0 – 100 %	100 %	Blends compressed and original sound ("parallel compression" when below 100 %).
Auto Gain	On / Off	Off	Adds make-up gain automatically, based on the threshold and ratio.
Detection	**Peak** / **RMS**	Peak	Peak reacts to every spike; RMS reacts to average loudness and sounds smoother.

> Tip: Start from the defaults, play the track, and lower the **Threshold** until the Gain Reduction meter moves by a few dB on the loudest parts. Then raise **Make Up** until the track is as loud as it was with the Compressor bypassed.

Noise Gate

What it is. The opposite of a compressor: it turns the sound down when it's *quiet*. Between notes it shuts out hum, hiss, headphone spill or the other drums bleeding into a mic. An **OPEN** / **CLOSED** light shows what it's doing, and a scrolling graph shows the level against the threshold (the dashed line).

When to use it. On noisy guitar amps, tom and kick mics, or to tidy breaths between vocal phrases.

Setting	Range	Default	What it does
Threshold	–80 to 0 dB	–45 dB	The level the sound must reach to open the gate. Set it just above the noise between notes.
Reduction	–100 to 0 dB	–100 dB	How far it turns the sound down when closed. Less than all the way sounds more natural.
Hysteresis	0 – 20 dB	4 dB	How far below the threshold the level must fall before the gate closes, so it doesn't chatter.

Setting	Range	Default	What it does
Attack	0.1 – 200 ms	1 ms	How quickly it opens.
Hold	0 – 500 ms	40 ms	How long it stays open after the level drops.
Release	1 – 2,000 ms	120 ms	How long it takes to close. Longer lets notes ring out.
Key Low Cut	20 Hz – 2 kHz	20 Hz	Ignores lows when deciding whether to open, so rumble or kick spill doesn't open it.

Factory presets: Clean Gate (the defaults), **Tight Drums**, **Vocal Breaths** (only turns breaths down by 15 dB), **Guitar Hum** (ignores lows below 150 Hz) and **Gated Snare** (the 1980s gated-reverb trick: a long hold and a very fast release).

Transient Shaper

What it is. A **transient** is the very start of a sound: the crack of a snare, the pluck of a string. The Transient Shaper turns the start of each hit up or down, and separately the ring after it, without a threshold to set. The picture shows a drum hit before (dashed) and after your changes.

When to use it. To make drums punchier or softer, or to dry out (or bring out) the room sound in a drum recording.

Setting	Range	Default	What it does
Attack	–100 to +100 %	0 %	Turns the start of each hit up (punchier) or down (softer, further away).
Sustain	–100 to +100 %	0 %	Turns the ring after each hit up (more room and body) or down (tighter, drier).
Output	–24 to +12 dB	0 dB	The level out.
Mix	0 – 100 %	100 %	Blends the shaped sound with the original.

Factory presets: Neutral, **Punchy Drums** (Attack +60 %, Sustain –15 %), **Tighten** (Sustain –60 %), **More Room** (Sustain +60 %) and **Soften Attack** (Attack –60 %).

Maximizer

What it is. A limiter for making a finished mix louder. **Gain** pushes the level up while the **Ceiling** stops any peak going over a set level. It also measures **loudness** the way streaming services do, in **LUFS** (Loudness Units relative to Full Scale).

When to use it. As the very last effect on your mix, when you want it to be as loud as commercial releases without clipping. Put it in the last **Audio FX** slot on the **Stereo Out** strip, so it works on the whole mix (see Chapter 7). The **Loud Master** preset is a good place to start there. For a master set up for the song, with EQ and compression too, use Mastering instead.

The readouts show:

- **Momentary**: loudness over the last 0.4 s.
- **Short-Term**: loudness over the last 3 s.
- **Integrated** (the big orange number): loudness of everything since you last reset it. Streaming services aim for about −14 LUFS.
- **True Peak**: the highest peak since the reset, including peaks between samples, in dBTP.

Click **Reset Loudness** and then play the song from the top to measure the whole song. The graph shows gain reduction over the last few seconds.

Setting	Range	Default	What it does
Gain	0 – 24 dB	0 dB	How hard the level is pushed into the limiter.
Ceiling	−6 to 0 dB	−1 dB	The highest the output can go. −1 dB leaves room for MP3 and AAC encoding.
Release	1 – 1,000 ms	60 ms	How quickly the gain comes back after a peak. Shorter is louder but can pump or distort.
True Peak	On / Off	On	Also catches peaks between samples, which can clip after conversion.

Factory presets: Safety Limiter (the defaults: no extra gain, just a −1 dB ceiling), **Gentle Lift** (+3 dB), **Loud Master** (+8 dB), **Very Loud** (+12 dB) and **Broadcast Safe** (+4 dB, −2 dB ceiling).

> Tip: Start with **Gentle Lift**, reset the loudness, play the whole song, and read **Integrated**. Raise **Gain** a little at a time; when the gain reduction graph is moving constantly, you've gone far enough.

The Maximizer looks slightly ahead to catch peaks, which delays its sound a little. Ongaku Studio compensates automatically (see "Plug-in delay compensation" in Chapter 7).

De-Esser

What it is. A compressor that only listens to the hissy part of a voice: the "ss", "sh" and "t" sounds (called **sibilance**), which can sound harsh, especially after EQ and compression. It turns down everything above its **Frequency** when that part gets too loud. The picture shows the band it's turning down, dipping as it works.

When to use it. On vocals and spoken word, usually after the compressor.

Setting	Range	Default	What it does
Frequency	2 – 12 kHz	6.5 kHz	Where the sibilance starts.
Threshold	−60 to 0 dB	−30 dB	How loud the sibilance must be before it's turned down. Lower catches more.
Range	0 – 24 dB	8 dB	The most it turns the sibilance down.
Listen	On / Off	Off	Plays only what's being taken out, to help you find the frequency.

Factory presets: Vocal (the defaults), **Female Vocal** (7.5 kHz), **Male Vocal** (5.5 kHz), **Gentle** and **Strong**.

> Tip: Turn on **Listen** and sweep **Frequency** until you hear only the "ss" sounds, then turn **Listen** off and lower **Threshold** until they're tamed but the voice still sounds natural.

Reverb

What it is. Reverb is the sound of a space: the echoes of a room, hall or plate blurring into a tail behind the sound. The Reverb's panel shows three pictures: the **Early echoes**, how the tail dies away (the **Decay** picture, with the highs dying sooner when **High Cut** is lower), and the **Tone** after the cuts.

When to use it. To put dry recordings and instruments in a believable space, or for a big, washy effect. It's usually best on an aux that several tracks send to (see Chapter 7), with **Mix** at 100 %.

Setting	Range	Default	What it does
Predelay	0 – 250 ms	25 ms	The gap before the reverb starts. Longer keeps vocals and drums clear of their reverb.
Decay	0.2 – 20 s	2.8 s	How long the reverb takes to die away (by 60 dB).
Size	0 – 100 %	75 %	How big the space is. Bigger spaces spread their early echoes further apart.
Diffusion	0 – 100 %	75 %	How quickly the echoes blur into a smooth wash. Low sounds like a bare room, high like a plate.
High Cut	1 – 20 kHz	7 kHz	Darkens the reverb, and makes its highs die away sooner.
Low Cut	20 Hz – 1 kHz	60 Hz	Thins the reverb so it doesn't muddy the bass.
Modulation	0 – 100 %	30 %	Gently wobbles the echoes, smoothing out metallic ringing.
Width	0 – 100 %	100 %	How wide the reverb spreads. 0 % is mono.
Mix	0 – 100 %	30 %	How much reverb against the dry sound.

Rooms. The Reverb's factory presets are its rooms. Pick a space from the header's preset menu: **Room**, **Chamber**, **Hall** (what a new Reverb starts as), **Plate** and **Cathedral**. Choosing a room sets every knob except **Mix**, so your balance stays as it was.

Delay

What it is. An echo effect: it repeats the sound after a set time. Each repeat comes back quieter, a little darker and a little thinner, like a tape echo. The picture shows each echo as a bar over the song's beats.

When to use it. For slap-back on vocals and guitars, rhythmic echoes on synths, or a sense of space that's clearer than reverb.

Tempo sync. With **Sync** on (the default), the delay time is a note value at the song's tempo, so the echoes stay in time if you change the tempo. The tempo it's following is shown at the top right (for example **120.0 BPM**). Turn **Sync** off to set the time in milliseconds instead. The **Note** knob then becomes **Time**. Note values with **T** are triplets; a dot means half as long again.

Setting	Range	Default	What it does
Sync	On / Off	On	Times the echoes in notes at the song's tempo.
Note (Sync on)	1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, 1/4., 1/2	1/8.	The time between echoes.
Time (Sync off)	1 – 2,000 ms	375 ms	The time between echoes.
Feedback	0 – 100 %	35 %	How much of each echo comes back as another. Higher repeats for longer.
Ping-Pong	On / Off	Off	Bounces the echoes from side to side.
Low Cut	20 Hz – 2 kHz	120 Hz	Thins the echoes, a little more with every repeat.
High Cut	500 Hz – 20 kHz	5 kHz	Darkens the echoes, a little more with every repeat.
Wow & Flutter	0 – 100 %	20 %	Wobbles the echoes' timing and pitch like a tape machine.
Drive	0 – 100 %	15 %	Saturates loud echoes softly, like tape pushed hard.
Mix	0 – 100 %	30 %	How much echo against the dry sound. Use 100 % on an aux send.

The Delay has no factory presets; save your favourite settings as your own presets.

Convolution Reverb

What it is. A reverb built from an **impulse response**: a recording of how a real space responds to a click. Convolution reverbs sound very natural. Ongaku's can make its own impulse response from **Decay**, **Size** and **Damping** (the panel says **Synthesized from Decay, Size and Damping**), or use a file you load.

When to use it. When you want a realistic room or hall, or the reverb of a specific real space you have an impulse response for.

Loading an impulse response:

1. Click **Load Impulse Response…** and choose an audio file (WAV or AIFF), or drag a file onto the picture.
2. The panel shows its name and length. The first 8 seconds of the file, and its first two channels, are used. If the file is longer or has more channels, an orange note under the picture says what was left out, for example "Only the first 8 s is used." or "Only the first two of its 6 channels are used."
3. **Decay**, **Size** and **Damping** are dimmed, since the file now decides the room's shape. The other knobs still work.

4. Click **Use Synthesized** to go back to the built-in response.

If a file can't be used, a message in orange says why.

Setting	Range	Default	What it does
Predelay	0 – 250 ms	12 ms	The gap before the reverb starts.
Decay	0.2 – 7 s	2.2 s	How long the reverb takes to die away (synthesized responses only).
Size	0 – 100 %	55 %	How big the space is: how far apart its first reflections are (synthesized only).
Damping	0 – 100 %	40 %	How fast the highs die away: soft rooms are damped, stone ones aren't (synthesized only).
Width	0 – 100 %	100 %	How wide the reverb spreads. 0 % is mono.
Low Cut	20 Hz – 1 kHz	120 Hz	Thins the reverb.
High Cut	1 – 20 kHz	12 kHz	Darkens the reverb.
Mix	0 – 100 %	30 %	How much reverb against the dry sound. Use 100 % on an aux send.

Factory presets: Medium Hall (the defaults), **Small Room**, **Studio Room**, **Concert Hall**, **Cathedral**, **Bright Plate** and **Dark Chamber**.

Like the Maximizer, the Convolution Reverb delays its sound slightly, and Ongaku Studio compensates for it automatically.

Saturation

What it is. Gentle distortion that adds warmth and harmonics, like recording through tape, a valve (tube) amp or a transistor circuit pushed hard. The level stays about the same as you turn up **Drive**, so you hear the colour rather than just "louder". The picture shows the curve for the type and drive.

When to use it. To warm up a thin synth or DI bass, glue a drum bus together, or deliberately dirty something up.

Setting	Range	Default	What it does
Type	**Tape**, **Tube**, **Transistor**	Tape	Tape is soft and warm, with a little low bump. Tube is rounder. Transistor is harder and grittier.
Drive	0 – 36 dB	6 dB	How hard the sound is pushed: warmer, then dirtier.
Tone	−100 to +100 %	0 %	Darker to the left, brighter to the right.
Output	−24 to +12 dB	0 dB	The level out.
Mix	0 – 100 %	100 %	Blends the saturated sound with the clean one.

Factory presets: Warm Tape (the defaults), **Tube Glow**, **Subtle Glue** (3 dB of drive at 60 % mix, a good start on a bus), **Crunch** and **Lo-Fi Drive**.

Doubler

What it is. Makes one performance sound like two or three: it adds two slightly late, slightly out-of-tune copies, one either side of the original, that drift like real second takes. The picture shows the original in the middle and the copies spread across the stereo field.

When to use it. To widen a lead vocal or a guitar part without recording it twice.

Setting	Range	Default	What it does
Delay	5 – 60 ms	18 ms	How far behind the copies are. 10–30 ms sounds like a second take; less is tighter.
Detune	0 – 50 cents	9 cents	How out of tune the copies are (one sharp, one flat). A lot sounds like chorus.
Drift	0 – 100 %	40 %	How much the copies' timing and tuning wander.
Width	0 – 100 %	80 %	How far left and right the copies sit.
Mix	0 – 100 %	60 %	How loud the copies are against the original.

Factory presets: Natural Double (the defaults), **Wide Guitars**, **Tight ADT**, **Chorused** and **Subtle**.

Vocal Strip

What it is. A whole vocal chain in one plug-in, with one simple control per stage. From left to right: **Low Cut**, **De-Ess**, **Compress**, **EQ** and **Output**. The De-Ess and Compress stages each show how much they're turning the voice down right now.

When to use it. When you want a vocal (or podcast voice) to sound finished quickly, without setting up four separate plug-ins.

Stage	Knob	Range	Default	What it does
Low Cut	**Frequency**	20 – 400 Hz	90 Hz	Cuts rumble, handling noise and pops below this.
De-Ess	**Amount**	0 – 100 %	40 %	Softens harsh "ss" and "t" sounds.
Compress	**Amount**	0 – 100 %	40 %	Evens out the level so quiet words are heard and loud ones don't jump out.
EQ	**Body**	–100 to +100 %	0 %	Warmth and weight, around 220 Hz.
EQ	**Presence**	–100 to +100 %	+20 %	Clarity and forwardness, around 4.5 kHz.
EQ	**Air**	–100 to +100 %	+20 %	Breathiness and sparkle, above 12 kHz.
Output	**Level**	–24 to +12 dB	0 dB	The level out.

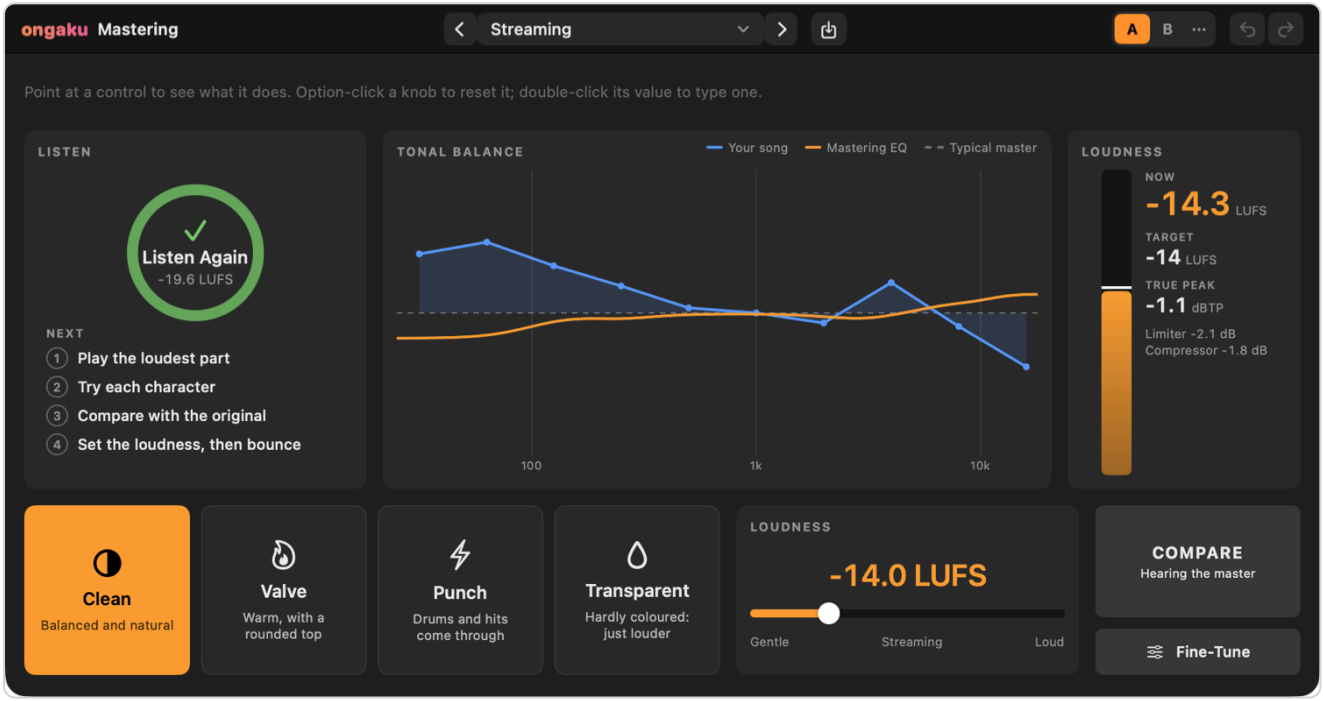
In the automation menu these appear as **Low Cut**, **De-Ess**, **Compress**, **Body**, **Presence**, **Air** and **Output**.

Factory presets: Lead Vocal (the defaults), **Natural**, **Backing Vocals**, **Podcast Voice** and **Rap Vocal**.

Mastering

What it is. Mastering is the last step before a song is released: a gentle EQ to even out its balance, a little compression to hold it together, and a limiter to make it as loud as other records without clipping. Ongaku's **Mastering** plug-in does all of that in one window, and sets most of it up for you. It **listens** to your song, compares it with a typical finished master, and chooses the EQ, compression and stereo width the song needs. You then choose a **character** and a **loudness** by ear, and use **Compare** to check that the master really is better than the mix.

When to use it. Once your mix is finished, on **Stereo Out**, as the very last plug-in, so it works on the whole song. It isn't meant for single tracks.



The Mastering window after listening, with the tonal balance graph

Putting Mastering on Stereo Out

Click **Master Song** at the bottom of the Stereo Out strip in the Mixer or the Inspector (or choose **File › Mastering…**). If Stereo Out doesn't have Mastering yet, it's added after any plug-ins already there, and its window opens. If it has, the window opens.

You can also add it like any other effect: click **+** under **Audio FX** on Stereo Out and choose **Ongaku › Mastering › Mastering**.

Mastering a song, step by step

1. **Listen.** Click the big **Listen** ring, then play your song: all of it, or at least its loudest part. The ring fills as it listens (a full minute fills it) and shows how long it has heard. Click **Done**. Mastering works out the settings for the song, and they're one step in the window's undo. It needs at least 10

seconds of music that isn't silent; if it heard less, it says so and changes nothing. Instead of playing the song, you can click **Analyze Whole Song** in the window's title bar; see Analyze Whole Song below.

2. **Follow the Next list.** Once it has listened, the Listen card shows what's left to do, and ticks each step off as you do it:
    1. **Play the loudest part.** The **Loudness** meter on the right should settle near your target. It ticks once the master plays within 1.5 LU of it.
    2. **Try each character.** Click the four character cards while the song plays, and keep the one you like (see the table below). **Clean** is the neutral starting point.
    3. **Compare with the original.** Click **Compare** to hear the song as it came in, at the same loudness and timing, then click it again for the master. If the master doesn't sound better, try another character, or click **Fine-Tune** and turn **EQ Amount** or **Dynamics** down.
    4. **Set the loudness, then bounce.** Drag the **Loudness** slider (see below), then bounce the song (**File › Bounce…**). Mastering stays on Stereo Out, so what you bounce is the master.

The ticks are only a guide: they start again whenever you listen again, and aren't kept when you close the window.

> Note: Until Mastering has listened, it doesn't turn the song up at all: it doesn't yet know how loud the song is. The Loudness card says **Listen first** until then.

The window

- **Listen** (left): the Listen ring, the song's measured loudness once it has listened, and the **Next** list.
- **Tonal Balance** (middle): your song's balance from low to high (blue) against a typical finished master (the dashed flat line), and the EQ Mastering applies (orange). Where the blue line is above the dashed line, the song has more of those frequencies than most masters; the orange curve dips there to even it out. The orange curve follows the character, **EQ Amount** and the EQ knobs as you change them.
- **Loudness** (right): how loud the master is now (over the last 3 seconds, the orange bar and the big number), your target (the white line), the **True Peak** (red if it goes over the ceiling), and how much the limiter and compressor are turning the sound down.
- **Character** cards, the **Loudness** slider, **Compare** and **Fine-Tune** along the bottom.

Character	What it sounds like
Clean	Balanced and natural: the EQ and compression as set for the song.
Valve	Warm, with a rounded top: a little tube warmth, and the Air and High bands 0.5 dB lower.
Punch	Drums and hits come through: a slower compressor attack, and 1 dB more low end.
Transparent	Hardly coloured, just louder: half the EQ and compression, and no widening.

The **Loudness** slider runs from −16 to −8 LUFS (LUFS measures loudness the way streaming services do). Its words say where you are: **Gentle** below −14.5, **Streaming** around −14, **Loud** from about −12.5, and **Very loud** above −9.5. Streaming services play songs at about −14 LUFS and turn louder songs down, so going louder only helps if you want the song to match loud records outside streaming.

Fine-Tune

Click **Fine-Tune** to replace the graph and meter with every setting; click **Overview** to go back. The window remembers which you were looking at. The first row is the overall controls; the second is what Listen sets for the song (listening again replaces them).

Setting	Range	Default	What it does
Character	Clean, Valve, Punch, Transparent	Clean	The flavour of the master (see above).
Loudness	–16 to –8 LUFS	–14 LUFS	How loud the master ends up.
EQ Amount	0 – 100 %	100 %	How much of the song's EQ is applied.
Dynamics	0 – 100 %	50 %	How firmly the compressor works: 50 % is as set for the song, less is gentler, more is firmer.
Width	0 – 100 %	0 %	Widens the stereo image above about 300 Hz. Listen sets 20 % or 40 % for a narrow mix, and none for a mix whose sides cancel in mono.
Low-End Mono	50 – 200 Hz	120 Hz	Makes the lows mono below this frequency, for a solid centre. At its lowest it's off.
Ceiling	–3 to 0 dBTP	–1 dBTP	The highest the output can go, between samples too. –1 dBTP leaves room for MP3 and AAC encoding.
Low, **Low-Mid**, **Presence**, **Air**, **High**	–6 to +6 dB	0 dB	The EQ: a low shelf at 100 Hz, bells at 250 Hz, 3 kHz and 7 kHz, and a high shelf at 10 kHz. Listen corrects at most 3 dB in each.
Threshold	–40 to 0 dB	–18 dB	Where the compressor starts working.
Ratio	1 : 1 to 4 : 1	1.5 : 1	How firmly it works above the threshold (scaled by Dynamics). Listen chooses 1 : 1 to 2 : 1, more for a mix with big peaks.
Attack	1 – 100 ms	20 ms	How quickly the compressor acts.
Release	20 – 1,000 ms	200 ms	How quickly it lets go.

Factory presets say where the song is going, and set only the **Loudness** and **Ceiling** (and turn Compare off). They leave the character and the settings Listen chose alone. **Streaming** (–14 LUFS, –1 dBTP; the defaults), **Gentle** (–16 LUFS), **Loud** (–10 LUFS) and **Club / CD** (–8 LUFS, –0.3 dBTP).

Analyze Whole Song

Playing the whole song for Listen takes as long as the song. **Analyze Whole Song** does the same listening offline, faster than the song plays, like a bounce:

1. Open Mastering's window and click **Analyze Whole Song** in its title bar, or choose **File › Analyze Whole Song for Mastering**. If Stereo Out doesn't have Mastering yet, it's added first.

2. Playback stops and a progress sheet appears. Mastering hears the whole song, from the start to the end of the last region, even if Cycle is on. Flexed audio is finished first, as for a bounce. Click **Cancel** to stop.
3. When it's done, the song's Mastering takes what it found: the graph, the EQ, the compressor, the width and the measured loudness. If its window is open, it shows the new settings straight away.

The result is one step called **Analyze Whole Song**, which **Edit › Undo** takes back. Analyze Whole Song is part of Ongaku Studio; in other apps, use **Listen**.

Things to know

- **Plug-ins before Mastering** on Stereo Out are part of what it hears and masters. The Stereo Out fader comes after it, so keep the fader at 0 dB when mastering, or the master won't reach the loudness you set.
- **Its settings are saved with the song**, including what Listen found, so the Tonal Balance graph is still there when you reopen the project.
- **Mastering delays the sound by 3.5 ms** (168 samples at 48 kHz) while it looks ahead and catches peaks between samples. Ongaku Studio compensates automatically (see "Plug-in delay compensation" in Chapter 7), and Compare is delayed to match.
- **Don't use the Maximizer after it.** Mastering already ends in a true-peak limiter, so a second one only squashes the song further.

Instruments

Ongaku's instruments go in a software instrument track's instrument slot. Each has a keyboard or pads in its window so you can hear it without a MIDI keyboard, and a hint line along the bottom. Double-click or Option-click any knob to reset it. Choose patches, kits and presets from the header's preset menu.

> Note: The plug-in windows name notes the way Ongaku Studio's Piano Roll and Logic do: middle C (MIDI note 60) is **C3**.

Retro Synth

What it is. A classic analog-style synthesizer: two oscillators make the raw sound, a 24 dB low-pass filter shapes its tone, two envelopes shape how each note moves, and an LFO adds wobble. It also has a sub oscillator, noise, FM, oscillator sync and ring modulation, and unison for huge, wide sounds.

Getting started. Choose a patch from the header's preset menu (or step with its **<** and **>** arrows): **Analog Saws** (the default), **Fat Bass**, **Soft Pad**, **Square Lead**, **Brass**, **Pluck**, **PWM Strings**, **Supersaw**, **Sync Lead**, **Ring Bells**, **FM Keys** or **Sub Bass**. Then change it. Play the keyboard at the bottom with your mouse (click or drag across the keys); **OCTAVE –** and **+** move its lowest C between C0 and C5.



The Retro Synth window

The tables give each knob's label on the panel; where the automation menu uses a different name, it's in brackets.

Top row

Control	Range	Default	What it does
VOICES (POLY / MONO)	Poly, Mono	Poly	Poly plays chords. Mono plays one note at a time and glides between notes played legato.
Glide	0 – 2,000 ms	0 ms	How long notes take to slide into each other.
Bend (Bend Range)	1 – 12 semitones	2	How far the pitch bend wheel bends.
Volume	–24 to +12 dB	0 dB	The synth's overall volume.

Oscillators

Control	Range	Default	What it does
Osc 1 wave	Saw, Pulse, Triangle, Sine	Saw	Where the sound starts: saw is bright, pulse hollow, triangle soft, sine pure.
Shape (Osc 1 Shape)	5 – 95 %	50 %	The pulse's width, or how far a saw, triangle or sine skews.
FM	0 – 100 %	0 %	Oscillator 2 bends oscillator 1's wave: from warm to bell-like and metallic.
Osc 1 ◂ ▸ Osc 2 (Osc Mix)	0 – 100 %	50 %	Blends the two oscillators.
Sub (Sub Osc)	0 – 100 %	0 %	A square wave an octave below oscillator 1, for weight in basses.

Control	Range	Default	What it does
Noise	0 – 100 %	0 %	Breath for pads, a chiff for plucks.
Osc 2 wave	Saw, Pulse, Triangle, Sine	Saw	Oscillator 2's wave.
Osc 2 mode (NORM / SYNC / RING)	Normal, Sync, Ring	Normal	Sync restarts oscillator 2 with oscillator 1 (sweep its Pitch for a tearing lead). Ring multiplies them, for metallic, bell-like tones.
Shape (Osc 2 Shape)	5 – 95 %	50 %	Oscillator 2's shape.
Pitch (Osc 2 Pitch)	–24 to +24 semitones	0	Tunes oscillator 2: +12 is an octave, +7 a fifth.
Fine (Osc 2 Fine Tune)	–50 to +50 cents	+7	Slight detuning, for a thick, chorused sound.

Unison and filter

Control	Range	Default	What it does
Voices (Unison Voices)	1 – 7	1	How many copies of each note play at once: more is thicker and wider.
Detune (Unison Detune)	0 – 100 cents	20	How far apart the copies are tuned.
Spread (Stereo Spread)	0 – 100 %	60 %	How widely the copies spread across the stereo field.
Cutoff	20 Hz – 20 kHz	2.5 kHz	The filter's brightness. Lower is darker and warmer.
Resonance	0 – 100 %	20 %	Boosts the sound at the cutoff, for a sharper, squelchy tone.
Env Amount (Filter Env Amount)	–100 to +100 %	+40 %	How much the filter envelope opens the filter on each note (below 0 closes it).
Key Track	0 – 100 %	50 %	Opens the filter more for higher notes, so they stay as bright as low ones.

Envelopes and LFO

An **envelope** is how something changes over the life of a note: **Attack** (how long it takes to rise), **Decay** (how long it then takes to fall), **Sustain** (where it stays while you hold the key) and **Release** (how long it takes to fade after you let go).

Control	Range	Filter Envelope default	Amp Envelope default
Attack	1 – 10,000 ms	5 ms	5 ms
Decay	1 – 10,000 ms	400 ms	300 ms
Sustain	0 – 100 %	30 %	80 %

Control	Range	Filter Envelope default	Amp Envelope default
Release	1 – 10,000 ms	300 ms	300 ms
Velocity (Amp Envelope only)	0 – 100 %		60 % (how much playing harder makes notes louder)

In the automation menu, the Filter Envelope's knobs are **Filter Attack**, **Filter Decay**, **Filter Sustain** and **Filter Release**; the Amp Envelope's are plain **Attack**, **Decay**, **Sustain** and **Release**.

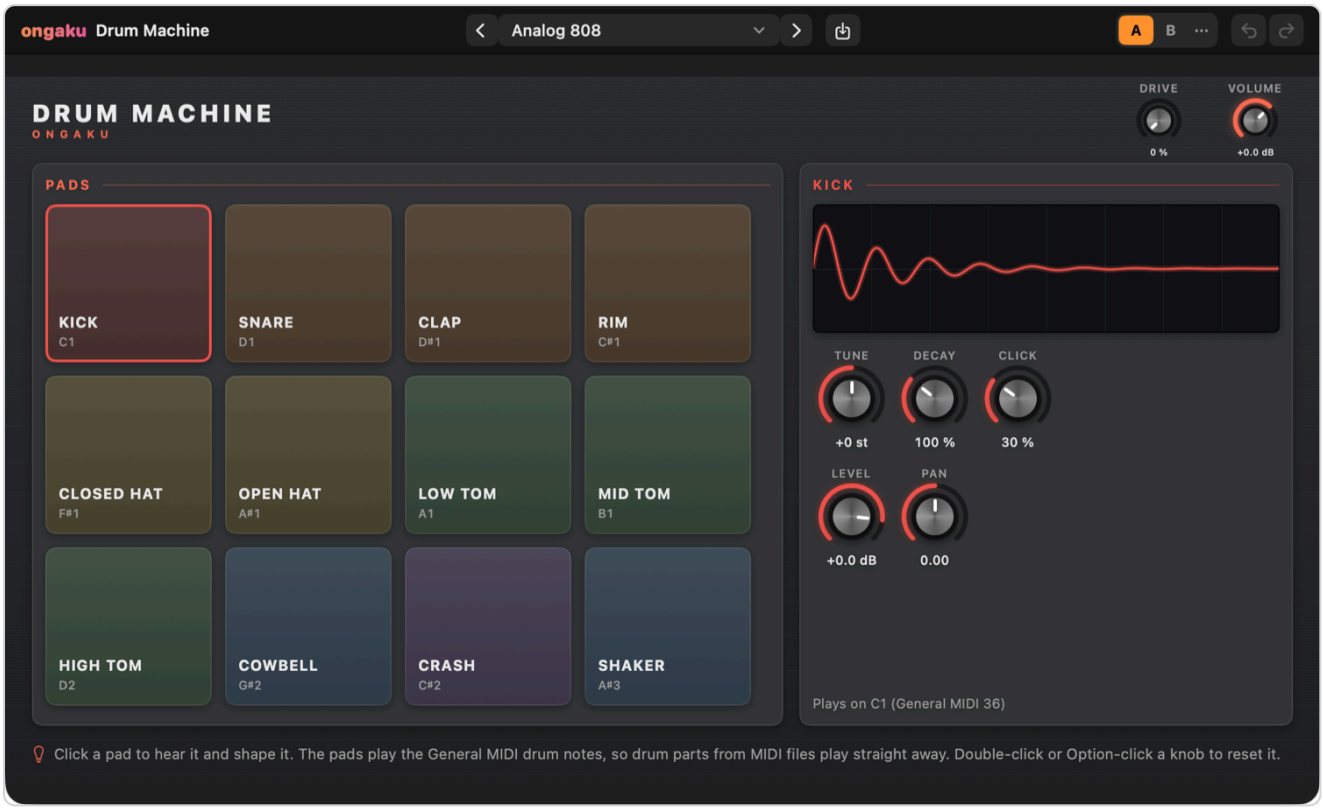
Control	Range	Default	What it does
LFO wave	Triangle, Square, Random	Triangle	A smooth wobble, a jump between two values, or a new value each cycle.
Rate (LFO Rate)	0.05 – 20 Hz	5 Hz	How fast the LFO wobbles.
Vibrato	0 – 100 cents	0	How much the LFO wobbles the pitch. The mod wheel adds more.
Wah (LFO Wah (to Filter))	0 – 100 %	0 %	How much the LFO moves the filter cutoff.
PWM (LFO PWM (to Shape))	0 – 100 %	0 %	How much the LFO moves both oscillators' shape, for a shimmering sound.

Drum Machine

What it is. A synthesized drum kit with twelve pads, in the spirit of classic drum machines. The pads play the standard General MIDI drum notes, so drum parts from MIDI files and the Drummer play on it straight away.

Using it.

1. Choose a kit from the header's preset menu: **Analog 808** (the default), **Punchy 909**, **Lo-Fi** or **Electro**.
2. Click a pad to hear it and select it. Pads light up as they play.
3. Shape the selected drum with the knobs on the right. The picture shows how it dies away.
4. **Drive** and **Volume** at the top right affect the whole kit.



The Drum Machine with its pads

Pad	Plays on	Its third knob	Starting pan
Kick	C1 (MIDI 36)	**Click**: how much click it starts with	Centre
Rim	C♯1 (37)	**Tone** (not used: use **Tune**)	Slightly left
Snare	D1 (38)	**Snappy**: how much the snares rattle	Centre
Clap	D♯1 (39)	**Brightness**	Centre
Closed Hat	F♯1 (42)	**Brightness**	Slightly right
Low Tom	A1 (45)	**Attack**: stick attack	Left
Open Hat	A♯1 (46)	**Brightness** (the closed hat cuts it off)	Slightly right
Mid Tom	B1 (47)	**Attack**	Centre
Crash	C♯2 (49)	**Brightness**	Slightly left
High Tom	D2 (50)	**Attack**	Right
Cowbell	G♯2 (56)	**Tone** (not used: use **Tune**)	Left
Shaker	A♯3 (70)	**Tone** (not used: use **Tune**)	Right

Note names here (and in the Drum Machine's window) put middle C, MIDI note 60, at C3, as in the Piano Roll.

Setting	Range	Default	What it does
Pad **Tune**	−12 to +12 semitones	0	Tunes the drum up or down.

Setting	Range	Default	What it does
Pad **Decay**	10 – 300 %	100 %	How long the drum rings: shorter is tighter, longer is boomier.
Pad third knob	0 – 100 %	30 % (kick and toms), 50 % (others)	See the pad table.
Pad **Level**	–36 to +6 dB	0 dB	How loud this drum is in the kit.
Pad **Pan**	–100 to +100	see the pad table	Where the drum sits, left to right.
Drive	0 – 100 %	0 %	Saturates the whole kit: warmer, then dirtier.
Volume	–24 to +12 dB	0 dB	The kit's overall volume.

In the automation menu, pad settings are named after the pad, for example **Snare Tune** or **Kick Click**.

Sampler

What it is. Plays any sound you give it from the keyboard: a single note of a real instrument, a vocal phrase, a drum hit, a sound effect. Higher keys play it faster and higher, lower keys slower and lower, unless you fix the pitch. It has a filter, envelopes and an LFO for shaping the sound, much like the Retro Synth.

Loading a sample.

1. Click **Load Sample…** and choose an audio file, or drag a file onto the waveform area (it says **Drag an audio file here** when empty).
2. The section title shows the sample's name, length and whether it's mono or stereo.
3. The start and end markers go back to the ends of the sample, and if the sound has a clear pitch, the **Root Note** is set to it for you.
4. Play it from the keyboard at the bottom. The dot on the keyboard marks the root note, where the sample plays at its own pitch.

The sample is saved inside your song (and inside the plug-in's settings in other apps too), so you don't need to keep the original file.

> Important: The Sampler keeps the **first 60 seconds** of a file, and its first two channels. When a file is cut to fit, an orange note on the waveform says so: "Only the first 60 s is used.", "Only the first two of its 6 channels are used.", or both. Trim long files first if you need a particular part.



The Sampler with a sample loaded

Markers and looping. Drag the **S** and **E** flags to set where the sample starts and ends; the parts outside are shaded. In **Classic** mode with **Loop** on, two more markers along the bottom set the part that repeats while you hold a key, so short sounds can sustain. A white line shows where each sounding note has got to.

Setting	Range	Default	What it does
Mode (CLASSIC / 1 SHOT)	Classic, One Shot	Classic	Classic plays while you hold the key, and can loop. One Shot plays the whole sample from one tap, for drums and hits.
Loop (OFF / FWD / ALT)	Off, Forward, Alternate	Off	Forward jumps from the loop's end back to its start; Alternate plays back and forth.
Direction (FWD / REV, "Reverse")	Forward, Reverse	Forward	Plays the sample backwards.
X-Fade (Loop X-Fade)	0 – 500 ms	20 ms	Blends the loop's end into its start so the jump is smooth.
Sample Start / End (S, E)	0 – 100 %	0 % / 100 %	Where the sample starts and ends.
Loop Start / End	0 – 100 %	25 % / 90 %	The part that repeats.
Root Note	C−2 – G8	C3 (MIDI 60)	The key that plays the sample at its own pitch.
Transpose	−24 to +24 semitones	0	Shifts the whole sample up or down.
Fine (Fine Tune)	−100 to +100 cents	0	Tunes the sample finely.

Setting	Range	Default	What it does
Follow Keys (FIXED / KEYS)	Fixed, Keys	Keys	Fixed plays every key at the sample's own pitch, for drums and sound effects.
VOICES, **Glide**, **Bend**, **Volume**	as the Retro Synth	Poly, 0 ms, 2, 0 dB	As the Retro Synth's top row.
Filter (OFF / ON)	Off, On	Off	Turns on the 24 dB low-pass filter.
Cutoff	20 Hz – 20 kHz	4 kHz	The filter's brightness.
Resonance	0 – 100 %	15 %	Boosts the sound at the cutoff.
Env Amount (Filter Env Amount)	–100 to +100 %	0 %	How much the filter envelope opens the filter.
Key Track	0 – 100 %	50 %	Opens the filter more for higher notes.
Filter Envelope A / D / S / R	as the Retro Synth	5 ms / 400 ms / 50 % / 300 ms	Shapes the filter over each note.
Amp Envelope A / D / S / R	as the Retro Synth	1 ms / 300 ms / 100 % / 150 ms	Shapes the volume over each note.
Velocity	0 – 100 %	60 %	How much playing harder makes notes louder.
LFO wave, **Rate**	as the Retro Synth	Triangle, 5 Hz	
Vibrato	0 – 100 cents	0	Wobbles the pitch. The mod wheel adds more.
Wah (LFO Wah (to Filter))	0 – 100 %	0 %	Moves the filter cutoff.
Tremolo	0 – 100 %	0 %	Wobbles the volume.

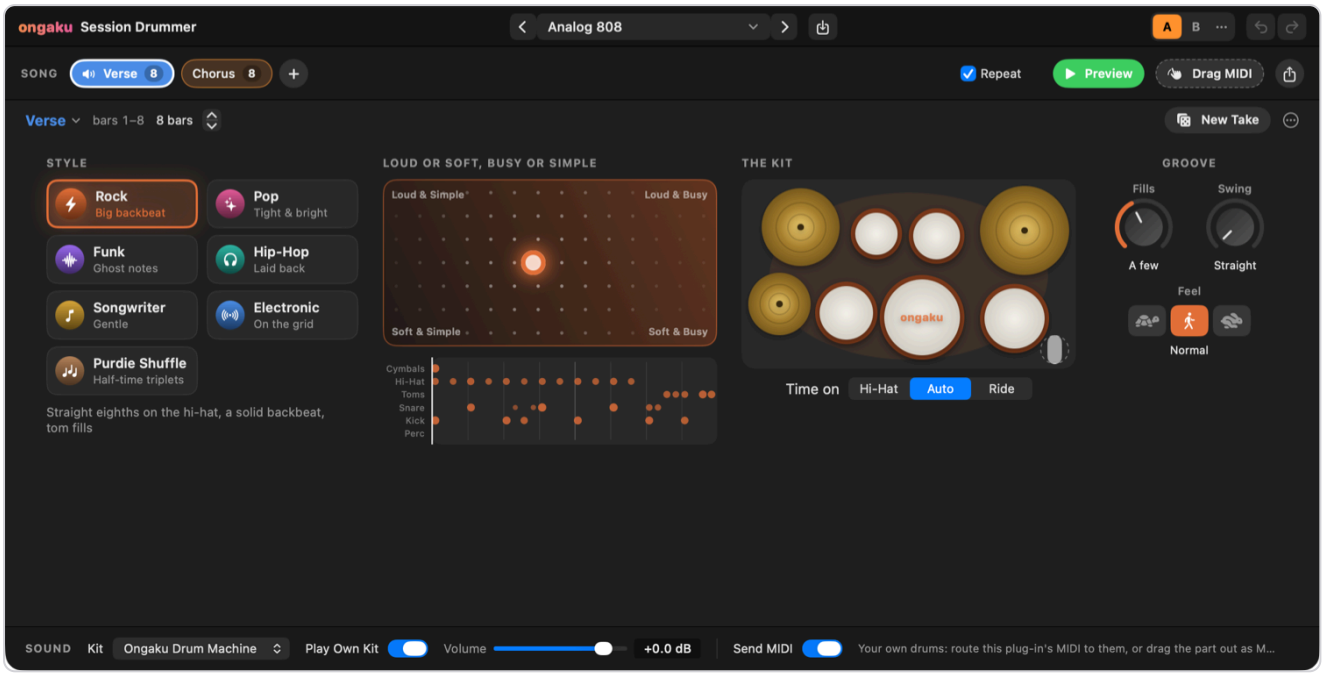
Presets, chosen from the header's preset menu, set how the sample plays, never the sample itself, its markers or its root note: **Classic**, **One Shot**, **Sustained Loop**, **Soft Pad**, **Pluck**, **Reverse Swell**, **Lo-Fi**, **Mono Lead** and **Fixed Pitch**.

> Tip: For a drum hit or sound effect, choose the **Fixed Pitch** preset. For a sustained instrument note, try **Sustained Loop**, then drag the loop markers to a steady part of the sound and raise **X-Fade** until the loop is smooth.

Session Drummer

What it is. The plug-in version of Ongaku Studio's Drummer (see Chapter 5). It plays a drum part that follows the host app's tempo and position, section by section. Other apps don't tell plug-ins about their song sections, so the Session Drummer keeps its own list, laid out from bar 1 of the host's timeline. It's mainly for using Ongaku's drummer in other apps; inside Ongaku Studio, a Drummer track does the same job with your real arrangement markers.

A new Session Drummer starts with a Rock **Verse** and **Chorus** of 8 bars each, repeating.



The Session Drummer window

The song strip (top).

- Each section is a coloured chip with its length in bars. Click one to select it; while the song plays, the playing section has a speaker icon and a white outline.
- **+** adds a section after the selected one: **Intro**, **Verse**, **Pre-Chorus**, **Chorus**, **Bridge**, **Solo**, **Breakdown** or **Outro**.
- **Repeat**: after the last section, start again from the first (otherwise the last one carries on).
- **Preview** / **Stop**: hear the drummer while your song is stopped. When the song plays, the drummer follows it.
- **Drag MIDI**: drag this into your song's tracks (in Ongaku Studio, or another app that accepts dropped MIDI files) to drop in the whole part as a MIDI region you can edit or play on another drum instrument.
- The export button (an arrow out of a box) saves the part as a MIDI file.

The section bar. Click the section's name to rename it; it's played to suit the new name. The stepper sets its length (1 to 64 bars), and "bars 9–16" shows where it falls. **New Take** plays the same settings a different way. The ⋯ menu has **Move Earlier**, **Move Later**, **Duplicate** and **Delete**.

How the selected section is played. These work as in the Drummer editor described in Chapter 5:

- **Style**: **Rock**, **Pop**, **Funk**, **Hip-Hop**, **Songwriter**, **Electronic** or **Purdie Shuffle**.
- **Loud or Soft, Busy or Simple**: drag the dot; right is busier, up is louder. The chart below shows the first two bars.
- **The Kit**: click a piece to leave it out or bring it back (cymbals, toms, kick and snare, and shaker go in and out together). **Time on** chooses what keeps time: **Hi-Hat**, **Auto** (opens up to the ride when the part gets loud) or **Ride**.
- **Groove**: **Fills** (None to Lots), **Swing** (Straight to Shuffle; in Purdie Shuffle this knob is **Ghosts**, for the ghost notes) and **Feel** (**Half Time**, **Normal**, **Double Time**). Double-click or Option-click **Fills** or **Swing** to put it back where a new section in that style starts.

The Sound bar (bottom). Choose what plays the part:

Control	Choices	What it does
Kit	**Ongaku Drum Machine**, **Apple Standard Kit**, **Apple Power Kit**, **Apple Electronic Kit**, **Apple TR-808 Kit**, **Apple Brush Kit**, **Apple Orchestra Kit**	Ongaku's Drum Machine, or one of the General MIDI kits built into macOS. With the Drum Machine, choose its kit (Analog 808, Punchy 909, Lo-Fi, Electro) from the header's preset menu.
Play Own Kit	On / Off (On)	Plays the part on the chosen kit. Turn it off to hear only your own drums through MIDI.
Volume (Kit Volume)	–40 to +6 dB (0 dB)	The kit's level.
Send MIDI	On / Off (On)	Sends the part out as MIDI, for apps that can route a plug-in's MIDI to another drum instrument.

The song and all its sections are saved with the plug-in's settings, and the header's undo and A/B include them.

MIDI FX

What MIDI FX are

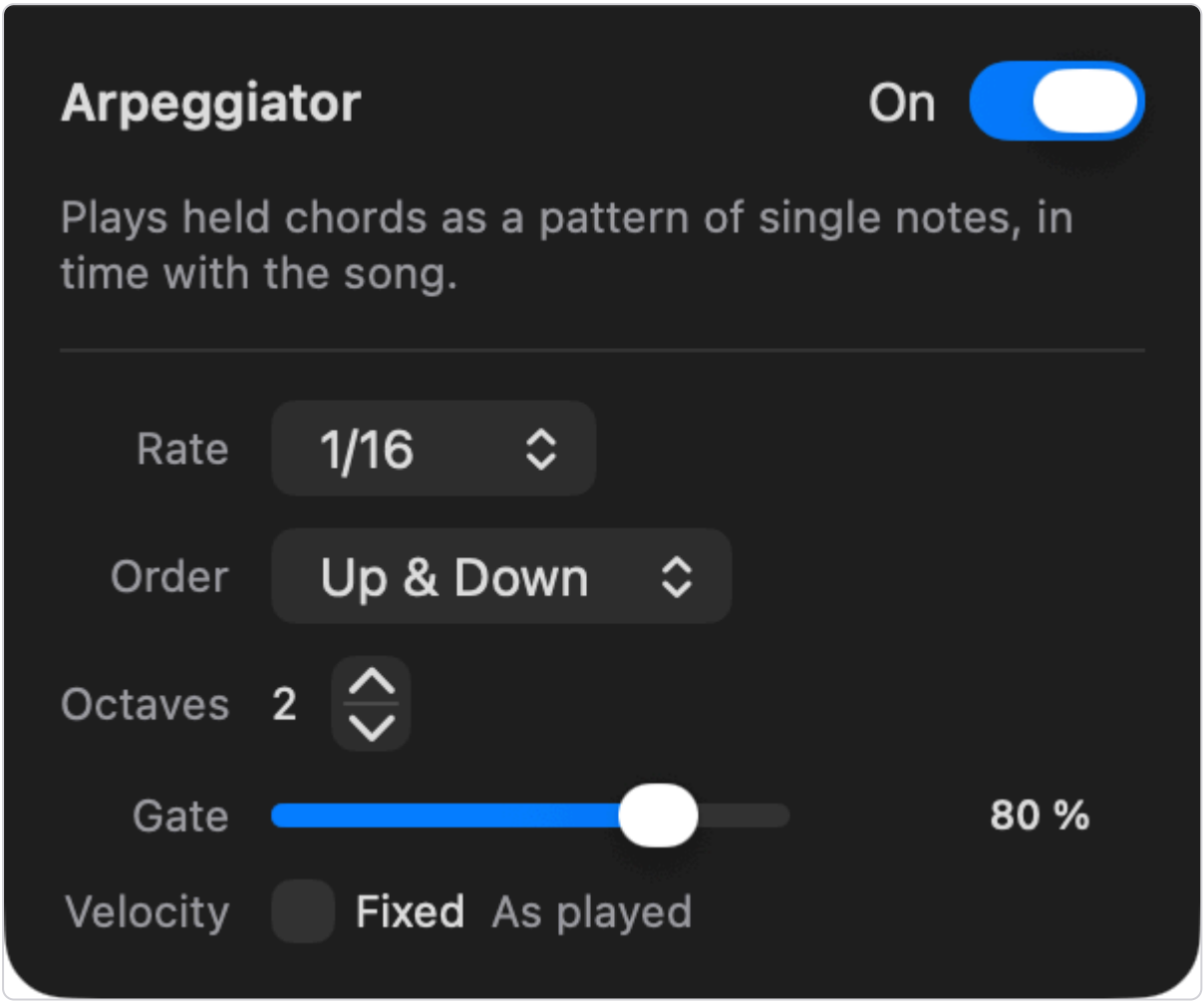
A MIDI effect changes **notes**, not sound. It sits before the instrument on a software instrument track and changes what's played: turning a held chord into an arpeggio, adding a chord to every note, forcing notes into a key, and so on.

- **Your recordings stay as you played them.** MIDI FX are applied as the song plays, so the notes in your regions are your own playing. You can change, bypass or remove an effect at any time and the region is untouched. The Piano Roll always shows the notes you played, not the effect's output.
- **They work live.** When you play a MIDI keyboard into the track, you hear your playing through the effects. While the song plays, the Arpeggiator and Note Repeater stay on the song's grid. The sustain pedal and pitch bend pass straight through.
- **Order matters.** Up to eight MIDI FX can be stacked; notes go through them top to bottom. A Chord Trigger above a Scale Quantize gives chords that stay in key.
- **They're included** when the track is bounced or frozen. While a track is frozen, its MIDI FX can't be changed.

Adding and editing MIDI FX

1. Open the Mixer (**X**) or Inspector (**I**) and find the software instrument track's strip.
2. Click **+** under **MIDI FX** and choose an effect: **Arpeggiator**, **Chord Trigger**, **Note Repeater**, **Randomizer**, **Scale Quantize**, **Transposer** or **Velocity Processor**.

3. Click the effect's slot to open its editor, a small popover with a one-line description, an **On** switch and its settings. Changes are heard straight away, and each one can be undone with ⌘Z (a slider's change counts when you let go).
4. Click and **hold** the slot, or **Control-click** it, for its menu: **Edit *name*…**, **Bypass**, **Move Up**, **Move Down** and **Remove**. A bypassed effect is dimmed.



The Arpeggiator's editor, opened from its MIDI FX slot

Note values used below: **1/4** is a quarter note, **1/8** an eighth, and so on; a **T** after one means a triplet (**1/8T** is an eighth-note triplet).

Arpeggiator

Plays held chords as a pattern of single notes, in time with the song. Hold a chord and it steps through the notes; let go of every key and the pattern starts again from the beginning next time.

Setting	Choices	Default	What it does
Rate	1/4, 1/4T, 1/8, 1/8T, 1/16, 1/16T, 1/32, 1/32T	1/16	How fast the pattern steps.
Order	Up, Down, Up & Down, Down & Up, As Played, Random, Chord	Up	The order of the notes. **As Played** follows the order you pressed the keys; **Chord** plays the whole chord on every step.

Setting	Choices	Default	What it does
Octaves	1 – 4	1	How many octaves the pattern climbs through.
Gate	5 – 100 %	80 %	How long each note plays, as a share of a step. Low is staccato.
Velocity	**Fixed** (1 – 127) or As played	As played	Tick **Fixed** to play every note at one velocity (it starts at 100).

Chord Trigger

Plays a whole chord from each note you play, with your note as the root.

Setting	Choices	Default
Chord	Major, Minor, Dominant 7th, Major 7th, Minor 7th, Sus 4, Add 9, Power (5th), Octaves	Major

Note Repeater

Repeats each note like an echo, fading (or growing) and optionally climbing as it goes.

Setting	Range	Default	What it does
Rate	1/4 to 1/32T (as the Arpeggiator)	1/8	The time between repeats.
Repeats	1 – 16	3	How many repeats follow each note.
Velocity	–100 to +100 %	–25 %	How much each repeat's velocity changes from the last. Negative fades out.
Transpose	–12 to +12 semitones	0	How far each repeat moves from the last, for rising or falling runs.

Randomizer

Changes notes a little at random so a part sounds less mechanical. The changes come out the same way every time the song plays (and bounces), so your mix doesn't change from one play to the next.

Setting	Range	Default	What it does
Velocity	± 0 – 64	± 15	How far velocities can move either way.
Timing	± 0 – 120 ticks	± 0	How far notes can move earlier or later. 120 ticks is a 1/32 note.
Pitch	± 0 – 12 semitones	± 0	How far pitches can move…
How Often	0 – 100 %	0 %	…and how often a note's pitch is changed.

Click **New Random Pattern** for a different set of random changes.

Scale Quantize

Moves every note to the nearest note of a scale, so nothing is out of key. When a note is exactly between two scale notes, it goes to the lower one.

Setting	Choices	Default
Root	C, C♯, D, D♯, E, F, F♯, G, G♯, A, A♯, B	C
Scale	Major, Natural Minor, Harmonic Minor, Dorian, Mixolydian, Major Pentatonic, Minor Pentatonic, Blues	Major

Transposer

Shifts every note up or down.

Setting	Range	Default
Transpose	−48 to +48 semitones	0

Notes pushed outside the MIDI range are dropped.

Velocity Processor

Makes notes louder or softer, keeps them within a range, or fixes them all at one velocity. Each note's velocity is multiplied by **Scale**, then **Offset** is added, then the result is kept inside **Range**.

Setting	Range	Default	What it does
Fixed	Off, or 1 – 127	Off (As played)	Every note at one velocity. Overrides Scale and Offset.
Scale	0 – 200 %	100 %	Multiplies each velocity. Below 100 % evens out a heavy-handed part.
Offset	−127 to +127 (steps of 5)	0	Adds to each velocity.
Range	1 to 127 (steps of 5)	1 to 127	The lowest and highest velocity allowed.

Limitations

- **The Sampler keeps only the first 60 seconds** (and the first two channels) of a file, and the Convolution Reverb the first 8 seconds (and two channels) of an impulse response. Each says so when it cuts a file short.
- **Channel EQ, Compressor and Delay have no factory presets.** Save your own with the header's save button.
- **Plug-in undo and A/B are per window.** They're cleared when the window closes; after that, use **Edit › Undo** in the main window.
- **Plug-in settings show in their own units in the automation strip only while the plug-in is loaded** (see Chapter 7); otherwise they show as a percentage of their range. Instrument settings can't be automated in Ongaku Studio.
- **MIDI FX only work inside Ongaku Studio**, on software instrument tracks.
- **The Delay assumes 120 BPM** until the app hosting it has told it the song's tempo. If the tempo readout at the top right doesn't match your song, synced echoes won't either.

- **There's no way to turn MIDI FX output into ordinary notes.** The effects always play live from your original notes.

Quick reference

Action	How
Add an Ongaku effect	Channel strip › **+** under **Audio FX** › **Ongaku** › category
Choose an Ongaku instrument	Instrument name on the track header or strip › **Ongaku**
Open a plug-in window	Click the insert slot; instrument › **Show Instrument Window…**
Turn a knob	Drag up or down
Type a value (effects)	Double-click the value, type, press Return
Reset a knob or value	Option-click it (instruments: double-click works too)
Choose a preset, patch, kit or room	Header preset menu, or the **< >** arrows
Save, rename or delete your preset	Header save button; preset menu › **Rename** / **Delete**
Compare two settings	**A** / **B** in the header; ⋯ › **Copy A to B**
Undo or redo in a plug-in window	Header arrows, or ⌘Z / ⇧⌘Z
Load a sample	Sampler › **Load Sample…**, or drag a file onto the waveform
Load an impulse response	Convolution Reverb › **Load Impulse Response…**, or drag a file onto the picture
Take the Session Drummer's part as MIDI	Drag **Drag MIDI** into the tracks, or click the export button
Add a MIDI effect	Channel strip › **+** under **MIDI FX**
Edit a MIDI effect	Click its slot
Bypass, move or remove a MIDI effect	Click and hold its slot, or Control-click it

9. Library, Bounce, MIDI Files and Settings

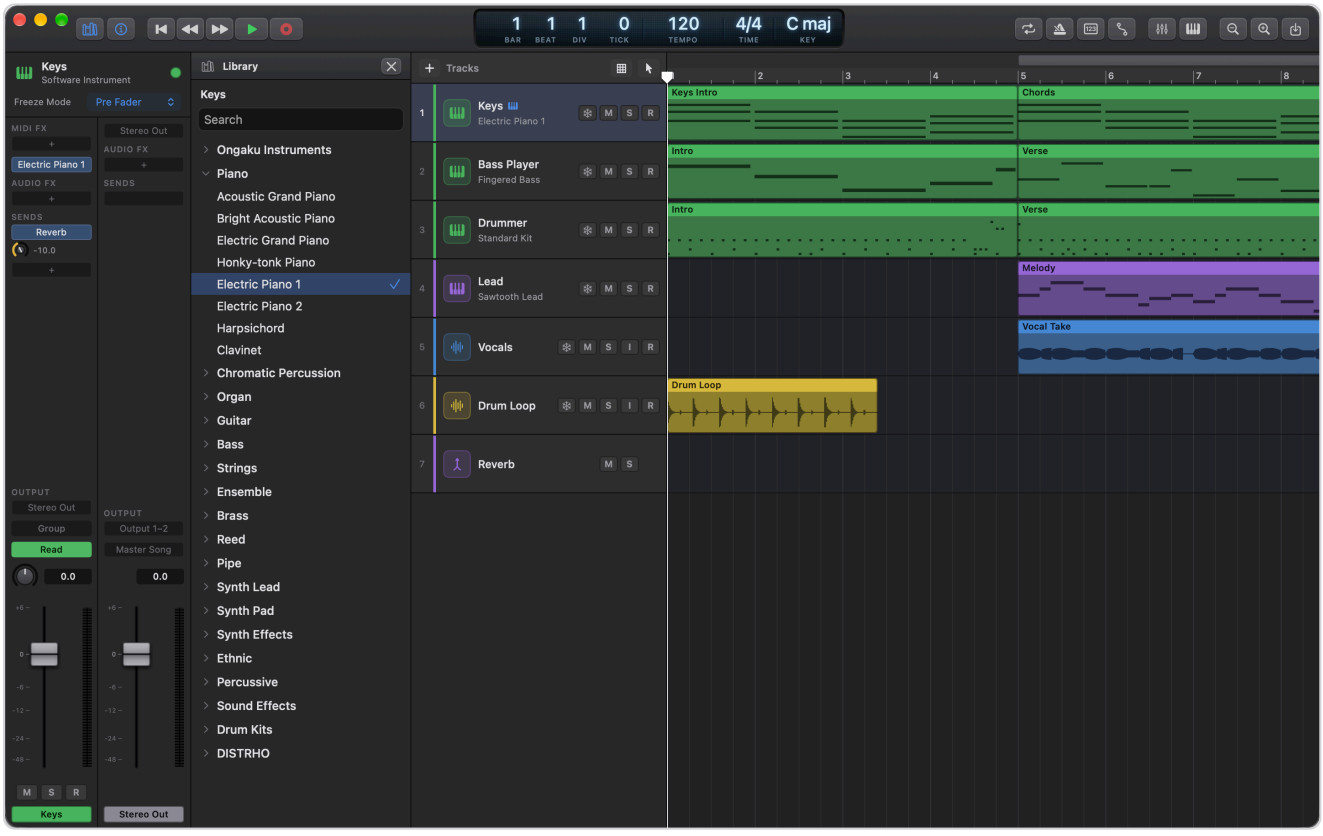
This chapter covers getting sounds in and music out. You'll use the Library to pick the sound a track plays, Bounce to turn your song into an audio file you can share, MIDI files to bring parts in from other apps (or send them out), and the Settings window to set up how Ongaku Studio behaves on your Mac.

The Library

What it is

The Library is a panel on the left of the project window that lists every sound a software instrument track can play. Click a sound and the selected track switches to it straight away. You don't need to confirm anything, so you can keep playing your MIDI keyboard while you click down the list and hear each sound in turn. It's the quickest way to audition sounds.

The Library only deals with **instrument sounds** for software instrument tracks. It doesn't list effects, and it doesn't list presets inside a plug-in. (Effects go on the mixer's insert slots, covered in Chapter 7. Presets for Ongaku's own plug-ins are in each plug-in's header, covered in Chapter 8.)



The Library open beside the tracks, with the Keys track's sound ticked

Opening and closing the Library

Use any of these:

- Press **Y**.

- Click the **Library** button at the far left of the control bar. It's lit while the Library is showing.
- Choose **View › Show Library** (the item changes to **Hide Library** while it's open).
- Click the close button in the Library's own header (**Hide Library**).

The Library opens next to the Inspector if the Inspector is showing (press **I** for the Inspector).

Choosing a sound for a track

1. Click the header of a software instrument track to select it. The Library shows that track's name at the top. If you select an audio or aux track, the Library says **No Instrument Track Selected**, because those tracks don't have an instrument.
2. Click a category name to open it. The category that holds the track's current sound opens by itself, and the current sound is highlighted with a tick.
3. Click a sound. The track plays it straight away.
4. Play a few notes (or press **Space** to play the song) to hear it. If you don't like it, click another sound. You can also press **⌘Z** to go back, because each change is a normal undoable edit called "Change Instrument".

What's in the list

The categories always appear in this order:

Category	What it holds
Ongaku Instruments	Ongaku's own instruments: **Retro Synth**, **Drum Machine**, **Sampler** and **Session Drummer** (see Chapter 8).
General MIDI families	The 128 standard General MIDI sounds from the Mac's built-in sound set, in 16 families: **Piano**, **Chromatic Percussion**, **Organ**, **Guitar**, **Bass**, **Strings**, **Ensemble**, **Brass**, **Reed**, **Pipe**, **Synth Lead**, **Synth Pad**, **Synth Effects**, **Ethnic**, **Percussive** and **Sound Effects**.
Drum Kits	Seven General MIDI drum kits: **Standard Kit**, **Power Kit**, **Electronic Kit**, **TR-808 Kit**, **Brush Kit**, **Orchestra Kit** and **Sound Effects Kit**.
One category per maker	Any other Audio Unit instruments installed on your Mac, grouped under their manufacturer's name. An instrument you install while Ongaku Studio is open appears here (and in the instrument menus) by itself, without opening a new window.

Searching

Type in the **Search** field at the top of the Library to filter the list. Every word you type has to appear in the sound's name or its category's name, and capitals and accents don't matter. So "bass" finds everything in the Bass family plus any sound with "bass" in its name, and "synth pad" narrows it to the Synth Pad family. While you're searching, every matching category stays open. If nothing matches, the Library tells you so. Clear the field to see everything again.

Another way: the track's instrument menu

Each software instrument track header shows its current sound's name under the track name. Click it (or right-click the header and choose **Instrument**) to get the same sounds as menus: **Ongaku**, **General MIDI**, **Drum Kits** and **Audio Units**. For a plug-in instrument there's also **Show Instrument Window…**. The menu is handy when the Library is closed. The Library is better for auditioning, because you can click through many sounds in a row.

Tips and limitations

- A new choice replaces the track's instrument completely. If you had adjusted a plug-in instrument's settings, choosing a different sound starts fresh. Press ⌘**Z** if you didn't mean to.
- If a sound you expect isn't there, check that you selected a *software instrument* track and not an audio track. If you've just installed it, close and open the Library (**Y** twice): it looks for new instruments each time it opens.

Bounce: turning your song into an audio file

What it is

"Bouncing" means recording the whole mix, with every track, effect, fader and automation move, into a single stereo audio file. You'd bounce to send a song to a friend, upload it, burn it to a CD, or put it on your phone. Ongaku Studio renders the bounce faster than real time, so you don't have to sit and listen while it works.

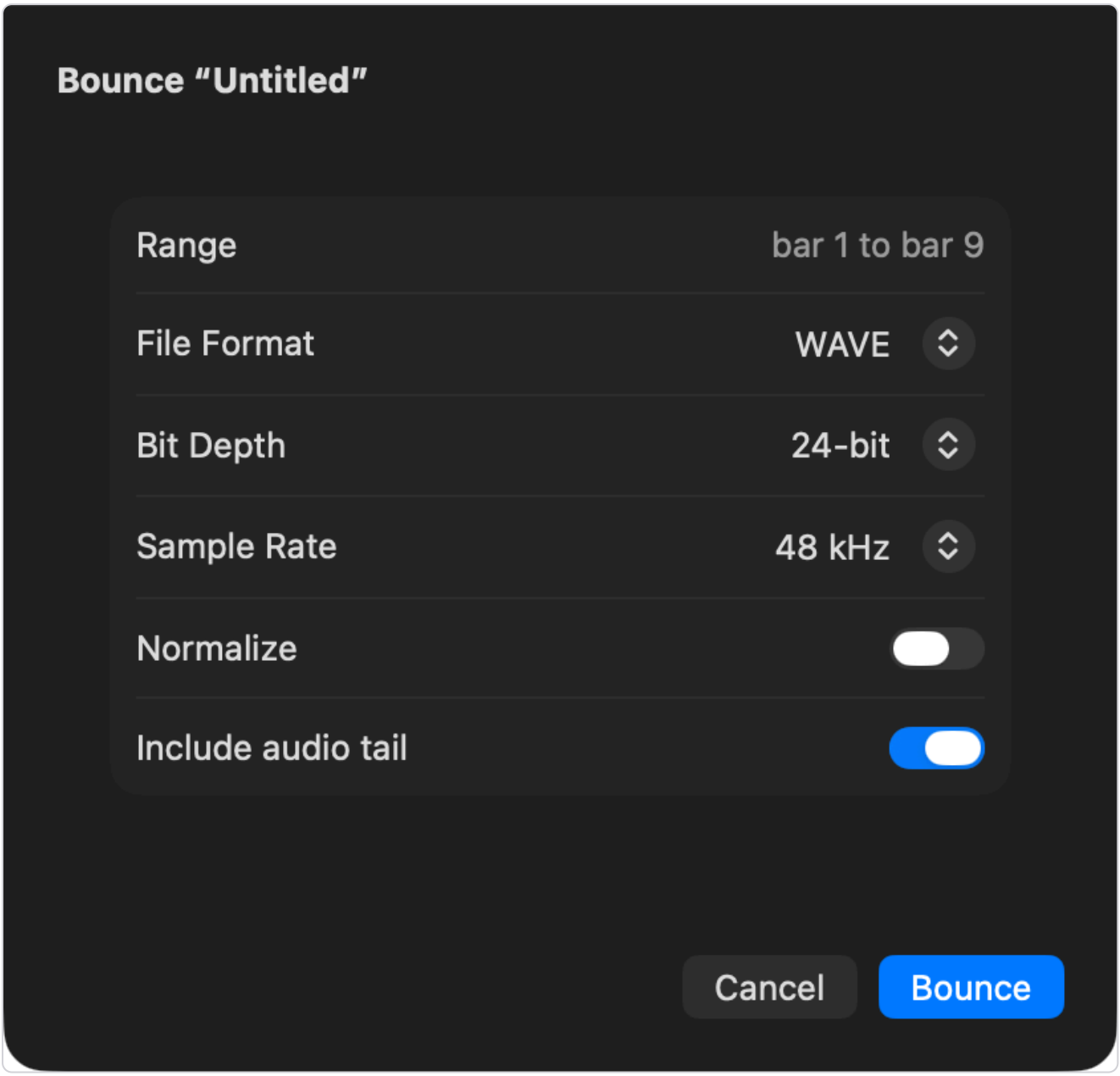
Mute and solo count: whatever you'd hear when you press play is what gets bounced, including any mastering plug-ins on the **Stereo Out** strip (Chapter 7). The metronome is never bounced. Plug-in windows you still have open are included with their current settings.

What part of the song is bounced

Bounce decides the range for you, the way Logic does:

- **If Cycle is on** (press **C**), it bounces exactly the cycle range. This is how you bounce just a section, such as the chorus.
- **If Cycle is off**, it bounces from the very start of the project to the end of the last region, including the repeats of looped regions.

The dialog shows the range it will use at the top, for example "bar 1 to bar 33" or "Cycle range, bar 5 to bar 9". If the project has no regions at all, you'll see "There's nothing to bounce: the project has no regions." instead of the dialog.



The Bounce dialog with its options

Bouncing step by step

1. If you only want part of the song, set the cycle range in the ruler and turn Cycle on. Otherwise turn Cycle off.
2. Choose **File › Bounce…** or press **⌘B**. Playback stops and the Bounce dialog opens, titled with your project's name.
3. Choose the options (each is explained below).
4. Click **Bounce** (or press **Return**). A progress bar shows the render. If flexed audio regions are still being processed (Chapter 6), the dialog first says **Finishing processing flexed audio…** and waits for them, so the bounce includes every flex edit. Click **Cancel** (or press **Escape**) to stop it.
5. When it's done, the dialog shows the bounce's **Length** and its **Peak** level. Read any warning (see "Checking the result" below).
6. Click **Save…** (or press **Return**). A save panel opens, suggesting your project's name and the right file extension. Choose a folder, such as Desktop or Music, and click **Save**. The dialog closes.

If you'd rather change something first, click **Bounce Again** to return to the options, or **Discard** to throw the render away and close the dialog.

The options

Option	Choices	What it does
File Format	**WAVE**, **AIFF**, **AAC (M4A)**	WAVE and AIFF are full-quality, uncompressed files that every music app reads. AAC (M4A) is a much smaller compressed file, good for phones, email and sharing.
Bit Depth	**16-bit**, **24-bit**, **32-bit Float** (WAVE); **16-bit**, **24-bit** (AIFF)	How much detail each sample keeps. 24-bit is a safe choice. Use 16-bit for CDs. 32-bit Float is for sending to another studio app for further mixing or mastering.
Quality	Shown for AAC only: **256 kbit/s**	AAC files are always encoded at 256 kbit/s, which is high quality for a compressed file. There's nothing to choose.
Sample Rate	**44.1 kHz**, **48 kHz**, **88.2 kHz**, **96 kHz** (AAC: 44.1 or 48 kHz only)	Samples per second. 44.1 kHz is CD and most streaming. 48 kHz is video. Higher rates make bigger files.
Normalize	On or off	Raises or lowers the whole bounce so its loudest moment sits just below full scale (–0.1 dBFS, or –1 dBFS for AAC, which needs a little more room). It's an easy way to get a good level, and it stops clipping. It doesn't change the balance or squash the dynamics: the whole file goes up or down by the same amount.
Include audio tail	On or off	When on, the bounce keeps going after the last note until reverb and delay have faded away, so the ending isn't cut off. It stops once the sound has stayed silent for half a second, and never runs more than 20 seconds past the end. Turn it off if you need the file to stop exactly at the end of the range, for example for a loop.

Some notes on these:

- Ongaku Studio remembers your last choices for the next bounce, even after you quit.
- If you switch to a format that can't store the bit depth or sample rate you'd picked, the dialog shows the nearest one it can store, and that's what gets written. For example, 32-bit Float becomes 24-bit for AIFF.
- 16-bit bounces automatically get *dither*: a tiny amount of noise that keeps quiet fades smooth instead of grainy. You don't need to do anything.

Checking the result

After rendering, the dialog may warn you:

- **"The mix goes over 0 dBFS and will clip."** Part of the mix is louder than a digital file can hold, which sounds like harsh crackling. Lower your faders, or turn on **Normalize**, then click **Bounce Again**.

- **"The bounce is silent."** Nothing was heard. Check that your tracks aren't muted, or that a solo isn't silencing everything.

Where the file goes

Until you click **Save…**, the bounce lives in a temporary folder that Ongaku Studio cleans up itself. **The file only ends up somewhere you can find it once you save it** in the save panel. If you click **Discard** or close the dialog, the render is deleted.

Limitations

- You can't type a custom start and end. Use the cycle range to choose a section.
- Bounce always renders the whole mix. To render a single track, solo it first. (To save processor power on a heavy track, use **Track › Freeze Tracks** instead. That renders the track for playback inside the project, not as a file you can share.)

MIDI files

What they are

A Standard MIDI File (".mid") holds notes and controller moves, not sound. Almost every music app can read and write them, so they're the easiest way to move a part or a whole song between Ongaku Studio and other software, or to start from a MIDI file you've downloaded.

Importing a MIDI file

1. Choose **File › Import MIDI File…**.
2. Pick one or more ".mid" files and click **Open**.

You can also **drag MIDI files from the Finder** onto the tracks area. You can drag audio and MIDI files together, and each goes the right way.

Each imported track gets a new software instrument track, added below your existing tracks. The whole import is one step, so ⌘**Z** undoes it.

What happens to the file's contents

In the file	In Ongaku Studio
Each track	A software instrument track with one MIDI region, named after the file's track name (or "Track 1", "Track 2"… if it has none). The region runs from the start of the bar with the first note to the end of the bar with the last one.
A file track that uses several MIDI channels	One Ongaku track per channel, named like "Piano (Ch 2)".
Channels with only controllers and no notes	Left out.

In the file	In Ongaku Studio
Program changes	The first one on each channel picks the track's General MIDI sound. Channel 10 becomes a General MIDI drum kit. Later program changes are ignored.
Controllers, pitch bend, aftertouch	Kept in the region. You can see and edit them in the Piano Roll's MIDI Draw lane (Chapter 4).
Tempo and time signature changes	Used only if your project is empty (see below).
System exclusive and other data	Ignored.

Importing into an empty project vs. an existing one

This is the part that most often surprises people:

- **Into a project with no regions yet** (for example a brand-new project), Ongaku Studio takes over the file's **tempo, time signatures and key signatures**, including every tempo change and key change, and places the notes exactly where they are in the file. Use this when you want to start a song from a MIDI file.
- **Into a project that already has regions**, your project keeps **its own** tempo and meter. The file's tempo is ignored, and its parts are placed starting at **the bar containing the playhead** (or the bar where you dropped the file, if you dragged it in). The notes keep their positions in bars and beats, so they'll play at your song's tempo.

If you import several files at once into an empty project, the first file sets the tempo and the rest follow your new song's tempo.

Track colours

By default every imported track gets the colour set for new instrument tracks (green, unless you've changed it in **Settings › Display**). If you'd like each track in a different colour, so a big arrangement is easier to read, change **Settings › MIDI › Importing MIDI Files › Track colors** to **A different color for each track**. See "Settings" below.

Limitations

- Only format 0 and format 1 files with normal (beat-based) timing are supported. Format 2 files and SMPTE-timed files show an error.
- A file with no notes shows "doesn't contain any notes" and adds nothing.
- Imported tracks play the Mac's built-in General MIDI sounds. Use the Library to swap in a better instrument afterwards.

Exporting the project as a MIDI file

Use this to take your notes into another app, or to give a part to someone who uses different software.

1. Choose **File › Export Project as MIDI File…**.
2. Choose a name and folder (the project's name is suggested) and click **Save**.

What gets written:

- A first "conductor" track with your song's **tempo changes and time signatures**.
- One track for **each software instrument track**, named after it, with its notes and MIDI controllers. Tracks using a General MIDI sound or drum kit also get a program change, so the other app picks a similar sound.
- What you hear is what you get: **muted regions and muted notes are left out**, and notes are cut off at the end of their region.
- **Looped regions** are written with their repeats, as far as they play: to the next region on the track, the loop end you dragged, or the end of the song.
- **Take folders** are written as their comp plays: each part from the take chosen for it.

Limitations

- Audio tracks and aux tracks aren't exported, because they don't hold MIDI.
- Mixer automation, markers and the chord track aren't written to the file. The key and key changes are, as key signatures.
- Tracks playing an Ongaku instrument or another plug-in get no program change, so the receiving app picks its own default sound.
- MIDI effects aren't applied: the file has the notes you played, not what an arpeggiator (say) makes of them.

Importing audio files

You can bring in recordings, loops and samples with **File › Import Audio Files…** (⇧⌘I), the **Import Audio** button at the right of the control bar, or by dragging files from the Finder onto the tracks area.

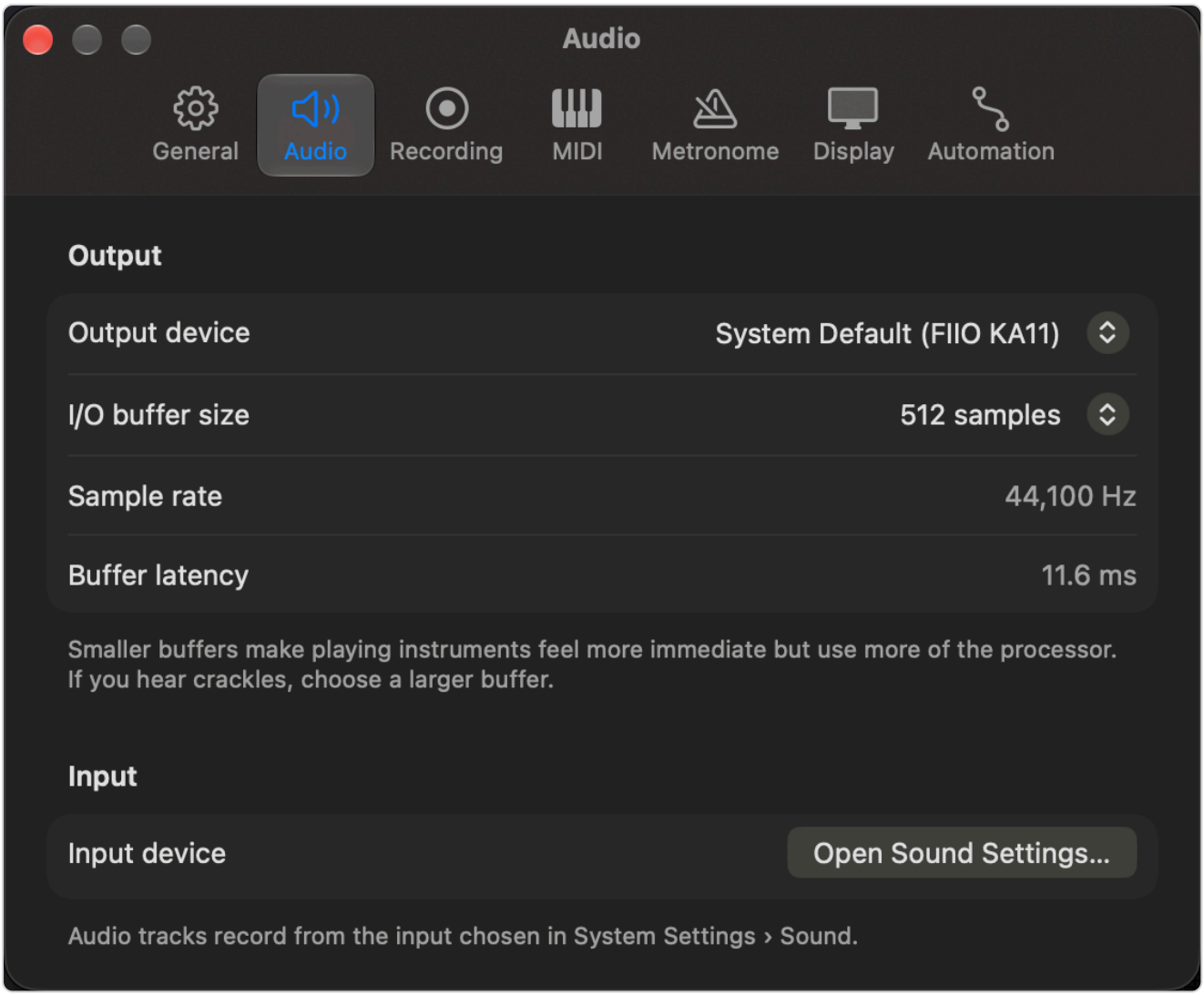
- From the menu or button, files go in at the playhead (snapped to the grid, as dropped files are). If an audio track is selected, they're placed one after another on that track. Otherwise each file gets its own new audio track.
- When you drag files in, they go where you drop them (snapped to the grid), onto the audio track under the pointer if there is one.
- The files are copied into your project when you save, so the project keeps working even if you move or delete the originals.

See Chapter 2 for working with audio regions once they're in.

Settings

What it is

The Settings window holds the choices that apply to the whole app rather than to one song: your audio interface, your MIDI keyboards, the metronome sound, and so on. Open it with **Ongaku Studio › Settings…** (⌘,). The window has a tab for each area. Changes take effect straight away in every open project, and there's no OK button. Just close the window when you're done.



The Settings window, Audio tab

General

Setting	What it does
New projects start with	What **File › New** gives you: **One software instrument track** (the default), **One audio track**, or **No tracks**. Pick the one that matches how you usually start.
Snap regions to	What dragging regions, resizing them and clicking in the tracks area lines up to. **Automatic (follows the zoom)** (the default) snaps to bars when you're zoomed out and finer divisions as you zoom in. You can also fix it to **Bars**, **Beats**, **1/8 Triplets** or **1/16 Triplets**, or turn it **Off** for free placement. It's the same setting as the **Snap** button above the track headers, so changing one changes the other. Automation points snap to it too. The Piano Roll has its own **Grid** menu and **Snap** button for notes.
Scroll to follow the playhead during playback	When on (the default), the tracks area scrolls along as the song plays so the playhead stays in view. Turn it off if you want to look at another part of the song while it plays. When you jump the playhead while stopped, the view always follows.

Audio

Setting	What it does
Output device	Where Ongaku Studio plays: **System Default** (whatever macOS uses, shown in brackets), or a specific interface or headphones. If the device you chose is unplugged, it shows as **Not Connected** and stays chosen, and the system default plays in the meantime.
I/O buffer size	How many samples the Mac processes at a time: 32 to 2048 (default **512 samples**), limited to what your device accepts. **Smaller** buffers make playing a keyboard feel more immediate but work the processor harder. **Larger** buffers are safer for big mixes. If you hear crackles or dropouts, choose a larger buffer. This only affects Ongaku Studio, not your other apps.
Sample rate	Shown for information: the output device's current rate.
Buffer latency	Shown for information: how long one buffer takes, in milliseconds. This is the delay the buffer adds between pressing a key and hearing it.
Input device	The **Open Sound Settings…** button opens macOS's Sound settings. Audio tracks record from the input chosen there.

Recording

These settings are explained more fully in Chapter 3.

Setting	What it does
Count-in	Clicks before recording starts, so you can get ready: **None**, **1 bar** or **2 bars**. The **Count-in** button in the control bar switches between the length you chose here and **None**.
Click while recording, even with the metronome off	Turns the metronome on during recording only, so you can keep it off for listening back.
File type	The format of new audio recordings: **CAF**, **WAV** or **AIFF**.
Bit depth	**16-bit**, **24-bit** (the default) or **32-bit float**.

Recordings are saved inside the project, at the output device's sample rate.

MIDI

Setting	What it does
Inputs	A switch for each connected MIDI keyboard or controller. Ongaku Studio plays and records from the inputs that are on. Switch one off to ignore it, for example a control surface that sends stray notes. If nothing is connected, it says so.
Track colors (under **Importing MIDI Files**)	**All the same color**: imported tracks get the colour set for new software instrument tracks on the Display tab (the swatch beside it shows that colour). **A different color for each track**: each track gets the next colour in the rainbow, starting again for each file.

Your MIDI keyboard plays the frontmost project window: every record-enabled instrument track, or, if none is armed, the selected track. A keyboard icon after a track's name shows which tracks it's playing (Chapter 3).

Metronome

Setting	What it does
Sound	**Click**, **Woodblock**, **Hi-Hat** or **Beep**.
Volume	How loud the metronome is, from 0% to 100%.
Accent the first beat of each bar	Makes beat one stand out, so you can always tell where the bar starts.

Switch the metronome itself on with **K** or the **Metronome** button in the control bar; whether it's on is saved with each project. The count-in is set on the Recording tab.

Display

Setting	What it does
LCD color	The colour of the display in the middle of the control bar: **Blue**, **Amber**, **Green** or **White**. A sample shows the result.
Colors of New Tracks	The colour new **Software instrument tracks** (green by default), **Audio tracks** (blue) and **Aux tracks** (purple) get. Tracks you already have keep their colours.

Automation

Automation is explained in Chapter 7.

Setting	What it does
Mode for new tracks	The automation mode new tracks start in: **Off**, **Read** (the default), **Touch** or **Latch**. Read plays a track's automation. Touch and Latch also record it while you move a fader or pan control during playback.
Thin recorded automation	Moving a control records lots of points. Thinning removes the ones that don't change the shape of the curve, so it's easier to edit later. **Off (keep every point)**, **Light**, **Normal** (the default) or **Heavy**.

Agents

AI agents are explained in Chapter 11.

Setting	What it does
Let AI agents make music in Ongaku Studio	Switches on the MCP server that agents such as Claude Code connect to. Off by default. **Status** shows whether it's ready and which agents are connected.
Port	The port the server listens on, 7317 unless you change it.
Address	What agents connect to.
Token	The secret agents must send. **Copy** copies it; **New Token** replaces it (agents using the old one stop working).
Copy Setup Command	Copies a Terminal command that adds Ongaku Studio to Claude Code.

About Ongaku Studio

Choose **Ongaku Studio › About Ongaku Studio** to see which version you have. That's useful to know when you report a problem.

Quick reference

To…	Do this
Show or hide the Library	**Y**, the **Library** button, or **View › Show Library**
Change a track's sound	Select the track, then click a sound in the Library
Find a sound	Type in the Library's **Search** field
Bounce the song to an audio file	**File › Bounce…** (⌘**B**), **Bounce**, then **Save…**
Bounce only a section	Set the cycle range and turn Cycle on (**C**), then bounce
Avoid clipping in a bounce	Turn on **Normalize**, or lower the faders
Import a MIDI file	**File › Import MIDI File…**, or drag it onto the tracks area
Keep a MIDI file's tempo	Import it into a project with no regions yet
Export notes as a MIDI file	**File › Export Project as MIDI File…**
Import audio	**File › Import Audio Files…** (⇧⌘**I**), or drag files in
Open Settings	**Ongaku Studio › Settings…** (⌘**,**)
Fix crackles	**Settings › Audio › I/O buffer size**: choose a larger buffer
Change what regions snap to	The **Snap** button above the track headers, or **Settings › General › Snap regions to**
Switch a MIDI keyboard off	**Settings › MIDI › Inputs**

10. Key Commands

This chapter lists every keyboard shortcut in Ongaku Studio, plus the mouse gestures that change when you hold a modifier key. Many shortcuts are the same as Logic Pro's, so if you know Logic they'll feel familiar. If you don't, start with the handful in "The keys you'll use most" and pick up the rest as you go.

How focus affects which keys work

Ongaku Studio has two kinds of shortcut, and they behave a little differently.

Single keys (Space, Return, R, C, K, E and so on) are the Logic-style shortcuts. They work anywhere in the project window, wherever you last clicked, with two exceptions:

- **While you're typing in a text field** (a track name, the tempo, the Library's search field, a chord name), keys type text instead. Press **Return** to finish typing or **Escape** to cancel. Clicking in the tracks area or an editor also ends typing, so the shortcuts work again straight away.
- **They only work in the project window.** If a plug-in window or the Settings window is in front, click the project window first. Space won't start playback from a plug-in window.

They're shown next to their commands in the menus, the same as ⌘ shortcuts: for example **Transport › Play** shows Space, and **View › Show Editor** shows E. Choosing the menu item does the same as pressing the key.

Shortcuts with ⌘ (⌘B, ⌘T, ⌘R…) are ordinary menu shortcuts. They're shown next to their menu items and work whenever a project window is in front.

Where you clicked last matters. The project window remembers whether you last clicked in the **tracks area** (the regions at the top) or the **editor pane** (the Piano Roll and the other editors at the bottom). While the editor pane is open, the area your keys act on has a **thin blue outline**; click in the other area to move it. That decides what some keys act on:

- **Delete** removes the selected *notes* (or selected MIDI Draw points) if you last clicked in the Piano Roll. If you last clicked in another editor (the Step Editor, Step Sequencer, Drummer and so on), it does nothing. If you last clicked a chord on the Chord track, it removes that chord. Otherwise it removes the selected *regions*. With nothing selected, it does nothing. Delete never deletes tracks; use ⌘⌫ for that.
- **⌘X**, **⌘C**, **⌘V**, **⌘A** and **⌘R** act on *notes* if you last clicked in the Piano Roll (with a MIDI region open in it). Otherwise they act on *regions*. The Edit menu even renames **Repeat Regions** to **Repeat Notes** to show which one you'll get.
- ⌥← and ⌥→ nudge the selected *notes* if you last clicked in the Piano Roll, and the selected *regions* if you last clicked in the tracks area.
- **Q**, ⌥↑ and ⌥↓ only work in the Piano Roll, and only after you've clicked in it.
- **T** opens the Tool menu for the area the pointer is over. If the pointer is over neither area, you get the menu for the one you clicked last.

The keys you'll use most

Shortcut	Command	Notes
Space	Play or stop	
Return	Go to the beginning	
R	Record	Press again (or Space) to stop.
C	Cycle on or off	
⌘Z	Undo	
⇧⌘Z	Redo	
E	Show or hide the editor	Opens whichever editor suits the region.
X	Show or hide the mixer	

File and project

Shortcut	Command	Notes
⌘N	**File › New**	What a new project contains is set in **Settings › General**.
⌘O	**File › Open…**	
⌘S	**File › Save**	
⇧⌘S	**File › Duplicate**	Standard macOS. Hold ⌥ in the File menu for **Save As…**.
⌘W	**File › Close**	
⇧⌘I	**File › Import Audio Files…**	Also the **Import Audio** button in the control bar.
(none)	**File › Import MIDI File…**	No shortcut. You can also drag MIDI files onto the tracks area.
⌘B	**File › Bounce…**	See Chapter 9.
(none)	**File › Export Project as MIDI File…**	No shortcut.
(none)	**File › Mastering…**	Opens Mastering on Stereo Out, adding it first if it isn't there. See Chapter 8.
(none)	**File › Analyze Whole Song for Mastering**	Mastering listens to the whole song offline. See Chapter 8.
⌘,	**Ongaku Studio › Settings…**	
⌘Q	**Ongaku Studio › Quit Ongaku Studio**	

Transport

Shortcut	Command	Notes
Space	Play / Stop	**Transport › Play** or **Stop**.

Shortcut	Command	Notes
Return or Enter	Go to Beginning	Moves the playhead to the start of the project.
, (comma)	Back One Bar	From inside a bar, the first press goes to the start of that bar.
. (period)	Forward One Bar	
R	Record / Stop Recording	
C	Turn Cycle On / Off	
K	Turn Metronome On / Off	The sound is set in **Settings › Metronome**. Saved with the project, and undoable.
(none)	Count-in on / off	The **Count-in** button in the control bar. No key.
⌥'	Create Marker	Adds a marker at the playhead (snapped to the grid) and shows the global tracks.
⌥,	Go to Previous Marker	
⌥.	Go to Next Marker	

Editing

These work on regions in the tracks area, or on notes in the Piano Roll if you clicked there last (see "How focus affects which keys work"). While you're typing in a text field, ⌘X, ⌘C, ⌘V and ⌘A cut, copy, paste and select text as usual.

Shortcut	Command	Notes
⌘Z	Undo	Every edit can be undone.
⇧⌘Z	Redo	
⌘X	Cut	Regions or notes.
⌘C	Copy	Regions or notes.
⌘V	Paste	Regions or notes.
⌘A	Select All	All regions, or all notes in the open region.
Delete or Forward Delete	**Edit › Delete Selection**	Deletes the selected regions, or the selected notes if you clicked in the Piano Roll last. With nothing selected, it does nothing. It never deletes tracks.
⌘R	Repeat Regions / Repeat Notes	Places a copy straight after the selection.
⌘T	Split Regions at Playhead	Splits the selected regions, or, with none selected, every region under the playhead on the selected track.
⌘J	Join Regions	Needs at least two selected regions.

Shortcut	Command	Notes
⌥← / ⌥→	Nudge regions	Moves the selected regions one snap-grid step earlier or later, in the tracks area. One undo step per press. (In the Piano Roll, these nudge notes.)
⌥⌘F	Pack Take Folder	Turns the selected regions into one take folder. It's in the **Edit** menu, not the Track menu. Needs at least two selected regions.

Section Between Locators

These edit the whole song between the cycle locators, across every track. They're under **Edit › Section Between Locators**, and need a cycle range to be set.

Shortcut	Command	Notes
^⌘X	Cut Section	Removes the section and closes the gap.
^⌘Z	Insert Silence	Opens up an empty gap the length of the cycle range.
^⌘R	Repeat Section	Repeats the section straight after itself.

Tools

Press **T** to open the Tool menu, then press the tool's key. Each tool's key is shown in the menu. **Escape** closes the menu without changing tools. Any other key also closes it. You can also click a tool in the menu, or use the tool button at the top of the track headers (or of the Piano Roll).

The keys pressed after T mean something different from the same keys on their own. **T** then **E** picks the Eraser and doesn't open the editor, and **T** then **I** picks the Scissors and doesn't open the Inspector.

Tracks area tools

Shortcut	Command	Notes
T, then T	Pointer Tool	Select, move and resize regions.
T, then P	Pencil Tool	Click an empty instrument lane to add a region.
T, then E	Eraser Tool	Click a region to delete it.
T, then I	Scissors Tool	Click a region to split it there.
T, then G	Glue Tool	Click a selected region to join the selected regions.
T, then M	Mute Tool	Click a region to mute or unmute it.
T, then A	Fade Tool	Drag over an audio region's start or end to fade it, or where two regions meet or overlap to crossfade.

Piano Roll tools

Click in the Piano Roll (or point at it) before pressing T.

Shortcut	Command	Notes
T, then T	Pointer Tool	Select, move and resize notes. Drag in empty space to select several.
T, then P	Pencil Tool	Click to add a note, dragging to set its length.
T, then B	Brush Tool	Drag to paint notes along the grid.
T, then E	Eraser Tool	Click a note to delete it (or the selection, if it's selected).
T, then I	Scissors Tool	Click a note to split it there.
T, then G	Glue Tool	Click a selected note to join the selected notes of each pitch.
T, then M	Mute Tool	Click a note to mute or unmute it.
T, then V	Velocity Tool	Drag a note up or down to change its velocity (and the selection's).
T, then Q	Quantize Tool	Click a note to quantize it (or the selection) to the grid.
Hold ⌘	Pencil, temporarily	While the Pointer is chosen, holding ⌘ gives you the Pencil.

Tracks

Shortcut	Command	Notes
⌥⌘A	**Track › New Audio Track**	
⌥⌘S	**Track › New Software Instrument Track**	
⌥⌘U	**Track › New Drummer Track**	
(none)	**Track › New Drummer Track (Drum Machine)**	No shortcut. A drummer on the Ongaku Drum Machine.
(none)	**Track › New Aux Track**	No shortcut.
(none)	**Track › New Bass Player Track**, **New Keyboard Player Track**	No shortcut. See Chapter 5.
(none)	**Track › New Pattern Region**	No shortcut. On the selected instrument track, at the playhead's bar.
⌘D	**Track › Duplicate Track**	
⌘⌫ (⌘-Delete)	**Track › Delete Track**	Deletes every selected track (the menu says **Delete Tracks** when several are selected). A summing stack takes its subtracks with it. One ⌘Z brings them all back.
⇧⌘D	**Track › Create Summing Stack**	Groups the selected tracks into a stack.
(none)	**Track › Flatten Stack**	No shortcut.

Shortcut	Command	Notes
(none)	**Track › Freeze Tracks** / **Unfreeze Tracks**	No shortcut.
(none)	**Track › Add Arrangement Marker**	No shortcut.
(none)	**Track › Detect Chords**	No shortcut. Fills the Chord track from the selected MIDI regions.
G	**Track › Show Global Tracks** / **Hide Global Tracks**	

Views, panes and windows

Shortcut	Command	Notes
Y	Show / Hide Library	Also the **Library** button and **View › Show Library**.
I	Show / Hide Inspector	Also the **Inspector** button and **View › Show Inspector**.
E	Show / Hide Editor	Also the **Editor** button and **View › Show Editor**. Opens the selected region in whichever editor suits it: Piano Roll, Step Editor, Step Sequencer, Drummer, Player, Takes or Flex Pitch.
X	Show / Hide Mixer	Also the **Mixer** button and **View › Show Mixer**. The mixer and the editor share the bottom of the window, so showing one hides the other.
A	Show / Hide Automation	Also the **Automation** button and **View › Show Automation**.
G	Show / Hide Global Tracks	It's in the **Track** menu (**Track › Show Global Tracks**), not the View menu. There's no control bar button.
⌘=	Zoom In	Zooms the tracks area. Also the zoom buttons in the control bar.
⌘-	Zoom Out	
⌘M	Minimize window	Standard macOS.

Piano Roll

These work when the Piano Roll is showing a MIDI region and you last clicked in it. The general editing keys (⌘X, ⌘C, ⌘V, ⌘A, ⌘R, Delete) act on notes here too.

Shortcut	Command	Notes
Q	Quantize	Quantizes the selected notes to the grid, using the strength and swing set in the Piano Roll's Quantize controls. **With nothing selected, it quantizes every note in the region.**
⌥↑	Transpose up a semitone	Selected notes only.

Shortcut	Command	Notes
⌥↓	Transpose down a semitone	
⌥⇧↑	Transpose up an octave	
⌥⇧↓	Transpose down an octave	
⌥←	Nudge left one grid step	Selected notes only. The step is the Piano Roll's grid setting.
⌥→	Nudge right one grid step	
Delete	Delete selected notes	If points are selected in the MIDI Draw lane, Delete removes those instead. With nothing selected, it does nothing.

Step Editor and Step Sequencer

The Step Editor and Step Sequencer have no keys of their own. They're mouse-driven (see the gestures table below). The transport and view keys still work while they're showing. **Delete** does nothing while you're working in them (or in the Drummer, Player, Takes and Flex Pitch editors), so it can't remove Piano Roll notes you can't see. See Chapter 4.

Mixer

The mixer has no keys of its own apart from **X** to show and hide it. Its reset gestures are in the table below. See Chapter 7.

Dialogs, text fields and plug-in windows

Shortcut	Command	Notes
Return	Confirm	In the Bounce dialog this is **Bounce** and then **Save…**. It also confirms a new track name, tempo, chord, fade length or preset name.
Escape	Cancel	In the Bounce dialog this is **Cancel** or **Discard**. It also cancels typing a track name, tempo, fade length or plug-in value, and closes the Tool menu.
⌘Z	Undo (in a plug-in window)	In the window of an Ongaku plug-in, ⌘Z undoes your last change to that plug-in's settings, the same as its header's undo button.
⇧⌘Z	Redo (in a plug-in window)	

Mouse and modifier gestures

Tracks area

Shortcut	Command	Notes
Click	Select a region	With the Pointer or Pencil. Other tools act on the region instead.
⇧-click	Add a region to the selection, or take it away	
Drag a region	Move it (also to another track)	Snaps as set with the **Snap** button above the track headers (or **Settings › General › Snap regions to**).
^ while dragging	Move freely, off the grid	Press Control after the drag starts (a Control-click is a right-click). Works for regions, their edges, notes and automation points.
⌥-drag a region	Copy it	The pointer shows a +, and the original stays where it was. What counts is whether ⌥ is held when you let go.
Drag a region's left or right edge	Resize it	
Drag the upper half of a region's right edge	Loop it	The pointer changes to show the loop handle.
Drag across empty space	Select regions with a rubber band	Pointer only. Hold ⇧ to add to the selection.
Double-click a region	Open it in the editor	
Double-click an empty instrument lane	Create a MIDI region	The Pencil does this with a single click.
Drag in the lower half of an audio region	Flex Time	When Flex is on. See Chapter 6.
Double-click a flex marker	Delete the flex marker	Only while the track's Flex mode is on.
Fade Tool: drag where two audio regions meet or overlap	Crossfade them	See Chapter 2.
Right-click a region or lane	Region menu	Cut, Copy, Paste, Repeat, Split, Loop, New MIDI/Pattern/Drummer/Player Region, Take Folder and more.
Drag in the ruler's upper strip	Set the cycle range	Click the strip to turn cycle on or off.
Click the ruler's lower part	Move the playhead	

Track headers

Shortcut	Command	Notes
Click	Select a track	
⇧-click	Select every track between this one and the last one clicked	
⌘-click	Add or remove one track from the selection	
Double-click the track name	Rename	Return to confirm, Escape to cancel.
Right-click	Track menu	Rename, stack, freeze, instrument, **Color** and **Delete Track**. On one of several selected tracks, it's **Delete Tracks** and deletes them all.
Right-click the snowflake	**Freeze Mode**	Choose **Pre Fader** or **Source Only** (not while the track is frozen).
Double-click the divider between headers and lanes	Reset the header width	

Global tracks

Shortcut	Command	Notes
Double-click an empty spot on the Marker track	Add a marker	
Click a marker	Move the playhead there	Drag to move it, double-click to rename it.
Drag an arrangement section	Move it, with everything in it	Drag its right edge to resize it. Click to select it with its regions. Double-click to rename it.
Double-click an empty spot on the Chord track	Add a chord	Double-click a chord to change it, drag it along the grid, right-click for more.
Click a chord	Select it and move the playhead there	Then Delete (or **Edit › Delete Selection**) removes it.
Drag a tempo handle up or down	Change the tempo	Drag sideways to move the change.
Double-click an empty spot on the Tempo track	Add a tempo change	
Click a time signature	Open its box	Set the beats, the beat unit or a common meter, or **Delete** it.
Click **+** beside **Signature**	Add a time signature change at the next bar	Opens its box.
Drag a time signature change	Move it to another bar	Right-click to change or delete it.

Control bar and window

Shortcut	Command	Notes
Double-click the tempo	Type a new tempo	Return to confirm, Escape to cancel.
Click the time signature or key	Choose a new one	
Click **Count-in**	Count-in on or off	Uses the length set in **Settings › Recording**.
Click the **Fade In** or **Fade Out** number in the Region Inspector	Type a fade length	"250", "250 ms" or "1.5 s". Return to confirm, Escape to cancel.
Drag the divider above the editor or mixer	Resize the pane	Double-click it to go back to the default height.

Automation lanes

Shortcut	Command	Notes
Click the automation strip	Add a point	The strip is under the track's lane while automation is shown (A). Points snap to the grid.
Drag a point	Move it	Hold ^ after you start dragging to move it off the grid.
Double-click a point	Delete it	

Piano Roll

Shortcut	Command	Notes
Click a note	Select it (and hear it)	
⇧-click a note	Add it to the selection	
Drag a note	Move the selection (time and pitch)	Snaps to the grid unless the **Snap** button is off. Hold ^ after you start dragging to move freely.
⌥-drag a note	Copy the selection	The originals stay where they are.
Drag a note's start or end	Resize it (and the rest of the selection)	
Double-click empty space	Add a note	Pointer tool.
Drag across empty space	Select notes with a rubber band	Hold ⇧ to add to the selection.
Hold ⌘ and click or drag	Pencil tool	While the Pointer is chosen.
Click a key on the keyboard	Select every note of that pitch	⇧-click adds them to the selection.
Right-click	Note menu	Cut, Copy, Paste, Repeat, Delete, selection commands and note functions (Quantize, Transpose, Velocity, Humanize…).

MIDI Draw lane

Shortcut	Command	Notes
Drag a velocity stalk	Change a note's velocity	All selected notes move together.
Drag across empty space (velocity)	Sweep velocities to the pointer's height	
Click empty space (controllers)	Add a point	
Drag across empty space (controllers)	Draw freehand	
⇧-click	Add to the selection	
Drag the lane's top edge	Change the lane's height	

Step Editor, Step Sequencer and other editors

Shortcut	Command	Notes
Click a step (Step Editor)	Set an event to the height clicked	
Drag across steps (Step Editor)	Paint events	
Double-click a bar (Step Editor)	Remove it	
Click a step (Step Sequencer)	Turn it on or off	
Drag across steps (Step Sequencer)	Paint steps	The edit mode chooses what painting changes.
Drag across a take (Takes editor)	Make that take play there	Quick-swipe comping. Click a take's name to play it throughout.
Right-click a take (Takes editor)	Take menu	**Rename…** or **Delete Take**.
Drag a note (Flex Pitch)	Retune it	Double-click to put it back.

Mixer and plug-ins

Shortcut	Command	Notes
Click a channel strip	Select its track	⇧-click and ⌘-click select more tracks, as on the track headers.
⌥-click a fader	Reset to 0 dB	
Double-click or ⌥-click a knob	Reset to its default	Pan knobs go to the centre, send knobs to –6 dB.
Click an insert slot	Open the plug-in's window	Also on Stereo Out.

Shortcut	Command	Notes
Hold or ^-click an insert or MIDI FX slot	Slot menu	**Bypass**, **Move Up**, **Move Down**, **Remove** and more.
⌥-click a plug-in knob or value	Reset it to its default	In every Ongaku plug-in, effects and instruments. One step in the window's undo.
Double-click a plug-in value	Type a value	In Ongaku effects. Escape cancels.
Double-click a plug-in knob	Reset it	In Ongaku instruments (Retro Synth, Drum Machine, Sampler), and the Session Drummer's **Fills** and **Swing** knobs.

11. Making Music with AI Agents

Ongaku Studio can let an AI agent, such as Claude, work in your song with you. You describe what you want in plain words ("a 16-bar lo-fi groove in D minor with a walking bass", "make the chorus lift more", "mix this and bounce it"), and the agent builds it in the open project window while you watch: tracks appear, sections and chords fill in, the drummer starts playing, effects go on. You can play it, change anything by hand, or undo what the agent did, exactly as if you'd done it yourself.

This chapter explains what an agent can do, how to connect one, and how to work with it.

What it is

An AI agent is a program that uses an AI model to carry out tasks by calling *tools*. Ongaku Studio offers tools for making music through **MCP** (the Model Context Protocol), a standard way for agents to use apps. Claude Code, Claude Desktop and many other agents speak it.

When you switch it on, Ongaku Studio runs a small MCP server on your Mac. An agent that connects to it can:

- **Read the song**: tempo, key, sections, chords, tracks, regions and notes.
- **Set up the song**: tempo, time signature, key and key changes, the cycle, Arrangement sections, markers and the Chord track.
- **Write parts**: create tracks with any sound from the Library, write notes into regions, transform and quantize them, and copy and move regions.
- **Use the session players**: add the Drummer and the Bass and Keyboard players to sections, change their style and settings, or ask for a new take (see Chapter 5).
- **Mix**: set levels, pan, mute and solo; add effects and change their settings and presets; send tracks to an aux with a reverb; draw automation; add MIDI effects such as the Arpeggiator (see Chapter 7 and Chapter 8).
- **Listen, in its way**: an agent can't hear, so it can *measure* the mix instead. It renders the song (faster than real time, nothing is played) and gets its loudness, peaks, stereo width and tonal balance compared with a typical finished master, for the whole song and each section.
- **Finish**: master the song with Ongaku Mastering's Analyze Whole Song, bounce it to an audio file, and save.
- **Play and stop** the song, so you can listen to what it made.

An agent works on the song in the **frontmost project window**. If no song is open, it asks you to open one.

Is it safe?

- The server is **off** until you switch it on in Settings.
- Only programs on **your Mac** can connect, and only with a secret **token** that you give them.
- Every change an agent makes is a normal edit, one per step. The Edit menu names them, such as **Undo Agent: Add Drummer**, so you can always undo.

- Agents can't open or delete files. They can save the open song, and their bounces go only to **Music › Ongaku Studio › Bounces**.
- What an agent reads of your song (its notes, names, structure and measurements) goes to the AI service the agent uses, such as Anthropic for Claude. Ongaku Studio doesn't send your audio. An agent that can read files on your Mac, as Claude Code can, could still open a bounced file, so keep to agents you trust.

Setting it up

Switching on the server

1. Choose **Ongaku Studio › Settings…** (⌘,) and click **Agents**.
2. Turn on **Let AI agents make music in Ongaku Studio**. The **Status** line changes to **Ready for agents**.

The server listens on port **7317**. If another app already uses that port, the status says so; type another number in **Port**.

Connecting Claude Code

Claude Code is Anthropic's agent for the Terminal. If you don't have it, install it first, for example with Homebrew: `brew install --cask claude-code`. Then run `claude` once and sign in.

1. In **Settings › Agents**, click **Copy Setup Command**.
2. Open **Terminal**, paste the command and press Return. It adds Ongaku Studio to Claude Code with your token.
3. Open a song in Ongaku Studio.
4. In Terminal, run `claude` and ask for something, such as "Make a 16-bar lo-fi groove in Ongaku Studio".

Connecting other agents

Other MCP agents need two things from **Settings › Agents**: the **Address** (`http://127.0.0.1:7317/mcp`) and the **Token**, sent as the header `Authorization: Bearer <token>`. Click **Copy** next to the token to copy it.

Agents that can only start a program rather than connect to an address, such as Claude Desktop, can use a bridge such as `mcp-remote`. In Claude Desktop, choose **Settings › Developer › Edit Config** and add:

```
{
  "mcpServers": {
    "ongaku": {
      "command": "npx",
      "args": ["-y", "mcp-remote", "http://127.0.0.1:7317/mcp",
               "--header", "Authorization: Bearer YOUR_TOKEN"]
    }
```

```
  }
}
```

A new token

Click **New Token** to make a new secret, for example if you shared a setup command by mistake. Agents set up with the old token stop working until you set them up again.

Working with an agent

Seeing it work

While an agent is connected, a small badge with its name appears in the control bar, next to the LCD. It glows while the agent is working. Hold the pointer over it to see what it did last. The Settings window's **Status** line also lists the connected agents.

Asking well

Agents do best when you say what you're after musically and leave the clicking to them:

- **Start broad, then refine.** "A gentle indie-folk song, about two minutes, in G" is a good start. Then: "the verse drums are too busy", "give the chorus a brighter keys part", "end with the intro again".
- **Name things the way the song does.** "The Bass Player track", "the second chorus", "bar 17".
- **Ask it to listen.** "Analyze the mix: is the chorus louder than the verse?" makes the agent measure rather than guess.
- **Play it yourself.** The agent can press play, but only you can hear whether it works. Tell it what you think.

Built-in starting points

Agents that support MCP *prompts* list Ongaku Studio's ready-made requests (in Claude Code, type `/` to see them):

Prompt	What it does
Sketch a Song	Builds a whole song from a style and mood: sections, chords, drums, bass, keys, a melody and a rough mix.
Arrange This Idea	Turns what's in the song now (a loop or a few bars) into a full arrangement.
Write a Melody	Writes a melody over the chords of a section, on a new track.
Mix and Master	Balances the mix, adds EQ, compression and a shared reverb, checks it with Analyze Mix, masters it and bounces it.

The agent can also read this manual, so you can ask it how a feature works.

How agents describe music

You don't need to know this to use an agent, but it helps to read what it says:

- Positions are **bar:beat**: "5:3" is beat 3 of bar 5.
- Note lengths are in beats: 1 is a quarter note, 0.5 an eighth.
- Note names put middle C at **C4**. (Ongaku's Piano Roll, like Logic, calls middle C **C3**, so an agent's "C4" is the Piano Roll's C3.)

Tips and limitations

- **Undo what you don't like.** Each agent step is one undo, so ⌘Z takes back the last thing it did. You can also ask the agent to undo.
- **You can keep working.** Edit, play and mix while an agent is connected. It reads the song afresh each time, so it sees your changes.
- **The frontmost window.** If you have several songs open, the agent works on the one in front. Bring the right one forward before asking.
- **Plug-in windows.** If a plug-in's window is open while an agent changes the plug-in, close and reopen the window to see the new settings.
- **What agents can't do yet**: record audio, edit audio regions in detail (Flex, takes, comping), freeze tracks or import files. Ask them to tell you when you need to do something by hand.
- **Measurements aren't ears.** Analyze Mix tells an agent where a mix stands against a typical master, which is a good guide, not a verdict. Trust your ears.